Complex Variable Theory

Previous knowledge

You need to know how to do complex algebra before you start this course. In Cartesian coordinates we have a complex number, z, and its complex conjugate, $\bar{z}$, which has a modulus, $\mid z \mid$,

$$z \equiv x + iy \qquad \bar{z} = x - iy \qquad z\bar{z} = \mid z \mid^2 = x^2 + y^2$$

In polar coordinates we have a modulus, ρ, and an angle, ϕ, and

$$z = \rho e^{i\phi} \qquad \bar{z} = \rho e^{-i\phi} \qquad z\bar{z} = \mid z \mid^2 = \rho^2$$

To convert between representations, you need

$$z = x + iy = \rho e^{i\phi} = \rho \cos\phi + i\rho \sin\phi$$

When you have two numbers you need to be able to add them, best in Cartesian

$$z_1 + z_2 = x_1 + iy_1 + x_2 + iy_2 = x_1 + x_2 + i(y_1 + y_2)$$

multiply them, best in polar

$$z_1 z_2 = \rho_1 e^{i\phi_1} \rho_2 e^{i\phi_2} = \rho_1 \rho_2 e^{i(\phi_1 + \phi_2)}$$

and a mess in Cartesians

$$z_1 z_2 = (x_1 + iy_1)(x_2 + iy_2) = x_1 x_2 - y_1 y_2 + i(x_1 y_2 + y_1 x_2)$$

You need to be able to invert them

$$\frac{1}{z} = \frac{\bar{z}}{\mid z \mid^2} = \frac{x - iy}{x^2 + y^2}$$

You need to know trigonometry, but this is essentially *trivial,* since

$$e^{i\phi_1} e^{i\phi_2} = e^{i(\phi_1 + \phi_2)} = \cos(\phi_1 + \phi_2) + i \sin(\phi_1 + \phi_2) = (\cos\phi_1 + i \sin\phi_1)(\cos\phi_2 + i \sin\phi_2)$$

$$= \cos\phi_1 \cos\phi_2 - \sin\phi_1 \sin\phi_2 + i(\cos\phi_1 \sin\phi_2 + \sin\phi_1 \cos\phi_2)$$

is all there is! You need to know the roots of unity, ie the solutions to z^n=1, and the equation is factorisable into

$$z^n - 1 = (z - 1)(1 + z + ... + z^{n-1}) = 0$$

and the roots are

$$z_m = e^{i\frac{2\pi m}{n}}$$

as $m \in \{0, 1, ..., n-1\}$. You need to know what an Argand diagram is and you need to be able to differentiate and integrate elementary real functions.

From your mathematics courses you need to know Taylor's theorem.

From your vector calculus course you need to know Stoke's theorem

$$\int_C \mathbf{dr}.\nabla \times \mathbf{A}(\mathbf{r}) = \int_S \mathbf{dS}.\mathbf{A}(\mathbf{r})$$

where S is the open simply-connected two-dimensional surface that connects up the boundary $C \equiv \partial S$, and also that for a simply connected domain a field with zero curl may be described by a scalar potential

$$\nabla \times \mathbf{A}(\mathbf{r}) = \mathbf{0} \qquad \Rightarrow \qquad \mathbf{A}(\mathbf{r}) = \nabla \Phi(\mathbf{r})$$

Complex functions are not like real functions

Initially, the most important thing to understand is that complex functions have structure that real functions do not. This is fundamental and is *not* a small difference. This can be explained simply using Taylor's theorem, something that you should all know. We can approximate a function with a power-series and one can ask how good this approximation is. It is natural to take the series all the way to infinity, include all the terms, and then one might expect that we would recover the original function and the expansion is in some sense an exact representation. For real analysis this is simply not true and Taylor's series guarantees absolutely *nothing* about the function, whereas for complex functions, if the series converges, then the series is indeed the function as one might have expected! This fact needs some explanation which we will now give.

We start out with the real function $f(x)$ which is defined by $f(0)=0$ and

$$f(x) = \exp\left(-\frac{1}{x}\right) \qquad x > 0$$

and think about Taylor's series close to the origin. We can differentiate this function as many times as we desire, and we get

$$\frac{d^n f}{dx^n}(x) = P_n\left(\frac{1}{x}\right)\exp\left(-\frac{1}{x}\right)$$

for some polynomial $P_n(z)$. In the limit that $x \mapsto 0$, the exponential goes to zero faster than any polynomial, as $\frac{1}{x} \equiv z \mapsto \infty$, and so this quantity vanishes for all values of n as $x \mapsto 0$. This function has a Taylor's series and that Taylor's series *vanishes* at all orders. For this function it is clear that the function is *not* equal to its Taylor's series, as the function does not vanish. The function $f(x)$ is very flat close to the origin and so Taylor's series is doing quite a good job of describing it, so maybe this is good and not bad?

Once we have this idea, we can then invent a more sophisticated function, $F(x)$, which is defined to vanish when $x \in (-\infty, -1]$ and when $x \in [1, \infty)$ and is

$$F(x) = \exp\left(-\frac{x^2}{1-x^2}\right) \qquad x \in (-1, 1)$$

This function has the same incredibly flat behaviour at both 1 and -1, but at the origin $F(0)=1$, and so this is a 'bump' which connects infinitely smoothly to zero outside of its support. We can then think about

$$RF\left(\frac{2}{b-a}\left[x-\frac{a+b}{2}\right]\right)$$

which is the same bump rescaled to lie in the interval (a,b) and to go up to a height of R. The point to this construction is that a, b and R are completely arbitrary. If we have an infinitely differentiable function we can add such a bump of any height anywhere and we still have an infinitely differentiable function. Taylor's series knows absolutely nothing about such a bump, no matter how big it is nor how close it is to the current point of the expansion, and so infinitesimally close to the point of expansion the function could be *anything,* infinite smoothness and a Taylor's expansion tells us *nothing* about the function....

Once one understands this, then surely we can use the same argument for a complex function which must also be equally sick, but the point is that for complex functions the original function $f(x)$ is not differentiable at the origin and so is not technically an analytic function (technical jargon for differentiable) at the origin. It is the definition of differentiation which is slippery!

Differentiation of a complex function

The standard real-analysis definition of differentiation is

$$\frac{df}{dx}(x)=\lim_{\delta x\mapsto 0}\frac{f(x+\delta x)-f(x)}{\delta x}$$

and as well as the limit existing, there is also the issue of positive and negative values for δx and the limit must be the same. When we try to invoke the same formula for complex functions, the complex plane is two dimensional and there are many different ways that we could limit δz to zero. It is this observation that leads to the difference in structure between real and complex analysis. As we wish to make δz vanish, it is natural to use polar coordinates to describe $\delta z\equiv re^{i\theta}$ where r is small and so we need

$$\frac{df}{dz}(z)=\lim_{r\mapsto 0}\frac{f(z+re^{i\theta})-f(z)}{re^{i\theta}}$$

exists for all values of θ, and moreover gives the same answer. We can perform these limits using real-analysis, if we split the function up into a real part and an imaginary part, $f(z)=X(x,y)+iY(x,y)$, and then

$$\frac{df}{dz}=\lim_{r\mapsto 0}\frac{X(x+r\cos\theta,y+r\sin\theta)+iY(x+r\cos\theta,y+r\sin\theta)-X(x,y)-iY(x,y)}{re^{i\theta}}$$

and then using partial derivatives

$$\frac{df}{dz}(z)=\lim_{r\mapsto 0}\frac{1}{r}e^{-i\theta}\left[r\cos\theta\frac{\partial X}{\partial x}+r\sin\theta\frac{\partial X}{\partial y}+ir\cos\theta\frac{\partial Y}{\partial x}+ir\sin\theta\frac{\partial Y}{\partial y}\right]$$

and we can cancel the r to get

$$\frac{df}{dz}(z) = (\cos\theta - i\sin\theta)\left[\cos\theta\frac{\partial X}{\partial x} + \sin\theta\frac{\partial X}{\partial y} + i\cos\theta\frac{\partial Y}{\partial x} + i\sin\theta\frac{\partial Y}{\partial y}\right]$$

Elementary trigonometry allows us to recognise three possibilities, since

$$\cos^2\theta = \frac{1}{2}\left[1+\cos 2\theta\right] \quad \sin^2\theta = \frac{1}{2}\left[1-\cos 2\theta\right] \quad \cos\theta\sin\theta = \frac{1}{2}\sin 2\theta$$

and so

$$\begin{aligned}\frac{df}{dz}(z) = &\frac{1}{2}\left[\frac{\partial X}{\partial x} + \frac{\partial Y}{\partial y} + i\frac{\partial Y}{\partial x} - i\frac{\partial X}{\partial y}\right] \\ &+\frac{1}{2}\left[\frac{\partial X}{\partial x} - \frac{\partial Y}{\partial y} + i\frac{\partial Y}{\partial x} + i\frac{\partial X}{\partial y}\right]\cos 2\theta \\ &+\frac{1}{2}\left[\frac{\partial Y}{\partial x} + \frac{\partial X}{\partial y} - i\frac{\partial X}{\partial x} + i\frac{\partial Y}{\partial y}\right]\sin 2\theta\end{aligned}$$

and in order for this to not depend on θ, ie the direction that we limit at, we need that

$$\frac{\partial X}{\partial x} = \frac{\partial Y}{\partial y} \qquad \frac{\partial X}{\partial y} + \frac{\partial Y}{\partial x} = 0$$

which are known as *Cauchy-Riemann equations.*

Employing these equations allows us to rewrite the derivative as

$$\frac{df}{dz}(z) = \frac{\partial X}{\partial x} + i\frac{\partial Y}{\partial x} = \frac{\partial Y}{\partial y} - i\frac{\partial X}{\partial y} = \frac{\partial f}{\partial x} = -i\frac{\partial f}{\partial y}$$

and we can interpret these as derivatives along the x- and iy-axes respectively, two natural Cartesian choices out of all choices of angles. We can also recognise that

$$\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y} = 0$$

which is a sort of governing equation and is Cauchy-Riemann in disguise.

Hopefully, you remember d'Alembert's solution to the wave equation

$$\frac{1}{c^2}\frac{\partial^2 f}{\partial t^2} = \frac{\partial^2 f}{\partial x^2}$$

which is solved using the change of variables, $u \equiv x - ct$ and $v = x + ct$, and then the chain rule allows us to deduce that

$$\frac{\partial}{\partial x} = \frac{\partial u}{\partial x}\frac{\partial}{\partial u} + \frac{\partial v}{\partial x}\frac{\partial}{\partial v} = \frac{\partial}{\partial u} + \frac{\partial}{\partial v}$$

$$\frac{1}{c}\frac{\partial}{\partial t} = \frac{1}{c}\frac{\partial u}{\partial t}\frac{\partial}{\partial u} + \frac{1}{c}\frac{\partial v}{\partial t}\frac{\partial}{\partial v} = -\frac{\partial}{\partial u} + \frac{\partial}{\partial v}$$

which allows us to deduce that

$$\frac{1}{c^2}\frac{\partial^2 f}{\partial t^2} - \frac{\partial^2 f}{\partial x^2} = -4\frac{\partial^2 f}{\partial u \partial v} = 0$$

which then solves to provide

$$f(u,v) = F[u] + B[v] = F[x-ct] + B[x+ct] = f(x,t)$$

in terms of a forward moving, $F[z]$, and a backward moving, $B[z]$, piece. Analogous simpler equations offer

$$\frac{1}{c}\frac{\partial f}{\partial t} + \frac{\partial f}{\partial x} = 2\frac{\partial f}{\partial v} = 0 \quad \Rightarrow \quad f[u,v] = F[u] = F[x-ct] = f[x,t]$$

$$\frac{1}{c}\frac{\partial f}{\partial t} - \frac{\partial f}{\partial x} = -2\frac{\partial f}{\partial u} = 0 \quad \Rightarrow \quad f[u,v] = B[v] = B[x+ct] = f[x,t]$$

and we could describe pure forwards or backwards motion with a first order partial differential equation.

The previous governing equation is analogous to this. Mathematically this is quite subtle, but the analogue is to use a change of variables, $z \equiv x + iy$ and $\bar{z} \equiv x - iy$, in terms of which

$$\frac{\partial}{\partial x} = \frac{\partial z}{\partial x}\frac{\partial}{\partial z} + \frac{\partial \bar{z}}{\partial x}\frac{\partial}{\partial \bar{z}} = \frac{\partial}{\partial z} + \frac{\partial}{\partial \bar{z}}$$

$$\frac{\partial}{\partial y} = \frac{\partial z}{\partial y}\frac{\partial}{\partial z} + \frac{\partial \bar{z}}{\partial y}\frac{\partial}{\partial \bar{z}} = i\frac{\partial}{\partial z} - i\frac{\partial}{\partial \bar{z}}$$

and then the previous governing equation is

$$0 = \frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y} = 2\frac{\partial f}{\partial \bar{z}} \quad \Rightarrow \quad f(x,y) = F[z]$$

and suggests that the coordinates are related through $z \equiv x + iy$ and we are not permitted to use the conjugate, $\bar{z} \equiv x - iy$, if we want the derivative to exist.

Laplace's Equation

Under the assumption that the requisite partial derivatives exist, one can employ Cauchy-Riemann equations to show that both the real part and the imaginary part of an analytic function satisfy Laplace's equation. The equations we have are

$$\frac{\partial X}{\partial x} = \frac{\partial Y}{\partial y} \qquad \frac{\partial X}{\partial y} + \frac{\partial Y}{\partial x} = 0$$

and then

$$\frac{\partial^2 X}{\partial x^2} = \frac{\partial^2 Y}{\partial x \partial y} = \frac{\partial^2 Y}{\partial y \partial x} = -\frac{\partial^2 X}{\partial^2 y}$$

$$\frac{\partial^2 Y}{\partial x^2} = -\frac{\partial^2 X}{\partial x \partial y} = -\frac{\partial^2 X}{\partial y \partial x} = -\frac{\partial^2 Y}{\partial^2 y}$$

where we were explicit that the derivatives commute with each other.

You should have sufficient previous knowledge to be able to transform this partial differential equation into polar coordinates and then to solve it using separation of variables. Explicitly, $x \equiv \rho\cos\phi$ and $y \equiv \rho\sin\phi$, gives

$$\frac{\partial}{\partial\rho} = \frac{\partial x}{\partial\rho}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\rho}\frac{\partial}{\partial y} = \cos\phi\frac{\partial}{\partial x} + \sin\phi\frac{\partial}{\partial y}$$

$$\frac{\partial}{\partial\phi} = \frac{\partial x}{\partial\phi}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\phi}\frac{\partial}{\partial y} = -\rho\sin\phi\frac{\partial}{\partial x} + \rho\cos\phi\frac{\partial}{\partial y}$$

which invert to provide

$$\frac{\partial}{\partial x} = \cos\phi\frac{\partial}{\partial\rho} - \sin\phi\frac{1}{\rho}\frac{\partial}{\partial\phi}$$

$$\frac{\partial}{\partial y} = \sin\phi\frac{\partial}{\partial\rho} + \cos\phi\frac{1}{\rho}\frac{\partial}{\partial\phi}$$

and then

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \frac{\partial^2}{\partial\rho^2} + \frac{1}{\rho^2}\frac{\partial^2}{\partial\phi^2} + \frac{1}{\rho}\frac{\partial}{\partial\rho}$$

where the final term comes from the ϕ derivative applied to the coefficients of the derivative in front. There are better ways of doing this, as your curvilinear coordinates knowledge should supply, but this brute force and ignorance method is perfectly serviceable. Now applying the separation of variables idea is equivalent to

$$f(\rho,\phi) = R(\rho)\Phi(\phi)$$

and then

$$\frac{1}{f}\left[\frac{\partial^2 f}{\partial\rho^2} + \frac{1}{\rho}\frac{\partial f}{\partial\rho} + \frac{1}{\rho^2}\frac{\partial^2 f}{\partial\phi^2}\right] = \frac{1}{R}\frac{\partial^2 R}{\partial\rho^2} + \frac{1}{R}\frac{1}{\rho}\frac{\partial R}{\partial\rho} + \frac{1}{\Phi}\frac{1}{\rho^2}\frac{\partial^2\Phi}{\partial\phi^2} = 0$$

and this can be separated using a separation parameter, m, as

$$\frac{1}{R}\rho^2\frac{\partial^2 R}{\partial\rho^2} + \frac{1}{R}\rho\frac{\partial R}{\partial\rho} = m^2 = -\frac{1}{\Phi}\frac{\partial^2\Phi}{\partial\phi^2}$$

All of this should be very familiar to you, but the solving of these equations is crucial to us and so we will devote some time to thinking about this. The angular equation solves trivially as the equation is a second-order differential equation with constant coefficients. The solution is

$$\Phi(\phi) = A_{+}e^{im\phi} + A_{-}e^{-im\phi}$$

The radial equation is greatly simplified with the change of variables $\rho \equiv e^s$, which offers

$$\frac{\partial}{\partial s} = \frac{\partial\rho}{\partial s}\frac{\partial}{\partial\rho} = \rho\frac{\partial}{\partial\rho}$$

and

$$\frac{\partial^2}{\partial s^2} = \rho\frac{\partial}{\partial\rho}\rho\frac{\partial}{\partial\rho} = \rho^2\frac{\partial^2}{\partial\rho^2} + \rho\frac{\partial}{\partial\rho}$$

and so

$$\frac{\partial^2 R}{\partial s^2} = \rho^2 \frac{\partial^2 R}{\partial \rho^2} + \rho \frac{\partial R}{\partial \rho} = m^2 R$$

which solves to give

$$R(s) = a_+ e^{ms} + a_- e^{-ms} = a_+ \rho^m + a_- \rho^{-m} = R(\rho)$$

and power laws. This gives us four distinct types of solution

$$\rho^m e^{im\phi} \qquad \rho^m e^{-im\phi} \qquad \rho^{-m} e^{im\phi} \qquad \rho^{-m} e^{-im\phi}$$

We can now recognise the complex variable because

$$z = \rho e^{i\phi} \qquad \bar{z} = \rho e^{-i\phi}$$

and so the linear superposition of all our solutions is accomplished with

$$f(z) = \sum_m a_m z^m + \sum_m b_m \bar{z}^m$$

which would be the solution to Laplace's equation but we have the restriction to analytic functions which eliminates the dependence on $\bar{z}$ and leaves us with

$$f(z) = \sum_m a_m z^m$$

where the sum is over all relevant values of m, but what are the relevant values of m?

When you were previously taught how to solve partial differential equations using separation of variables, the physical problem provided boundary conditions, which restricted the allowed solutions, and permitted you to find the desired solution to a physical problem. Here we are investigating an area of mathematics and so we do not currently know which physical problems it will be applied to and so this choice is unclear! As the course proceeds we will include more choices for m as we deal with more sophisticated physical problems!

The most natural assumption (that will at some stage be rescinded!), is that we want functions that map smoothly onto the complex plane and so we need to have one answer for each z. If we track ϕ around the unit circle, then

$$z^m = \rho^m e^{im\phi}$$

is only single-valued if m is an integer and, if we want to avoid discontinuities as we circle, this restricts attention to integer values of m and what is known as a *Laurent Series*

$$f(z) = \sum_{m=-\infty}^{\infty} a_m z^m$$

and all analytic functions that map smoothly onto the complex plane have a Laurent expansion about each and every point. Technically, both the function and its derivative

do not exist at the origin if there are negative values of m in the sum, as such terms diverge as $\rho \mapsto 0$, but the function is analytic away from the origin provided that this sum converges. In order to be analytic at the origin we need to eliminate the negative values and then an analytic function in the vicinity of the origin satisfies the analogue of a Taylor's series

$$f(z) = \sum_{m=0}^{\infty} a_m z^m$$

In complex variable theory the central quantity is an *analytic function.* This quantity has a differential everywhere in some region of the complex plane and can be described uniquely by a *Taylor's series.* One of the most important inclusions to this theory are *extractable singular points,* isolated points where the function is not differentiable but has a singularity which is described locally by a *Laurent series.* The Taylor's series or indeed Laurent series do not have to converge and are local to the expansion point, but their failure to converge is evidence for some other singular point at which the series is not expected to converge. We will devote a lot of attention over the next few lectures to functions which only have extractable singularities and map the complex plane onto itself.

The governing equation

Alternatively, we could take Cauchy-Riemann as the governing equation and solve them using separation of variables. From the previous section

$$\frac{\partial}{\partial x} + i\frac{\partial}{\partial y} = e^{i\phi}\frac{\partial}{\partial \rho} + ie^{i\phi}\frac{1}{\rho}\frac{\partial}{\partial \phi}$$

which gives

$$\frac{1}{R}\rho\frac{dR}{d\rho} = -i\frac{1}{\Phi}\frac{d\Phi}{d\phi} = m$$

where m is the separation parameter. This is easier to solve and gives

$$R(\rho) = \rho^m \quad \Phi(\phi) = e^{im\phi}$$

as expected, and we can get our z^m in a simpler way.

Following the d'Alembert logic, Laplace's equation transforms to

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \left(\frac{\partial}{\partial z} + \frac{\partial}{\partial \bar{z}}\right)^2 - \left(\frac{\partial}{\partial z} - \frac{\partial}{\partial \bar{z}}\right)^2 = 4\frac{\partial^2}{\partial z \partial \bar{z}}$$

and a final solution of

$$f(x, y) = F[z] + \bar{F}[\bar{z}]$$

in sympathy with all the other ideas.

Analytic functions from Laplace's equation

If one has a solution to Laplace's equation then one can construct an analytic function from this solution. The theory is based around the theorem in vector calculus

that, in a simply connected domain, if a field, $\mathbf{v}$, has a vanishing curl then it can be described by a potential, $\mathbf{v}=\nabla Y$, where Y is this potential. The way we use this theorem is by designing a field from our solution to Laplace's equation, $X(x,y)$, using

$$\mathbf{v}(\mathbf{r}) = \left(-\frac{\partial X}{\partial y}, \frac{\partial X}{\partial x}, 0\right)$$

and then

$$\nabla \times \mathbf{v}(\mathbf{r}) = \det \begin{bmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{bmatrix} = \det \begin{bmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -\frac{\partial X}{\partial y} & \frac{\partial X}{\partial x} & 0 \end{bmatrix} = \hat{\mathbf{z}}\left(\frac{\partial^2 X}{\partial x^2} + \frac{\partial^2 X}{\partial y^2}\right) = \mathbf{0}$$

provided that $X(x.y)$ satisfies Laplace's equation. Using the theorem then provides

$$\mathbf{v}(\mathbf{r}) = \nabla Y = \left(\frac{\partial Y}{\partial x}, \frac{\partial Y}{\partial y}, \frac{\partial Y}{\partial z}\right)$$

and so there is clearly no z-dependence and we recognise Cauchy-Riemann equations and so the associated potential is the imaginary part of a complex analytic function, $Z \equiv X + iY$.

In application, given $X(x,y)$, one can write down Cauchy-Riemann equations and solve them for $Y(x,y)$. Solving them involves integrating the two equations and setting the two solutions equal to each other. The constant of integration in each case is an arbitrary function of the other variable and when we set the two results equal we need to separate the x and y dependent pieces onto opposite sides of an equals sign and recognise that the resulting quantity does not depend on either variable so is an absolute constant. An example should help. Example sheet 1 has other examples.

Example: Show that $X(x,y)$=x+y-xy satisfies Laplace's equation and construct an analytic function from it.

$$\frac{\partial X}{\partial x} = 1 - y \quad \frac{\partial^2 X}{\partial x^2} = 0 \quad \frac{\partial X}{\partial y} = 1 - x \quad \frac{\partial^2 X}{\partial y^2} = 0 \quad \Rightarrow \quad \frac{\partial^2 X}{\partial x^2} + \frac{\partial^2 X}{\partial y^2} = 0$$

We already have what we need for Cauchy-Riemann

$$\frac{\partial Y}{\partial x} = -\frac{\partial X}{\partial y} = x - 1 \qquad \frac{\partial Y}{\partial y} = \frac{\partial X}{\partial x} = 1 - y$$

and integrating these equations is trivial

$$Y(x,y) = \frac{x^2}{2} - x + g(y) = y - \frac{y^2}{2} + f(x)$$

and separating the two dependences on x and y gives

$$f(x) - \frac{x^2}{2} + x = g(y) - y + \frac{y^2}{2} = c$$

where c is the absolute constant. Eliminating either of the two functions, $f(x)$ or $g(y)$ then provides

$$Y(x,y) = \frac{x^2}{2} - x + y - \frac{y^2}{2} + c$$

and then the analytic function is

$$Z = X + iY = x + y - xy + \left(\frac{x^2}{2} - x + y - \frac{y^2}{2} + c\right) i = \frac{z^2}{2} i + z - zi + ci$$

Example: Show that

$$X(x,y) = \frac{x}{x^2 + y^2}$$

satisfies Laplace's equation and construct an analytic function from it.

$$\frac{\partial X}{\partial x} = \frac{1}{x^2+y^2} - \frac{2x^2}{(x^2+y^2)^2} \qquad \frac{\partial^2 X}{\partial x^2} = -\frac{2x}{(x^2+y^2)^2} - \frac{4x}{(x^2+y^2)^2} + \frac{8x^3}{(x^2+y^2)^3}$$

$$\frac{\partial X}{\partial y} = -\frac{2xy}{(x^2+y^2)^2} \qquad \frac{\partial^2 X}{\partial y^2} = -\frac{2x}{(x^2+y^2)^2} + \frac{8xy^2}{(x^2+y^2)^3}$$

and so

$$\frac{\partial^2 X}{\partial x^2} + \frac{\partial^2 X}{\partial y^2} = 0$$

Cauchy-Riemann provide

$$\frac{\partial Y}{\partial x} = -\frac{\partial X}{\partial y} = \frac{2xy}{(x^2+y^2)^2}$$

$$\frac{\partial Y}{\partial y} = \frac{\partial X}{\partial x} = \frac{1}{x^2+y^2} - \frac{2x^2}{(x^2+y^2)^2}$$

and the first one is easy to integrate

$$Y(x,y) = -\frac{y}{x^2+y^2} + g(y)$$

but the second one is tough! If we differentiate our solution with respect to y we get

$$-\frac{1}{x^2+y^2} + \frac{2y^2}{(x^2+y^2)^2} = \frac{1}{x^2+y^2} - \frac{2x^2}{(x^2+y^2)^2}$$

which works! So

$$Y(x,y) = -\frac{y}{x^2+y^2} + f(x)$$

and we need $f(x)$=$g(y)$=c to get

$$Z = X + iY = \frac{x - iy}{x^2+y^2} + ic = \frac{1}{x+iy} + ic = \frac{1}{z} + ic$$

The integral *must* be related to each-other and so the easiest one is crucial.

Angle-integration into Complex Integration

We often meet integrals over angles in mathematical physics, for example

$$\int_0^{2\pi} \frac{d\phi}{2\pi} f\left(e^{i\phi}\right)$$

where $f\left(e^{i\phi}\right)$ is a complex function. It is important to be able to convert this type of integral into a complex integral. We will pass through an Argand diagram on the way. The idea is to consider the function to be a function of a complex variable, $f(z)$, where in polar coordinates, $z \equiv \rho e^{i\phi}$, and the angle is the polar angle in complex variable theory. To connect the two we need ρ=1 and so we are automatically on the unit circle in the associated Argand diagram. The original function has been generalised to this analytic function, as the original only needed to be evaluated on the unit circle but now we can investigate the function anywhere in the plane. In the Argand diagram we can consider line integrals with

$$\mathbf{dr} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} = (d\rho\cos\phi - \rho d\phi\sin\phi)\hat{\mathbf{x}} + (d\rho\sin\phi + \rho d\phi\cos\phi)\hat{\mathbf{y}} \equiv d\rho\hat{\boldsymbol{\rho}} + \rho d\phi\hat{\boldsymbol{\phi}}$$

using either Cartesian coordinates or polar coordinates. The appropriate vectors can be read off as

$$\hat{\boldsymbol{\rho}} \equiv \cos\phi\hat{\mathbf{x}} + \sin\phi\hat{\mathbf{y}} \qquad \hat{\boldsymbol{\phi}} \equiv -\sin\phi\hat{\mathbf{x}} + \cos\phi\hat{\mathbf{y}}$$

In complex variable theory we allow

$$\hat{\mathbf{x}} \leftrightarrow 1 \qquad \hat{\mathbf{y}} \leftrightarrow i$$

and so the analogue of $\mathbf{dr}$ is $dz = dx + idy$ and in polar coordinates

$$\hat{\boldsymbol{\rho}} \leftrightarrow e^{i\phi} \qquad \hat{\boldsymbol{\phi}} \leftrightarrow ie^{i\phi}$$

and so by analogy

$$dz \leftrightarrow e^{i\phi}d\rho + ie^{i\phi}\rho d\phi$$

On the unit circle ρ=1 so $d\rho$=0 and we can see that

$$dz = iz d\phi$$

and the original integral becomes

$$\int_0^{2\pi} \frac{d\phi}{2\pi} f\left(e^{i\phi}\right) = \int_C \frac{dz}{2\pi iz} f(z)$$

The final important quantity is the curve, C, which can be deduced by following the path mapped out in the Argand diagram as the angle is increased from zero to two pi. This is clearly a single traverse of the unit-circle in an anti-clockwise direction; a *closed curve.*

Analytic functions from Fourier series

One crucial question is; "Where might this theory prove useful?" and this is not immediately answerable. The theory is one of the most powerful tools in a Mathematical Physicist's arsenal, and almost all solvable problems are best solved using Complex Variable Theory, but one has to manipulate the problem into a form that is tractable and this is often quite subtle and convoluted. To give an example, one might be interested in quantum mechanics, and you should all be aware of Scrödinger's Equation

$$H\psi = E\psi$$

and to turn this into a Complex Variable Theory problem one tends to target a Green's function

$$G(E) = [E - H]^{-1}$$

the inverse of an energy dependent operator. To calculate this quantity one needs to choose a basis to describe state-space, and for real-spaces we tend to use plane-waves, e^{ikx}, as the basis, which leads to a Fourier transform and for angular problems we tend to use $e^{im\phi}$, where m is an integer, and Fourier series. If you attend the third year course on "Condensed Matter", then you will learn that electrons travelling about in periodic potentials are described by *Bloch momentum,* which is an analogue to momentum but is actually an angle and repeats. The associated Green's function (which will be carefully derived later in this course), when the energy is $E(k) \equiv -2t\cos k$, is then

$$G_n[E] = \int_0^{2\pi} \frac{dk}{2\pi} \frac{e^{ikn}}{E + 2t\cos k}$$

which then has to be converted into an analytic function if we want to use Complex variable Theory. For Fourier series, we employ the angle as the polar angle in complex numbers and then the radius is unity and we are dealing with complex numbers on the unit circle of the complex plane, $z \equiv \rho e^{ik}$. We want to replace the integral over k with an integral over z and since

$$z = \rho e^{ik} \quad \Rightarrow \quad dz = idk\rho e^{ik} = idkz$$

we obtain

$$G_n(E) = \oint_C \frac{dz}{2\pi iz} \frac{z^n}{E + t\left(z + \frac{1}{z}\right)}$$

where C is the unit circle and as k goes from zero to 2π this circle is traversed once in an anti-clockwise direction. Fourier series turn into complex integrals using these ideas, but integration is the topic of the next section and currently we are looking for complex analytic functions and this integrand provides us with our first example

$$f_n(z) = \frac{z^n}{tz^2 + Ez + t}$$

There are various algebraic processes that you will need to become proficient at; finding and classifying singularities and partial fractions, as the first two. Singularities

are points where the function diverges and are consequently points where the function is not analytic. One needs to decide if they are extractable, and if they are then there will be a Laurent series that you may be asked to find. If they are not you will need to be able to classify them further, but this will come later in the course. The current function has two clear singularities, where the quadratic denominator vanishes, and it should be no problem to solve this quadratic and discover that

$$z_{\pm} = -\frac{E}{2t} \pm \frac{1}{2t}\left[E^2 - 4t^2\right]^{\frac{1}{2}}$$

and we arrive at a critical technical point. The algebraic problem of finding the particular positions of the singularities is separate from the algebraic problem of processing the function into a more tractable form. It makes sense to separate the two procedures and do them one at a time. The method that we naturally use is *parameterisation,* we do not need to find the two roots to know that there are two roots, we can just give them a name, $z_{\pm}$, for the current example. We can process the analytic function in terms of this parameterisation and then find the particular explicit formulae for these quantities independently. The previous solving of the quadratic was this explicit solution for the current example. We can use the parameterisation to rewrite the function as

$$f_n(E) = \frac{z^n}{t(z-z_+)(z-z_-)} = \frac{1}{t(z_+ - z_-)}\left[\frac{z^n}{z-z_+} - \frac{z^n}{z-z_-}\right]$$

independently from whether we can solve the quadratic, using partial fractions. We can recognise the original function as a sum of two functions, where each function only has one of the two singularities arising from the quadratic! In principle we can split the original problem into the sum of two simpler problems. Depending on the sign of n, the origin may also be a singularity. For negative values of n, when the origin is a singularity, we can use our simplification, together with a geometric series, to actually find the Laurent expansion

$$f_{-n}(E) = \frac{1}{z^n}\frac{1}{t(z_+ - z_-)}\left[-\frac{1}{z_+}\frac{1}{1-\frac{z}{z_+}} + \frac{1}{z_-}\frac{1}{1-\frac{z}{z_-}}\right]$$

$$= \frac{1}{z^n}\frac{1}{t(z_+ - z_-)}\left[-\frac{1}{z_+}\sum_{m=0}^{\infty}\left(\frac{z}{z_+}\right)^m + \frac{1}{z_-}\sum_{m=0}^{\infty}\left(\frac{z}{z_-}\right)^m\right]$$

It should be clear to you that provided $\mid z \mid < \mid z_{\pm} \mid$ the series converges and so the series ceases to be mathematically useful outside of this circle, although the original formula is still valid.

We can play the same sort of trick on other integrals that you have probably met before, for example

$$I_n = \int_0^{2\pi} \frac{d\phi}{2\pi}\left[\cos\phi\right]^{2n}$$

transforms into

$$I_n = \oint_C \frac{dz}{2\pi i z}\frac{1}{2^{2n}}\left[z + \frac{1}{z}\right]^{2n}$$

and the binomial theorem immediately provides us with the Laurent series at the origin

$$f_n(z) \equiv \frac{1}{z}\frac{1}{2^{2n}}\left[z+\frac{1}{z}\right]^{2n} = \frac{1}{2^{2n}}\sum_{m=0}^{2n}\frac{2n!}{m!(2n-m)!}z^{2m-2n-1}$$

Parameterisation

Dealing with partial fractions is important enough for us to discuss it in some detail, but before we can do this we have to explain various complications that are equally as important and often tend to dominate. If electrons hybridise with the second-neighbours as well as to the neighbours, then the energy would be $E(k) = -2t_1\cos k - 2t_2\cos 2k$ and the associated Green's function would be

$$G_n(E) = \int_0^{2\pi}\frac{dk}{2\pi}\frac{e^{ikn}}{E+2t_2\cos k+2t_2\cos 2k}$$

which gives us another complex integral

$$G_n(E) = \oint_C \frac{dz}{2\pi iz}\frac{z^n}{E+t_1\left(z+\frac{1}{z}\right)+t_2\left(z^2+\frac{1}{z^2}\right)}$$

and a second analytic function

$$f_n(z) = \frac{z^{n+1}}{t_2z^4+t_1z^3+Ez^2+t_1z+t_2}$$

and now we have a quartic equation and quartics are not necessarily solvable. How should we proceed now? One might say that we should parameterise the roots, as before, but this is not a good choice because the problem has more structure than this and we can actually solve the quartic in this case. The first thing to appreciate is that we did not solve the previous example well enough! When we have a quadratic equation, we can analyse it in general and realise that

$$(z-z_+)(z-z_-) = z^2-(z_++z_-)z+z_+z_- \equiv z^2-a_1z+a_2$$

and so the two coefficients of the quadratic equation are

$$a_1 \equiv z_+ + z_- \qquad a_2 \equiv z_+z_-$$

and for the previous quadratic

$$tz^2+Ez+t=0 \quad \Rightarrow \quad a_1 = -\frac{E}{t} \equiv z_++z_- \quad a_2 = 1 \equiv z_+z_-$$

and the final equation tells us that the roots are reciprocals of each other, and so we can rewrite $z_- = \frac{1}{z_+}$ and only use one root to parametrise the answer

$$f_{-n}(E) = \frac{1}{z^n}\frac{1}{t\left(z_+-\frac{1}{z_+}\right)}\left[\frac{1}{z_+}\frac{-1}{1-\frac{z}{z_+}}+z_+\frac{1}{1-zz_+}\right]$$

This logic extends to the more complicated problem, since the denominator involves $\cos 2k = 2\cos^2 k - 1$ which can be written as a quadratic in $\cos k \equiv c$.

$$E + 2t_1 \cos k + 2t_2 \cos 2k = E + 2t_1 c + 2t_2(2c^2 - 1) \equiv 4t_2 \left(c^2 + \frac{t_1}{2t_2} c + \frac{E}{4t_2} - \frac{1}{2} \right)$$

$$\equiv 4t_2 (c - c_+)(c - c_-)$$

where we employed $c \equiv \cos k$ and then we parameterised the two possible solutions using $c_\pm$. We can now use partial fractions on these cosines before we ever try to use Complex Variable Theory

$$G_n(E) = \int_0^{2\pi} \frac{dk}{2\pi} \frac{e^{ikn}}{E + 2t_1 \cos k + 2t_2 \cos 2k} = \int_0^{2\pi} \frac{dk}{2\pi} \frac{e^{ikn}}{4t_2 (\cos k - c_+)(\cos k - c_-)}$$

$$= \frac{1}{4t_2(c_+ - c_-)} \left[\int_0^{2\pi} \frac{dk}{2\pi} \frac{e^{ikn}}{\cos k - c_+} - \int_0^{2\pi} \frac{dk}{2\pi} \frac{e^{ikn}}{\cos k - c_-} \right]$$

and the original problem

$$G_n[E] = \int_0^{2\pi} \frac{dk}{2\pi} \frac{e^{ikn}}{E + 2t \cos k} = \frac{1}{2t} \int_0^{2\pi} \frac{dk}{2\pi} \frac{e^{ikn}}{\cos k + \frac{E}{2t}}$$

has exactly the same form!

There are some very important lessons to be learned here. The most important is to think about the mathematical structure of what you are doing before you try to proceed. For the current problem, there is a single quantity

$$C_n(z_+) = \int_0^{2\pi} \frac{dk}{2\pi} \frac{e^{ikn}}{2\cos k - z_+ - \frac{1}{z_+}}$$

which controls this problem, and the quantity, z_+, must be chosen with great care. The nearest-neighbour problem can be rewritten as

$$G_n[E] = \int_0^{2\pi} \frac{dk}{2\pi} \frac{e^{ikn}}{E + 2t \cos k} = \frac{1}{t} C_n(z_+)$$

where

$$z_+ + \frac{1}{z_+} = -\frac{E}{t}$$

parameterises the energy and separates the control over the energy from the control over the integral. The next-nearest-neighbour problem is correspondingly parameterised by

$$G_n(E) = \int_0^{2\pi} \frac{dk}{2\pi} \frac{e^{ikn}}{E + 2t_1 \cos k + 2t_2 \cos 2k} = \frac{1}{t_2} \frac{C_n(z_+^{(+)}) - C_n(z_+^{(-)})}{z_+^{(+)} + \frac{1}{z_+^{(+)}} - z_+^{(-)} - \frac{1}{z_+^{(-)}}}$$

where

$$z_+^{(+)} + \frac{1}{z_+^{(+)}} + z_+^{(-)} + \frac{1}{z_+^{(-)}} = -\frac{t_1}{t_2} \quad \left(z_+^{(+)} + \frac{1}{z_+^{(+)}}\right)\left(z_+^{(-)} + \frac{1}{z_+^{(-)}}\right) = \frac{E}{t_2} - 2$$

parameterises the energy. It should be clear that unravelling the parameterisation is equally as important as doing the integrals and may in practice be more mathematically taxing!

Symmetric Polynomials

The previous argument can be generalised into a way of simplifying symmetric polynomials. If a polynomial is symmetric, ie a_{n-m}=a_m, for

$$P(z) = a_0 z^n + a_1 z^{n-1} + ... + a_{n-1} z + a_n$$

then the new variable $2c$=z+$\frac{1}{z}$, reduces the rank of the polynomial by half and simplifies the problem dramatically. The binomial theorem allows

$$\begin{aligned} 2c &= z + \frac{1}{z} \\ 4c^2 &= z^2 + 2 + \frac{1}{z^2} \\ 8c^3 &= z^3 + 3z + \frac{3}{z} + \frac{1}{z^3} \end{aligned}$$

and so on... If the polynomial is odd then z=-1 is a solution, which can be extracted, and the residual is then even. For an even polynomial, $2n$ say, we divide through by z^n and then we can eliminate z^m+$\frac{1}{z^m}$ in favour of c^m, order by order. The resulting equation is at least half the order of the original and should be more tractable. Once one has found c, one needs to solve

$$z + \frac{1}{z} = 2c \quad z = c \pm \left(c^2 - 1\right)^{\frac{1}{2}}$$

to find the original roots. Obviously if -1< c <1 we have a cosine and if $c = \cos\phi$, then $z = e^{\pm i\phi}$, otherwise we have two real roots which are reciprocals of each other.

Example:

$$z^3+2z^2+2z+1=0 \;\Rightarrow\; (z+1)(z^2+z+1)=0 \;\Rightarrow\; z+1+\frac{1}{z}=0=2c+1 \;\Rightarrow\; c=-\frac{1}{2}$$

Example:

$$z^4 + 2z^3 - 2z^2 + 2z + 1 = 0 \;\Rightarrow\; z^2 + 2z - 2 + \frac{2}{z} + \frac{1}{z^2} = 0 \;\Rightarrow\; 4c^2 + 4c - 4 = 0$$

$$\Rightarrow\; c = -\frac{1}{2} \pm \frac{\sqrt{5}}{2}$$

Example:

$$z^5+1=0 \;\Rightarrow\; (z+1)(z^4-z^3+z^2-z+1)=0 \;\Rightarrow\; z^2-z+1-\frac{1}{z}+\frac{1}{z^2}=0 \;\Rightarrow\; 4c^2-2c-1=0$$

$$\Rightarrow\; c=\frac{1}{4}\pm\frac{\sqrt{5}}{4}$$

Example:

$$z^6+1=0 \;\Rightarrow\; z^3+\frac{1}{z^3}=0 \;\Rightarrow\; 8c^3-3z-\frac{3}{z}=8c^3-6c=0 \;\Rightarrow\; c(4c^2-3)=0$$

$$\Rightarrow\; c=0,\pm\frac{\sqrt{3}}{2}$$

Roots of Unity

The last two examples could also be tackled using the roots of unity concept. For an even integer the roots of unity can be separated into two halves,

$$(z^{2n}-1)=(z^n-1)(z^n+1)=0$$

provides the roots of unity for n and the solutions to z^n=-1 are the roots half-way between the n'th roots on an Argand diagram. Even values of m in z=$e^{i\frac{2\pi m}{2n}}$ are the n'th roots and odd values of m are the solutions to z^n=-1.

Partial Fractions

All the previous examples of partial fractions required one quadratic and the formula

$$\frac{1}{(z-z_+)(z-z_-)}=\frac{1}{z_+-z_-}\left(\frac{1}{z-z_+}-\frac{1}{z-z_-}\right)$$

allows the separation into a sum of functions with only one singularity. One often meets quantities more complicated than a quadratic, for example

$$f(z)=\frac{1}{z^4-1}$$

how should we deal with this? The lessons of the previous section can be used

$$f(z)=\frac{1}{2}\left(\frac{1}{z^2-1}-\frac{1}{z^2+1}\right)=\frac{1}{2}\left(\frac{1}{2}\left[\frac{1}{z-1}-\frac{1}{z+1}\right]-\frac{1}{2i}\left[\frac{1}{z-i}-\frac{1}{z+i}\right]\right)$$

which recognises z^2 as a useful intermediate and then applies the previous formula twice. There is a second method, the "cover-up rule", which works even when the polynomial is not explicitly solvable and can only be parameterised. At quartic order we would need to process

$$\frac{1}{(z-z_1)(z-z_2)(z-z_3)(z-z_4)}$$

and the rule is; for each possible root chosen, z_c, cover up the particular chosen root, $(z - z_c)$, and replace $z \mapsto z_c$ in the rest and then the sum over all possible roots is the separation into individual singularities. For the current example the four roots are

$$\frac{1}{(z-1)(z+1)(z-i)(z+i)}$$

We start out with unity and we generate

$$\frac{1}{(1+1)(1-i)(1+i)} = \frac{1}{4}$$

as the coefficient, then minus unity gives

$$\frac{1}{(-1-1)(-1-i)(-1+i)} = -\frac{1}{4}$$

i provides

$$\frac{1}{(i-1)(i+1)(i+i)} = \frac{i}{4}$$

and -i offers

$$\frac{1}{(-i-1)(-i+1)(-i-i)} = -\frac{i}{4}$$

and building the final answer

$$\frac{1}{(z-1)(z+1)(z-i)(z+i)} = \frac{1}{4}\frac{1}{z-1} - \frac{1}{4}\frac{1}{z+1} + \frac{i}{4}\frac{1}{z-i} - \frac{i}{4}\frac{1}{z+i}$$

which does indeed agree with the previous method.

To understand the cover-up rule we need to appreciate that it works when

$$\frac{N(z)}{D(z)} = \sum_n \frac{a_n}{z - z_n} \qquad D(z) = \prod_n (z - z_n)$$

and the denominator, $D(z)$, is a product over known roots, z_n, and the numerator, $N(z)$, is an appropriate polynomial. To prove the theorem we can multiply by $(z - z_m)$ and take the limit $z \mapsto z_m$

$$\lim_{z \mapsto z_m} (z - z_m)\frac{N(z)}{D(z)} = \frac{N(z_m)}{\prod_{n \neq m}(z_m - z_n)} = \lim_{z \mapsto z_m} (z - z_m) \sum_n \frac{a_n}{z - z_n} = a_m$$

and so we need all the roots to be distinct and we need that the polynomial, $N(z)$, is of lower order than the denominator. If you have a situation where the numerator is higher order than the denominator, then you need to divide the denominator into the numerator to find an additive polynomial and then the remainder is the numerator that you can apply the theorem to. Examples should help.

Example:

$$\frac{z^2-4}{z^3-9z} = \frac{z^2-4}{z(z-3)(z+3)}$$

and we have three singularities at $z \in \{0, 3, -3\}$ and so

$$\frac{z^2-4}{z^3-9z} = \frac{-4}{(-3)(3)}\frac{1}{z} + \frac{9-4}{(3)(3+3)}\frac{1}{z-3} + \frac{9-4}{(-3)(-3-3)}\frac{1}{z+3}$$

$$= \frac{4}{9}\frac{1}{z} + \frac{5}{18}\frac{1}{z-3} + \frac{5}{18}\frac{1}{z+3}$$

Ugly Example:

$$\frac{1}{z^4+1} = \frac{1}{\left(z-\frac{1+i}{\sqrt{2}}\right)\left(z-\frac{1-i}{\sqrt{2}}\right)\left(z+\frac{1+i}{\sqrt{2}}\right)\left(z+\frac{1-i}{\sqrt{2}}\right)} =$$

$$\frac{1}{\left(\frac{2i}{\sqrt{2}}\right)\left(\frac{2(1+i)}{\sqrt{2}}\right)\left(\frac{2}{\sqrt{2}}\right)}\frac{1}{z-\frac{1+i}{\sqrt{2}}} + \frac{1}{\left(\frac{-2i}{\sqrt{2}}\right)\left(\frac{2}{\sqrt{2}}\right)\left(\frac{2(1-i)}{\sqrt{2}}\right)}\frac{1}{z-\frac{1-i}{\sqrt{2}}}$$

$$+\frac{1}{\left(\frac{-2(1+i)}{\sqrt{2}}\right)\left(\frac{-2}{\sqrt{2}}\right)\left(\frac{-2i}{\sqrt{2}}\right)}\frac{1}{z+\frac{1+i}{\sqrt{2}}} + \frac{1}{\left(\frac{-2}{\sqrt{2}}\right)\left(\frac{-2(1-i)}{\sqrt{2}}\right)\left(\frac{2i}{\sqrt{2}}\right)}\frac{1}{z+\frac{1-i}{\sqrt{2}}}$$

$$= \frac{1}{2\sqrt{2}(i-1)}\frac{1}{z-\frac{1+i}{\sqrt{2}}} + \frac{1}{2\sqrt{2}(-i-1)}\frac{1}{z-\frac{1-i}{\sqrt{2}}} + \frac{1}{2\sqrt{2}(1-i)}\frac{1}{z+\frac{1+i}{\sqrt{2}}} + \frac{1}{2\sqrt{2}(1+i)}\frac{1}{z+\frac{1-i}{\sqrt{2}}}$$

$$= \frac{-1-i}{4\sqrt{2}}\frac{1}{z-\frac{1+i}{\sqrt{2}}} + \frac{-1+i}{4\sqrt{2}}\frac{1}{z-\frac{1-i}{\sqrt{2}}} + \frac{1+i}{4\sqrt{2}}\frac{1}{z+\frac{1+i}{\sqrt{2}}} + \frac{1-i}{4\sqrt{2}}\frac{1}{z+\frac{1-i}{\sqrt{2}}}$$

a quite horrifying amount of work. Surely there is a better way.... There are always things that you can do if you are imaginative enough. For this example

$$\frac{d}{dz}\ln(z^4+1) = \frac{4z^3}{z^4+1} = \sum_n \frac{1}{z-z_n}$$

and multiply by $z/4$ to get

$$\frac{z^4}{z^4+1} = 1 - \frac{1}{z^4+1} = \frac{1}{4}\sum_n \frac{z}{z-z_n}$$

and rearrange to get

$$\frac{1}{z^4+1} = \frac{1}{4}\sum_n \frac{z-z_n}{z-z_n} - \frac{1}{4}\sum_n \frac{z}{z-z_n} = -\frac{1}{4}\sum_n \frac{z_n}{z-z_n}$$

which is clearly the same answer.... At least the algebra is correct....

$$\frac{1}{(z^2+1)(z^2+4)} = \frac{1}{(z-i)(z+i)(z-2i)(z+2i)}$$

but it is better to use partial fractions on z^2 to get

$$\frac{1}{3}\left(\frac{1}{z^2+1}-\frac{1}{z^2+4}\right)$$

before using partial fractions on the pieces

$$\frac{1}{6i}\left(\frac{1}{z-i}-\frac{1}{z+i}\right)-\frac{1}{12i}\left(\frac{1}{z-2i}-\frac{1}{z+2i}\right)$$

Example:

$$\frac{z^8}{z^4+1}=\frac{z^8+z^4-z^4-1+1}{z^4+1}=z^4-1+\frac{1}{z^4+1}$$

and the final term is one of the previous examples.

Differentiation: Logic

Analytic functions have a Taylor's expansion in z=x+iy, $f(z)$, at points where they are analytic. If a function is analytic at a point then it is infinitely differentiable at that point. Unlike real functions, the analytic function is equal to its Taylor's series, provided that the series converges. One can also have extractable singularities, lone points where the function is not analytic, and the function has a Laurent expansion about these points. Cauchy-Riemann equations are important, mathematically equivalent to the idea that the function does *not* depend on $\bar{z}$=x-iy. The real and imaginary parts of analytic functions satisfy Laplace's equation.

Functions with polynomial denominators may be simplified with partial fractions, which separate out the function into a regular piece and a sum of pieces with independent singularities. Parameterising the singularities is very useful as it separates the problem of finding the explicit positions of the singularities from simplifying and controlling the function. Usually finding the explicit positions of the singularities is more mathematically taxing than processing the function.

Integration

An Argand diagram is a two-dimensional representation of complex numbers where $1 \leftrightarrow \hat{\mathbf{x}}$ and $i \leftrightarrow \hat{\mathbf{y}}$. This provides an immediate interpretation of complex integrals in terms of line integrals in two-dimensional space, through $dz = dx$+$idy \leftrightarrow dx\hat{\mathbf{x}}$+$dy\hat{\mathbf{y}} \equiv \mathbf{dr}$. We can use integral theorems to deduce the properties of complex integrals.

Probably the most important result in Complex Variable Theory is Cauchy's theorem, which is simply explained as

$$\oint_C dz f(z) = 0$$

for any function, $f(z)$, which is analytic everywhere in the domain of the complex plane that the closed loop C encloses. This theorem can be established from Stoke's theorem

applied to a contour restricted to a two-dimensional plane within three-dimensional space

$$\oint_C d\mathbf{r}.\mathbf{A}(\mathbf{r}) = \int_S d\mathbf{S}.\nabla \times \mathbf{A}(\mathbf{r})$$

where we choose $\mathbf{A}(\mathbf{r}) \equiv X\hat{\mathbf{x}} + Y\hat{\mathbf{y}} + Z\hat{\mathbf{z}}$ and then the theorem becomes

$$\oint_C (dxX + dyY) = \int_S dxdy \left(\frac{\partial Y}{\partial x} - \frac{\partial X}{\partial y}\right)$$

For the complex variable analogue, we have $dz{=}dx{+}idy$, and $f{=}X{+}iY$ to provide

$$\oint_C dzf(z) = \oint_C (dxX - dyY + idxY + idyX) = \int_S dxdy \left(-\frac{\partial Y}{\partial x} - \frac{\partial X}{\partial y} + i\frac{\partial X}{\partial x} - i\frac{\partial Y}{\partial y}\right)$$

$$= 0$$

where we used the previous theorem twice on the real and imaginary parts identically, and then we used Cauchy-Riemann to show that the resulting integrals vanish.

Cauchy's theorem is usually the starting point for a course on Complex Variable Theory, because integration is the main target for such a course and proving things using the "solving partial differentials" approach is more volatile and requires more assumptions. The issues are still relevant for the integral approach but are more subtle. We used Stoke's theorem in our derivation and there is a subtle topological issue with Stoke's theorem; the theorem is only true for a *simply connected domain.* This might appear irrelevant mathematician gobble-di-gook, but for us it is immensely important! We found special singular points where the function was not analytic and where a Laurent series was the appropriate description. We would need to exclude these points if we want to use Cauchy's theorem, and excluding a single point transforms a simply connected domain to not a simply connected domain and the theorem would then not be applicable. This failure is not just cosmetic but is actually very real! When we have a Laurent series the vanishing of the integral for a loop that goes around the singularity is not always true! Let us follow the natural line of reasoning and uncover the difficulties. What we would like to do is focus on each singularity separately, and to do this it makes sense to try to eliminate a small circle around each singularity and try to use Cauchy's theorem on the rest of the space. The logic is then to include the path around the little circle in a clockwise direction to try to cancel out the effect of the singularity and leave behind an integral that would vanish. The sum of two integrals on the little circle, one going anti-clockwise and one going clockwise, obviously cancels out so we would include the clockwise one with the original integral and leave the anticlockwise paths as the afterthought to assess what happens at each of the singularities. We would then have to show that the original integral together with the sum of clockwise circles vanishes.

The simplest example is a single singularity and an original anticlockwise contour, C, that encloses the singularity. We would then have an annulus region enclosed by the original contour and the little circle. To try to use Cauchy's theorem we would need a simply connected domain and so we would draw a line from one contour to the other and construct a new contour from the two original contours and the additional line traversed twice. Start somewhere on the original contour and follow that contour in

an anti-clockwise direction until we arrive at the added line. Then follow the line until we hit the little circle. Follow the little circle in the clockwise direction and circle the singularity until we arrive back at the added line. Now follow the line back to the original contour and track on around the original contour in an anti-clockwise direction until we arrive back at the starting point and we can then close the contour. This new contour encloses a simply connected domain and so Cauchy's theorem applies and we can show that this integral vanishes. *If* the two integrals along the added line cancel, then the original contour in an anticlockwise direction plus the integral along the little circle in a clockwise direction vanish and we can replace the original contour with the little circle in an anticlockwise direction. It is analogous to collapsing the contour down from the original configuration onto the little circle, and this is the physical picture that you should have for applications of Cauchys theorem. The contour is collapsed down to a collection of little circles around each of the singularities, oriented in the same direction as the original contour.

There are still two issues: Firstly, when we solved the differential equation we observed the possibility of non-integer values of m and this possibility is not covered by the current analysis. The function would not join up as we circle the singularity and so we would *require* an added line for the discontinuity and then this added line could be used in our argument and the two integrals along this line in opposite directions would not cancel and our argument would fail. This occurs later in the course and the added line would be referred to as a *branch cut.* Secondly we need to deal with more than one singularity, but this is very easy, we can do one singularity at a time. For each singularity we can connect up the little circle with the original contour and generate a new contour which avoids this one singularity but still encloses the others. We keep going until we generate a contour that encloses none. Whether the added lines cross each-other or not is immaterial, but they should not be chosen to pass through one of the other singularities.

The residue theorem

The next step is to look at the little circle around a singularity and to find out what we get when the function is describable with a Laurent series. The key to this is to realise that

$$\int dz z^n = \frac{z^{n+1}}{n+1}$$

for all $n \neq$-1, including negative integers, and that

$$\int dz \frac{1}{z} = \ln z$$

and so

$$\int dz \sum_{m=-\infty}^{\infty} a_m z^m = \sum_{m\neq -1} a_m \frac{z^{m+1}}{m+1} + a_{-1} \ln z$$

Using the polar representation

$$\frac{z^{m+1}}{m+1} = \frac{\rho^{m+1} e^{i(m+1)\phi}}{m+1} \qquad \ln z = \ln \rho + i\phi$$

and so, provided that m is an integer, when we go around the little circle $\phi \mapsto \phi + 2\pi$ and for all cases except the logarithm we get back where we started and the integral vanishes, but for the logarithm we get $2\pi i$ and the integral does not vanish! This then gives us the residue theorem

$$\int_C dz f(z) = 2\pi i \sum_{z_e} R[z_e] W[z_e; C]$$

where the sum is over all extractable singularities enclosed by the contour, C, $R[z_e] \equiv a_{-1}$ is the coefficient of $\frac{1}{z-z_e}$ in the Laurent series associated with that singularity and $W[z_e; C]$ is the winding number, the number of times the contour winds around C in an anti-clockwise direction. We did not explain this before, but the contour can go around a singularity in either direction and can circuit the singularity multiple times. Each such circuit would leave behind a little circle and the winding number is just the total number of such little circles, with a plus unity if it is anticlockwise and minus unity if it is clockwise.

We now find a third procedure that you need to be proficient in, as well as finding singularities and classifying them, you will also need to be able to find any residue associated with the singularity. An extractable singularity is known as a *pole,* and in the Laurent expansion, if the series starts at a finite negative number, n say, then we call the singularity an n'th order pole, which amounts to classifying the singularity. We also use the language 'elementary pole' or 'simple pole' for first order.

We now have the main tool of the course; the residue theorem. For functions with extractable singularities we can calculate the integral of the function on a closed curve by using the residue theorem. Most of the assessed problems and exam questions will require you to do just that!

Examples: We may assume that we have collapsed the contour down to a little circle around the singularity and so all we need to be able to do is find a residue to use the residue theorem.

$$\frac{3z^2-1}{z^3-z} = \frac{3z^2-1}{z(z-1)(z+1)} = \frac{1}{z}\frac{-1}{(-1)(1)} + \frac{1}{z-1}\frac{2}{(1)(2)} + \frac{1}{z+1}\frac{2}{(-1)(-2)}$$

$$= \frac{1}{z} + \frac{1}{z-1} + \frac{1}{z+1}$$

and all three possible residues are unity for this example, and the cover-up rule is what we really needed.

$$\frac{z^2+1}{z^3-z} = \frac{z^2+1}{z(z-1)(z+1)} = \frac{1}{z}\frac{1}{(-1)(1)} + \frac{1}{z-1}\frac{2}{(1)(2)} + \frac{1}{z+1}\frac{2}{(-1)(-2)}$$

$$= -\frac{1}{z} + \frac{1}{z-1} + \frac{1}{z+1}$$

and now the origin has minus unity while the other two have unity for the residues.

Finding Residues

One might think that the previous partial fractions idea controls finding residues, but in practice, for simple poles, finding residues is *easier* than partial fractions. If we know that the function has an elementary pole, has only one singular term in the Laurent expansion that corresponds to the residue term, then we can multiply by this singularity and create an analytic function with a regular Taylor's series from it, $(z-z_e)f(z)$, if z_e is the position of the singularity. The residue is then the value of this new function at the singularity on the original function and so for this special case

$$R(z_e) = \lim_{z \mapsto z_e} (z-z_e) f(z)$$

and we can employ the cover-up rule directly without finding the regular piece. If we know that there is an order n pole at the singularity, ie the Laurent series starts at $\frac{a_{-n}}{(z-z_e)^n}$ then the same logic tells us that we can make an analytic function, $(z-z_e)^n f(z)$, but the residue is now the n-1 term in the Taylor's series so we need to use

$$R(z_e) = \lim_{z \mapsto z_e} \frac{1}{(n-1)!} \left(\frac{d}{dz} \right)^{n-1} (z-z_e)^n f(z)$$

for multiple poles.

Partial Fractions versus Taylor's Theorem

There is an alternative method of finding residues in simple cases, using Taylor's theorem on the denominator instead. If we know that there is a single root to the denominator and we are dealing with an elementary simple pole then we can use

$$D[z] = D[z_n] + (z-z_n) \frac{dD}{dz}[z_n] + ...$$

and then $D[z_n]$=0 as z_n is a root and we can pull out the root and use differentiation to find the product of the other roots. This is very useful when it is very easy to differentiate the denominator. The partial fractions issue with making sure that the numerator is lower order than the denominator remains, and we also need to evaluate the numerator at the root. Obviously, when finding residues we do not need to extract the regular piece. To show the value of this idea, let us look at the previous "ugly example" with this idea

$$\frac{1}{z^4+1} \equiv \frac{1}{D[z]}$$

for any root

$$D[z] = D[z_n] + (z-z_n) \frac{dD}{dz}[z_n] + ... \quad \Rightarrow \quad D[z] = (z-z_n) 4 z_n^3 + ...$$

and so

$$\frac{1}{z^4+1} = \sum_n \frac{1}{z-z_n} \frac{1}{4 z_n^3} = -\sum_n \frac{1}{z-z_n} \frac{z_n}{4}$$

where we used z_n^4=-1 to obtain the same answer that we got before, but with a lot less work. Note that here we are effectively parameterising, because we do not need to be able to solve the equation in order to find the final answer.

Example: Find the residue of the function

$$f(z) = \frac{1}{(z^3+1)^2}$$

at the point z=-1. We can employ the direct Taylor's theorem idea, using

$$z^3 + 1 = (z+1)(z^2 - z + 1)$$

to get

$$\frac{d}{dz}(z+1)^2 f(z) = \frac{d}{dz}\frac{1}{(z^2-z+1)^2} = \frac{(-2)(2z-1)}{(z^2-z+1)^3}$$

and then evaluating at z=-1 we find the residue is $\frac{2}{9}$. Would the Taylor's theorem on the denominator help? For this we would need to go to higher order

$$D[z] = D[z_0] + (z-z_0)\frac{dD}{dz}(z_0) + \frac{1}{2}(z-z_0)^2\frac{d^2D}{dz^2}(z_0) + ... = (z-z_0)3z_0^2 + (z-z_0)^2 3z_0 + ...$$

$$\mapsto 3(z+1) - 3(z+1)^2 + ...$$

and so

$$\frac{1}{D[z]^2} \mapsto \frac{1}{9(z+1)^2}\left[1 - (z+1) + ...\right]^{-2} = \frac{1}{9(z+1)^2}\left[1 + 2(z+1) + ...\right]$$

and we can deduce that the residue is $\frac{2}{9}$ again. Not much of a saving, if any, but if we wanted to find the residues at the other non-trivial poles then it would be much easier

$$\frac{1}{D[z]^2} \mapsto \frac{1}{9z_0^4(z-z_0)^2}\left[1 + \frac{1}{z_0}(z-z_0) + ...\right]^{-2} = \frac{1}{9z_0^4(z-z_0)^2}\left[1 - \frac{2}{z_0}(z-z_0) + ...\right]$$

and the residue is -$\frac{2}{9z_0^5}$=-$\frac{2z_0}{9}$.

Fourier Series

Hopefully, in one of your previous mathematics courses, you have encountered Fourier series. A periodic function may be represented as a sum of trigonometric functions. The simplest variant is

$$f(\phi) = \sum_m a_m e^{im\phi}$$

and this basis is particularly useful for dealing with the derivative, since

$$-i\frac{\partial}{\partial\phi}e^{im\phi} = me^{im\phi}$$

is diagonal. The inverse transform is

$$a_m = \int_0^{2\pi} \frac{d\phi}{2\pi} e^{-im\phi} f(\phi)$$

and interestingly this has a complex variable theory analogue. We can use polar coordinates $z \equiv \rho e^{im\phi}$ and then the integral is an integral around the unit circle and

$$a_m = \oint_C \frac{dz}{2\pi i z} z^{-m} f(z)$$

where

$$f(z) = \sum_m a_m z^m$$

is the natural generalisation from the angular function to an analytic function, analytic everywhere except perhaps at the origin. This is a simple Laurent expansion and then the residue theorem establishes the validity of the result, independently from your previous course.

The same idea allows us to calculate trigonometric integrals, recall

$$I_n = \int_0^{2\pi} \frac{d\phi}{2\pi} [\cos\phi]^{2n}$$

which transformed into

$$I_n = \oint_C \frac{dz}{2\pi i z} \frac{1}{2^{2n}} \left[z + \frac{1}{z} \right]^{2n}$$

and the binomial theorem immediately provided us with the Laurent series at the origin

$$f_n(z) \equiv \frac{1}{z} \frac{1}{2^{2n}} \left[z + \frac{1}{z} \right]^{2n} = \frac{1}{2^{2n}} \sum_{m=0}^{2n} \frac{2n!}{m!(2n-m)!} z^{2m-2n-1}$$

and now the residue theorem picks out the term with $m{=}n$ to give

$$I_n = \frac{1}{2^{2n}} \frac{2n!}{n!n!}$$

quite a simple method.

Example: You can do elementary trigonometric integrals using the residue theorem

$$\int_0^{2\pi} \frac{d\phi}{2\pi} \cos 4\phi \sin 2\phi \sin\phi \cos\phi$$

$$= \oint_C \frac{dz}{2\pi i z} \frac{1}{2} \left(z^4 + \frac{1}{z^4} \right) \frac{1}{2i} \left(z^2 - \frac{1}{z^2} \right) \frac{1}{2i} \left(z - \frac{1}{z} \right) \frac{1}{2} \left(z + \frac{1}{z} \right)$$

and we just need the constant term in

$$-\frac{1}{16} \left(z^6 - z^2 + \frac{1}{z^2} - \frac{1}{z^6} \right) \left(z^2 - \frac{1}{z^2} \right)$$

and we can read off that the constant term is -$\frac{1}{8}$. To do the integral using trigonometry would require

$$\sin\phi\cos\phi = \frac{1}{2}\sin 2\phi \qquad \sin 2\phi\sin 2\phi = \frac{1}{2}\left(1-\cos 4\phi\right) \qquad \cos 4\phi\cos 4\phi = \frac{1}{2}\left(\cos 8\phi + 1\right)$$

which gives the same answer but with more algebra and knowledge.

Laplace Transforms

In order to have a real physical problem for you to tackle, how to deal with linear ordinary differential equations in time will now be introduced. The formal details will be considered later, but the theory itself is surprisingly straightforward and easy to grasp and get control of. The central quantity is the Laplace transform, which is defined to be

$$F[z] = \int_0^\infty dt e^{-zt} f(t)$$

where $z \geq \Lambda$ and Λ is chosen to be large enough to make the integral converge. z is a complex number and the integral is usually done using real analysis. It is the inverse transform

$$f(t) = \oint_{C[\Lambda]} \frac{dz}{2\pi i} e^{zt} F[z]$$

that gives us useful examples to get you competent with finding residues and doing complex integrals. The contour, $C[\Lambda]$ is called the Bromwich contour and starts on the real axis at the point Λ, goes upwards parallel to the y-axis to the point $\Lambda+i\mu$, where $R^2{=}\Lambda^2+\mu^2$ is sent off to infinity at the end, then the contour circles around at radius R in an anti-clockwise direction until it arrives at Λ-$i\mu$ and it then goes upwards parallel to the y-axis until it closes when it meets the starting point again. This provides a closed contour that catches all of the singularities to the left of the point Λ and circles them each once in an anti-clockwise direction. The winding number is consequently always unity, the $2\pi i$ is the denominator in the integral and so the answer to this integral is simply a sum over all the residues of all the singularities to the left of Λ. Although it is currently not obvious, Λ is chosen so that it is to the right of *all* the singularities and so the final answer to this integral is the sum of all the residues of all the poles in the function $e^{zt}F[z]$.

To apply the theory one simply calculates the Laplace transform of the differential equation requiring solutions, probably the simplest example is nuclear decay

$$\frac{dN}{dt}(t) = -\lambda N(t)$$

and we need to calculate

$$\int_0^\infty dt e^{-zt}\frac{dN}{dt}(t) = -\lambda\int_0^\infty dt e^{-zt}N(t)$$

and we can recognise the Laplace transform of the number on the right hand side; $n[z]$. The first term must be performed using integration by parts

$$\int_0^\infty dt e^{-zt}\frac{dN}{dt}(t) = \left[e^{-zt}N(t)\right]_0^\infty + z\int_0^\infty dt e^{-zt}N(t)$$

and we feed in the initial conditions as we do the calculation! The value of z is chosen so that the behaviour at infinity is irrelevant and we find

$$-N(0) + zn[z] = -\lambda n[z] \quad \Rightarrow \quad n[z] = \frac{N(0)}{z+\lambda}$$

The number as a function of time is deduced from the inverse transform

$$N(t) = \int_C \frac{dz}{2\pi i} e^{zt} n[z] = \int_C \frac{dz}{2\pi i} e^{zt} \frac{N(0)}{z+\lambda}$$

and a simple application of the residue theorem provides

$$N(t) = N(0)e^{-\lambda t}$$

as expected.

More common application is to oscillating problems, for example

$$\frac{d^2 f}{dt^2} + f = e^{-t}$$

the forced harmonic oscillator. We get

$$\int_0^\infty dt e^{-zt} \frac{d^2 f}{dt^2}(t) + \int_0^\infty dt e^{-zt} f(t) = \int_0^\infty dt e^{-zt} e^{-t}$$

The final integral is trivial real integration

$$\int_0^\infty dt e^{-zt} e^{-t} = \left[\frac{e^{-zt-t}}{-z-1} \right]_0^\infty = \frac{1}{z+1}$$

The second integral is simply the quantity $F[z]$ and the first integral we do using integration by parts

$$\int_0^\infty dt e^{-zt} \frac{d^2 f}{dt^2}(t) = \left[e^{-zt} \frac{df}{dt}(t) \right]_0^\infty + z \int_0^\infty dt e^{-zt} \frac{df}{dt}(t)$$

and then we integrate by parts a second time

$$z \int_0^\infty dt e^{-zt} \frac{df}{dt}(t) = \left[e^{-zt} z f(t) \right]_0^\infty + z^2 \int_0^\infty dt e^{-zt} f(t)$$

and we can recognise the quantity $F[z]$ again. In total we get

$$(z^2+1)F[z] + \left[e^{-zt} \left(\frac{df}{dt}(t) + z f(t) \right) \right]_0^\infty = \frac{1}{z+1}$$

In this type of problem we know the initial conditions, where the particle is and how fast it is going at the start, which we can feed in to the boundary terms and Λ is chosen to be large enough that the terms at infinity vanish and so we are left with

$$(z^2+1)F[z] = \frac{df}{dt}(0) + z f(0) + \frac{1}{z+1}$$

and we can find the function, $F[z]$, and determine the residues of $e^{zt}F[z]$. For the current example, the particular integral comes from

$$F[z] = \frac{1}{(z+1)(z^2+1)} + \frac{df}{dt}(0)\frac{1}{z^2+1} + f(0)\frac{z}{z^2+1}$$

The final two terms are generic to all forced simple harmonic oscillator problems and the partial fractions idea gives

$$\frac{1}{z^2+1} = \frac{1}{2i}\left(\frac{1}{z-i} - \frac{1}{z+i}\right) \qquad \frac{z}{z^2+1} = \frac{1}{2i}\left(\frac{i}{z-i} - \frac{-i}{z+i}\right) = \frac{1}{2}\left(\frac{1}{z-i} + \frac{1}{z+i}\right)$$

which gives as the residues

$$\frac{df}{dt}(0)\frac{e^{it}-e^{-it}}{2i} + f(0)\frac{e^{it}+e^{-it}}{2} = \frac{df}{dt}(0)\sin t + f(0)\cos t$$

which you should recognise as the correct solution to the problem with no forcing. The forcing term requires more work, with

$$\frac{1}{(z+1)(z-i)(z+i)} = \frac{1}{z+1}\frac{1}{(-1-i)(-1+i)} + \frac{1}{z-i}\frac{1}{(1+i)(2i)} + \frac{1}{z+i}\frac{1}{(1-i)(-2i)}$$

$$= \frac{1}{2}\frac{1}{z+1} - \frac{1}{4}\frac{1+i}{z-i} - \frac{1}{4}\frac{1-i}{z+i}$$

and final residues of

$$\frac{1}{2}e^{-t} - \frac{1}{2}\cos t + \frac{1}{2}\sin t$$

which you can readily check does not disturb the previous boundary conditions and does indeed solve the original problem.

Example: An oscillatory force

$$\frac{d^2f}{dt^2} + f = \sin \nu t$$

The Laplace transform is analogous to last time with

$$(z^2+1)F[z] - \frac{df}{dt}(0) - zf(0) = \int_0^\infty dt e^{-zt}\frac{1}{2i}\left(e^{i\nu t} - e^{-i\nu t}\right)$$

$$= \frac{1}{2i}\left[\frac{e^{i\nu t - zt}}{-z+i\nu} - \frac{e^{-i\nu t - zt}}{-z-i\nu}\right]_0^\infty = \frac{1}{2i}\left(\frac{1}{z-i\nu} - \frac{1}{z+i\nu}\right) = \frac{\nu}{z^2+\nu^2}$$

and then the particular integral is

$$\int_C \frac{dz}{2\pi i}e^{zt}\frac{\nu}{(z^2+1)(z^2+\nu^2)}$$

and partial fractions offer

$$\frac{\nu}{1-\nu^2}\left(\frac{1}{z^2+\nu^2} - \frac{1}{z^2+1}\right) = \frac{\nu}{1-\nu^2}\left(\frac{1}{2i\nu}\left[\frac{1}{z-i\nu} - \frac{1}{z+i\nu}\right] - \frac{1}{2i}\left[\frac{1}{z-i} - \frac{1}{z+i}\right]\right)$$

and so the residues are

$$\frac{1}{1-\nu^2}\left[\frac{1}{2i}\left(e^{i\nu t}-e^{-i\nu t}\right)-\frac{\nu}{2i}\left(e^{it}-e^{-it}\right)\right]=\frac{\sin\nu t-\nu\sin t}{1-\nu^2}$$

which is indeed the solution which does not disturb the boundary conditions. There is an interesting pathology, $\nu\mapsto 1$, which corresponds to resonance, and we can deduce the answer by defining ν=1+δ and limiting $\delta\mapsto 0$

$$\frac{\sin(t+\delta t)-(1+\delta)\sin t}{-\delta(2+\delta)}=\frac{\sin t\cos\delta t+\cos t\sin\delta t-(1+\delta)\sin t}{-\delta(2+\delta)}\mapsto\frac{t\cos t-\sin t}{-2}$$

$$=\frac{1}{2}\left(\sin t-t\cos t\right)$$

Double poles and beyond

The previous resonant example can be tackled directly, with

$$\frac{d^2f}{dt^2}+f=\sin t$$

transforming to

$$(z^2+1)F[z]-\frac{df}{dt}(0)-zf(0)=\int_0^\infty dte^{-zt}\frac{1}{2i}\left(e^{it}-e^{-it}\right)$$

$$=\frac{1}{2i}\left[\frac{e^{it}}{-z+i}-\frac{e^{-it}}{-z-i}\right]_0^\infty=\frac{1}{2i}\left[\frac{1}{z-i}-\frac{1}{z+i}\right]=\frac{1}{z^2+1}$$

and the particular integral is now controlled by the function

$$\frac{1}{(z^2+1)^2}$$

an analytic function with a double pole. How should we handle double poles? Is there an analogue to the cover-up rule?

Sadly, multiple poles are much more difficult to handle than single poles. There is an analogue to the cover-up rule, you cover up the double pole and the rest, evaluated at the position of the pole, gives you the coefficient of the double pole. The problem is that one only cares about the residue and this is the next term in the series!

The most pedestrian method of proceeding is to multiply by the correct power of $z-z_e$ to turn the function into a Taylor's series and then expand the Taylor's series to one less order in order to deduce the residue. It is slightly better to translate to the pole, z=z_e+ζ, and then to expand using your mathematical ingenuity. One tends to be better at expanding simple functions rather than differentiating them. The previous example is

$$\frac{e^{zt}}{(z-i)^2(z+i)^2}\mapsto\frac{e^{it}e^{\zeta t}}{\zeta^2(2i+\zeta)^2}=e^{it}\frac{1}{-4\zeta^2}e^{\zeta t}\left(1-i\frac{\zeta}{2}\right)^{-2}$$

$$\mapsto -e^{it}\frac{1}{4\zeta^2}\left(1+\zeta t+...\right)\left(1+i\zeta...\right)$$

and so the residue is -$e^{it}\frac{t+i}{4}$. The direct Taylor's series idea says

$$f(z)=\frac{e^{zt}}{(z+i)^2} \quad \Rightarrow \quad \frac{df}{dz}(z)=-2\frac{e^{zt}}{(z+i)^3}+t\frac{e^{zt}}{(z+i)^2} \quad \Rightarrow \quad \frac{df}{dz}(i)=-2e^{it}\frac{1}{(2i)^3}+te^{it}\frac{1}{-4}$$

$$=-e^{it}\frac{t+i}{4}$$

agreeing with the previous result. The other residue, at -i, is -$e^{-it}\frac{t-i}{4}$ and so the residue of $e^{zt}F[z]$ is

$$\frac{1}{2}\left(-t\cos t+\sin t\right)$$

in agreement with the previous limit.

Example:

$$\frac{d^2f}{dt^2}+f=\cos t$$

subject to the boundary conditions $f(0)$=0 and $\frac{df}{dt}(0)$=0. This Laplace transforms into

$$F[z]=\frac{z}{(z^2+1)^2}$$

and so the answer comes from

$$f(t)=\int_C\frac{dz}{2\pi i}e^{zt}\frac{z}{(z^2+1)^2}$$

At first sight this example looks equally as awful as the previous one, but

$$-\frac{1}{2}\frac{d}{dz}\frac{1}{z^2+1}=\frac{z}{(z^2+1)^2}$$

and so integration by parts gives

$$f(t)=\frac{t}{2}\int_C\frac{dz}{2\pi i}e^{zt}\frac{1}{z^2+1}=\frac{t}{2}\sin t$$

Integration by parts allows one to reduce a double pole to a single pole! This is such a great saving that you should always check whether it works before finding residues. There are never any boundary terms from a closed contour.

Example: In nuclear physics, nuclear decay can be quite complicated. For example; nucleus one, with number $N_1(t)$, can decay into nucleus two, with number $N_2(t)$, with rate λ_1, but also, simultaneously, into nucleus three, with number $N_3(t)$, with rate λ_2. Nucleus two is also unstable and can decay into nucleus three, with rate λ_3, and into nucleus four with rate λ_4. Nucleus three and nucleus four are both stable. How would one solve this problem, and in particular if we start out only with nucleus one, what fraction would end up as nucleus four?

We solve can this type of problem with Laplace transforms. First we build the decay equations using physical intuition

$$\begin{aligned}
\frac{dN_1}{dt} =& -\lambda_1 N_1 - \lambda_2 N_1 \\
\frac{dN_2}{dt} =& \lambda_1 N_1 - \lambda_3 N_2 - \lambda_4 N_2 \\
\frac{dN_3}{dt} =& \lambda_2 N_1 + \lambda_3 N_2 \\
\frac{dN_4}{dt} =& \lambda_4 N_2
\end{aligned}$$

and then we employ the Laplace transform of each equation, which requires

$$\int_0^\infty dt e^{-zt} \frac{dN_\alpha}{dt}(t) = \left[e^{-zt} N_\alpha(t)\right]_0^\infty + z\int_0^\infty dt e^{-zt} N_\alpha(t) = z n_\alpha[z] - N_\alpha(0)$$

and this gives us a chain of algebraic equations

$$\begin{aligned}
z n_1[z] =& N - \lambda_1 n_1[z] - \lambda_2 n_1[z] \qquad \Rightarrow \qquad n_1[z] = \frac{N}{z+\lambda_1+\lambda_2} \\
z n_2[z] =& \lambda_1 n_1[z] - \lambda_3 n_2[z] - \lambda_4 n_2[z] \qquad \Rightarrow \qquad n_2[z] = \frac{\lambda_1 n_1[z]}{z+\lambda_3+\lambda_4} \\
z n_3[z] =& \lambda_2 n_1[z] + \lambda_3 n_2[z] \qquad \Rightarrow \qquad n_3[z] = \frac{\lambda_2 n_1[z]}{z} + \frac{\lambda_3 n_2[z]}{z} \\
z n_4[z] =& \lambda_4 n_2[z] \qquad \Rightarrow \qquad n_4[z] = \frac{\lambda_4 n_2[z]}{z}
\end{aligned}$$

These are best processed using partial fractions because the inverse Laplace transform formula for a single isolated elementary pole is

$$\int_C \frac{dz}{2\pi i} e^{zt} \frac{R}{z+\lambda} = R e^{-\lambda t}$$

and each residue is multiplied by the associated time dependence and summed to give the final answer. It is slightly better to do the partial fractions sequentially, as one only needs to separate a pair of poles at a time. The initial decay is trivial

$$N_1(t) = N e^{-(\lambda_1+\lambda_2)t}$$

and then for $N_2(t)$ we need our first partial fractions

$$\begin{aligned}
n_2[z] &= \frac{N\lambda_1}{(z+\lambda_1+\lambda_2)(z+\lambda_3+\lambda_4)} \\
&= \frac{N\lambda_1}{\lambda_3+\lambda_4-\lambda_1-\lambda_2}\left(\frac{1}{z+\lambda_1+\lambda_2} - \frac{1}{z+\lambda_3+\lambda_4}\right)
\end{aligned}$$

and so

$$N_2(t) = \frac{N\lambda_1}{\lambda_3+\lambda_4-\lambda_1-\lambda_2}\left(e^{-(\lambda_1+\lambda_2)t} - e^{-(\lambda_3+\lambda_4)t}\right)$$

The next round requires partial fractions with $\frac{1}{z}$ and we can use the previous results to get

$$n_3[z] = \frac{N\lambda_2}{z(z+\lambda_1+\lambda_2)} + \frac{N\lambda_1\lambda_3}{\lambda_3+\lambda_4-\lambda_1-\lambda_2}\left(\frac{1}{z(z+\lambda_1+\lambda_2)} - \frac{1}{z(z+\lambda_3+\lambda_4)}\right)$$

$$= \left(N\lambda_2 + \frac{N\lambda_1\lambda_3}{\lambda_3+\lambda_4-\lambda_1-\lambda_2}\right)\frac{1}{\lambda_1+\lambda_2}\left(\frac{1}{z} - \frac{1}{z+\lambda_1+\lambda_2}\right)$$

$$-\frac{N\lambda_1\lambda_3}{\lambda_3+\lambda_4-\lambda_1-\lambda_2}\frac{1}{\lambda_3+\lambda_4}\left(\frac{1}{z} - \frac{1}{z+\lambda_3+\lambda_4}\right)$$

The algebra is now horrendous but we can read off that

$$N_3(t) = \left(N\lambda_2 + \frac{N\lambda_1\lambda_3}{\lambda_3+\lambda_4-\lambda_1-\lambda_2}\right)\frac{1}{\lambda_1+\lambda_2} - \frac{N\lambda_1\lambda_3}{\lambda_3+\lambda_4-\lambda_1-\lambda_2}\frac{1}{\lambda_3+\lambda_4}$$

$$-\left(N\lambda_2 + \frac{N\lambda_1\lambda_3}{\lambda_3+\lambda_4-\lambda_1-\lambda_2}\right)\frac{1}{\lambda_1+\lambda_2}e^{-(\lambda_1+\lambda_2)t}$$

$$+\frac{N\lambda_1\lambda_3}{\lambda_3+\lambda_4-\lambda_1-\lambda_2}\frac{1}{\lambda_3+\lambda_4}e^{-(\lambda_3+\lambda_4)t}$$

The final calculation requires no new separation so

$$n_4[z] = \frac{N\lambda_1\lambda_4}{\lambda_3+\lambda_4-\lambda_1-\lambda_2}\left(\frac{1}{z(z+\lambda_1+\lambda_2)} - \frac{1}{z(z+\lambda_3+\lambda_4)}\right)$$

$$= \frac{N\lambda_1\lambda_4}{\lambda_3+\lambda_4-\lambda_1-\lambda_2}\frac{1}{\lambda_3+\lambda_4}\left(\frac{1}{z} - \frac{1}{z+\lambda_3+\lambda_4}\right)$$

$$-\frac{N\lambda_1\lambda_4}{\lambda_3+\lambda_4-\lambda_1-\lambda_2}\frac{1}{\lambda_1+\lambda_2}\left(\frac{1}{z} - \frac{1}{z+\lambda_1+\lambda_2}\right)$$

and we can read off that

$$N_4(t) = \frac{N\lambda_1\lambda_4}{\lambda_3+\lambda_4-\lambda_1-\lambda_2}\frac{1}{\lambda_1+\lambda_2} - \frac{N\lambda_1\lambda_4}{\lambda_3+\lambda_4-\lambda_1-\lambda_2}\frac{1}{\lambda_3+\lambda_4}$$

$$-\frac{N\lambda_1\lambda_4}{\lambda_3+\lambda_4-\lambda_1-\lambda_2}\frac{1}{\lambda_1+\lambda_2}e^{-(\lambda_1+\lambda_2)t} + \frac{N\lambda_1\lambda_4}{\lambda_3+\lambda_4-\lambda_1-\lambda_2}\frac{1}{\lambda_3+\lambda_4}e^{-(\lambda_3+\lambda_4)t}$$

The level of algebra is appalling but we can read off that the fraction of N_4 at the end is

$$\frac{\lambda_1\lambda_4}{(\lambda_1+\lambda_2)(\lambda_3+\lambda_4)}$$

Physical insight can be very useful in calculations. It is fairly obvious that the number of nuclei is invariant, they simply change type. This means in practice that the sum of the nuclei is a constant

$$N_1(t) + N_2(t) + N_3(t) + N_4(t) = N$$

and if one is feeling lazy, then one can calculate $N_1(t)$, $N_2(t)$ and $N_4(t)$ using the theory and deduce $N_3(t)$ from this relationship. If one is not lazy then this is an excellent internal check to ensure that you have done all the calculations correctly.

Moving the contour

Cauchy's theorem provides the power to move the contour to wherever you might want it. The logic is that one can create a contour which is part of the original contour, the part you want to move, and is part where you want to move it to. If the constructed contour encloses a simply connected region in which the integrand is analytic, then we can use Cauchy's theorem to show that the integral around this constructed contour vanishes. We can then add this contour to the previous contour with the sign that cancels the part you want to move and the bit where it has been moved to remains and reconnects the contour to form a new contour. We can add pieces outside the contour to enlarge the contour and we can add pieces inside the contour to shrink the contour. This is the physical logic behind the little circles argument. We want to try to shrink the contour down to zero, but the extractable singularities are not points of analyticity so we cannot include them. We can shrink the contour past them if we leave the little circle behind.

A second, mildly bizarre, reason for moving the contour is to *create* singularities that can then be used to deduce things through the residue theorem. A common physical example comes from the sinc function

$$S[z] = \frac{\sin z}{z} = \frac{e^{iz} - e^{-iz}}{2iz}$$

which is analytic in the whole complex plane. One might desire to break this function up into the sum of the two parts, each of which does have a singularity, for which the origin is not analytic. If the original contour passes through the origin then we cannot split the function up in this fashion, because the resulting pieces are mathematically meaningless, but if we move the contour off the origin then we can split it up. We will see how this type of argument is used later.

Closing the contour

Probably more important than moving the contour is closing the contour. When we do real analysis, we often end up with real integrals that have limits,

$$a_m = \int_0^{2\pi} \frac{d\phi}{2\pi} e^{-im\phi} \psi(\phi)$$

for example, and this is not a closed contour in ϕ. We previously saw that employing $z = \rho e^{i\phi}$ and thinking about the integral being on the unit circle allows us to convert this integral into a closed complex integral which can be analysed using complex variable theory. The example

$$I = \int_{-\infty}^{\infty} dz \frac{1}{z^4 + 1}$$

is a clear real-integral with distinct end-points and there is no obvious way to interpret this as a closed contour. The key here is to realise that the function is very small as

$z \mapsto \infty$, and we can show that a large semi-circle at radius R, $C(R)$ say, provides a vanishingly small contribution

$$\lim_{R \mapsto \infty} \int_{C(R)} dz \frac{1}{z^4 + 1} = 0$$

and then we can construct a closed contour for each radius R, by going along the real axis, from -R to R and then using the semicircle upwards in an anti-clockwise direction to complete the contour. In the limit that $R \mapsto \infty$, the semi-circle piece vanishes and the integral along the real-axis is the quantity we want to calculate. We can then use the residue theorem on the closed contour to calculate the real integral. Arguments based around this logic have two clear inputs; we need to show that the piece we want to add vanishes in the appropriate limit and we need to perform the limit on the finite contour.

The first task can be accomplished using the triangle inequality

$$|| x | - | y ||< | x + y |< | x | + | y |$$

so

$$| \int_{C(R)} dz \frac{1}{z^4 + 1} |< \int_{C(R)} | dz || \frac{1}{z^4 + 1} |< \int_{C(R)} | dz | \frac{1}{|| z |^4 - 1 |} = \pi R \frac{1}{R^4 - 1}$$

which clearly limits to zero as $R \mapsto \infty$.

The mathematical logic is the triangle inequality for the numerator

$$| N_0 + N_1 |< | N_0 | + | N_1 |$$

and for the denominator two triangle inequalities

$$| D_0 + D_1 - D_1 |< | D_0 + D_1 | + | D_1 | \qquad | D_0 + D_1 - D_0 |< | D_0 + D_1 | + | D_0 |$$

which combine to give

$$|| D_0 | - | D_1 ||< | D_0 + D_1 | \Rightarrow \frac{1}{| D_0 + D_1 |} < \frac{1}{|| D_0 | - | D_1 ||}$$

which combine to give

$$| \frac{N_0 + N_1}{D_0 + D_1} |< \frac{| N_0 | + | N_1 |}{|| D_0 | - | D_1 ||}$$

If the numerator and denominator both have a dominant piece in the limit, N_0 and D_0, then in the limit we can bound the function with this and *ignore* the small pieces, N_1 and D_1, to replace the function with a "bound in the limit" of

$$B_l =| \frac{N_0}{D_0} |$$

The integral is then controlled by

$$| \int_C dz \frac{N}{D} | < \int_C | dz | B_l = L(C) B_l$$

where $L(C)$ is the length of the contour. For us this is πR and then finally we limit $R \mapsto \infty$. Clearly, if the limiting bound beats the length then we can close the contour.

The second issue involves increasing the radius R and arguing that nothing changes, but this is the logic of moving the contour in the last section, and so provided there are no singularities close to infinity then this is totally safe.

The current logic is good when the real integral ranges from minus infinity to infinity and we can incorporate a large semi-circle at a huge radius, but what do you do if one of the limits is at a finite value? Usually, it is game over and you are screwed, but the final possibility to look for is *symmetry.* If the integrand has some symmetry that allows you to see the same integral again somewhere else in the complex plane, but starting at the same finite limit, then sometimes you can use use this second integral to extend the original contour off to infinity in two different directions, and then one is back in the previous situation and closure at infinity is then possible. The simplest example is inversion symmetry, if the integrand satisfies $f(-x)=f(x)$ then we can use $x \mapsto y = -x$ to show that

$$\int_0^\infty dx f(x) = -\int_0^{-\infty} dy f(-y) = \int_{-\infty}^0 dy f(y)$$

and then

$$\int_0^\infty dx f(x) = \frac{1}{2} \int_{-\infty}^\infty dx f(x)$$

Other symmetries usually involve rotations, for example

$$I \equiv \int_0^\infty dx \frac{1}{x^3+1}$$

can be attempted by rotating through $\frac{2\pi}{3}$, using a non-trivial cube root of unity, $\omega \equiv e^{i\frac{2\pi}{3}}$, and then $z \mapsto \omega z \equiv z'$ gives

$$I = \int_0^\infty dz \frac{1}{z^3+1} = \frac{1}{\omega} \int_0^{\omega\infty} dz' \frac{1}{\omega^{-3} z'^3 + 1} = \frac{1}{\omega} \int_0^{\omega\infty} dz \frac{1}{z^3+1}$$

and then

$$I\,(1-\omega) = \int_0^\infty dz \frac{1}{z^3+1} + \int_{\omega\infty}^0 dz \frac{1}{z^3+1}$$

We can close the contour at infinity using the previous argument, since $R^3 \gg \frac{2\pi R}{3}$ to get

$$I = \frac{1}{1-\omega} \oint_C dz \frac{1}{z^3+1}$$

where the contour C goes around the third segment of a large circle going from the x-axis to a line $\frac{2\pi}{3}$ away in an anti-clockwise direction, circling the contour in an anti-clockwise direction. Now (finally) we can use the residue theorem to get the answer, there is a single relevant pole at $\sqrt{\omega}$ and using the Taylor's theorem method, because the denominator is easy to differentiate, we get

$$I = 2\pi i \frac{1}{1-\omega} \frac{1}{3(\sqrt{\omega})^2} = \frac{2\pi i}{3(\omega - \omega^2)} = \frac{2\pi}{3\sqrt{3}}$$

Probably the most important thing to realise, is that the residue theorem is the key, but the algebraic skill required to do all the transformations and find all the residues correctly is the difficult thing!

Example: Calculate

$$I = \int_0^\infty dx \frac{1}{x^6 + 1}$$

It is clear that this is a symmetric function and so we can use the previous arguments to send the contour off to infinity and then close it at infinity.

$$I = \frac{1}{2} \int_C dz \frac{1}{z^6 + 1}$$

where the contour encloses a huge semi-circle above the real axis and rotates in an anti-clockwise direction. The calculation is accomplished by calculating the residues of the three poles captured by this contour. Although this is a natural method, that most students would use, it is not the best! The function also has the symmetry that $z \mapsto \Lambda z$, with Λ^6=1 and so $z \mapsto \Lambda z = z'$ yields

$$I = \frac{1}{\Lambda} \int_0^{\Lambda\infty} dz' \frac{1}{\Lambda^{-6} z'^6 + 1} = \frac{1}{\Lambda} \int_0^{\Lambda\infty} dz \frac{1}{z^6 + 1}$$

just like the last example, and so

$$(1 - \Lambda) I = \int_C dz \frac{1}{z^6 + 1}$$

where C is the contour which wraps around the sixth segment of a very large circle, ranging from ϕ=0 to ϕ=$\frac{\pi}{3}$. Closing at infinity is essentially the same problem as before but with only a third of the length. We then capture only the single pole at $\sqrt{\Lambda}$ which involves only a single residue. It is a choice whether to minimise the number of residues to calculate but with a tricky contour or have an elementary contour and many residues. The same trick of Taylor's expansion gives

$$I = 2\pi i \frac{1}{1-\Lambda} \frac{1}{6(\sqrt{\Lambda})^5} = \frac{2\pi i}{6(\Lambda^{\frac{5}{2}} - \Lambda^{\frac{7}{2}})} = \frac{2\pi i}{6(\Lambda^{\frac{5}{2}} - \Lambda^{-\frac{5}{2}})} = \frac{\pi}{6 \sin \frac{5\pi}{6}} = \frac{\pi}{6 \sin \frac{\pi}{6}} = \frac{\pi}{3}$$

where we used $\Lambda^6 = 1$ and $\Lambda = e^{i\frac{\pi}{3}}$ and $\sin(\pi - x) = \sin x$. Note that there are better and worse ways of doing the calculation, we have Λ^6=1 and Λ=$e^{i\frac{\pi}{3}}$, so

$$I = 2\pi i \frac{1}{1-\Lambda} \frac{1}{6(\sqrt{\Lambda})^5} = 2\pi i \frac{1}{1-\Lambda} \frac{(\sqrt{\Lambda})^7}{6} = 2\pi i \frac{1}{1-\Lambda} \frac{\sqrt{\Lambda}(\sqrt{\Lambda})^6}{6} = 2\pi i \frac{1}{1-\Lambda} \frac{\sqrt{\Lambda}(-1)}{6}$$

$$= \frac{2\pi i}{6\left(\sqrt{\Lambda} - \frac{1}{\sqrt{\Lambda}}\right)} = \frac{\pi}{6\sin\frac{\pi}{6}} = \frac{\pi}{3}$$

a clearer more simple and direct method.

Calculating the residues of all three poles using the Taylor trick gives $\frac{1}{6x^5}$ for each root x and

$$I = \frac{2\pi i}{2}\left(\frac{1}{6e^{i\frac{5\pi}{6}}} + \frac{1}{6e^{i\frac{15}{6}}} + \frac{1}{6e^{i\frac{25\pi}{6}}}\right) = \frac{i\pi}{6}\left(e^{-i\frac{5\pi}{6}} + e^{-i\frac{5\pi}{2}} + e^{-i\frac{\pi}{6}}\right)$$

$$= \frac{\pi}{6}\left(e^{-i\frac{\pi}{3}} + 1 + e^{i\frac{\pi}{3}}\right) = \frac{\pi}{6}\left(1 + 2\cos\frac{\pi}{3}\right) = \frac{\pi}{3}$$

and the same answer with a very different set of calculations....

Momentum space representation in quantum mechanics

In most early undergraduate quantum mechanics courses the real-space wavefunction takes centre stage. The momentum is an operator

$$\mathbf{p} \equiv -i\hbar\nabla$$

and Schrödinger's equation is generated through $H\psi(\mathbf{r})$=$E\psi(\mathbf{r})$, and for non-relativistic problems we use

$$H = \frac{\mathbf{p}.\mathbf{p}}{2m} + V(\mathbf{r}) \mapsto -\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r})$$

Quantum mechanics may also be set up in momentum-space instead of real-space and then $\mathbf{r}$ becomes an operator and the momentum wavefunction takes centre stage. The real-space wavefunctions that have definite momentum are plane waves and

$$-i\hbar\nabla e^{i\mathbf{k}.\mathbf{r}} = \hbar\mathbf{k}e^{i\mathbf{k}.\mathbf{r}}$$

shows that the momentum is simply $\hbar\mathbf{k}$. To convert between representations requires the Fourier transform, which in one-dimension is

$$\psi(x) = \int_{-\infty}^{\infty} \frac{dx}{\sqrt{(2\pi)}} e^{ikx}\tilde{\psi}(k)$$

and inverse transform

$$\tilde{\psi}(k) = \int_{-\infty}^{\infty} \frac{dk}{\sqrt{(2\pi)}} e^{-ikx}\psi(x)$$

Complex Variable Theory can be useful in performing Fourier transforms.

Example: An attractive δ-function potential.
Schrödinger's equation for such a problem is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} - \nu\delta(x)\psi(x) = E\psi(x)$$

which can be solved away from the origin with

$$\psi(x) = N\left(\theta(x)e^{-\kappa x} + \theta(-x)e^{\kappa x}\right)$$

where we used symmetry to relate the two pieces, we used θ-functions to separate the two regions, N is a normailsation constant and we have

$$-\frac{\hbar^2\kappa^2}{2m} = E$$

in order to satisfy Schrödinger's equation. Finally, integrating across the δ-finction tells us that

$$-\frac{\hbar^2}{2m}\left[\frac{d\psi}{dx}(0+) - \frac{d\psi}{dx}(0-)\right] - \nu\psi(0) = 0$$

where we have assumed that $\psi(x)$ is continuous at the origin, which symmetry required. This then provides

$$\frac{\hbar^2\kappa}{m} = \nu$$

which specifies κ in terms of known quantities and leads to the energy

$$E = -\frac{\nu^2 m}{2\hbar^2}$$

For this example we can determine the momentum-space wavefunction

$$\tilde{\psi}(k) = \frac{N}{\sqrt{(2\pi)}}\left[\int_0^\infty dx e^{ikx-\kappa x} + \int_{-\infty}^0 dx e^{ikx+\kappa x}\right] = \frac{N}{\sqrt{(2\pi)}}\left[\frac{1}{\kappa - ik} + \frac{1}{\kappa + ik}\right]$$

$$= \frac{N}{\sqrt{(2\pi)}}\frac{2\kappa}{k^2+\kappa^2}$$

and, as a function of k, we recognise an analytic function with two extractable singularities at $k{=}\pm i\kappa$. We cannot perform the inverse transform of this wavefunction using real-analysis but it is susceptible to the residue theorem.

Fourier Transforms

Space is not like time and one can go backwards as well as forwards. The analogue of Laplace transforms for space is Fourier transforms

$$\tilde{f}(k) = \int_{\mathbf{R}} \frac{dx}{\sqrt{(2\pi)}} e^{-ikx} f(x)$$

$$f(x) = \int_{\mathbf{R}} \frac{dk}{\sqrt{(2\pi)}} e^{ikx} \tilde{f}(k)$$

and neither of these integrals is closed, so to use Complex Variable Theory we need to close the contour and for this we need Jordan's Lemma or some analogue.

Jordan's Lemma

This is a mathematical trick that can be used to close the contour if the function limits to zero on a large circle of radius R, in the limit that $R \mapsto \infty$. When $x > 0$ we close the contour upwards and we include the exponential in the integral

$$| \int_{C(R)} dz e^{ixz} f(z) | \leq \int_0^{\pi} d\phi R e^{-xR\sin\phi} B_l(R) = 2\int_0^{\frac{\pi}{2}} d\phi R e^{-xR\sin\phi} B_l(R)$$

where $B_l(R)$ is the current bound for the function at radius R, $C(R)$ is the semi-circle at radius R, circled in an anti-clockwise direction and x is positive.

We can now bound this integral using real analysis

$$\frac{2\phi}{\pi} \leq \sin\phi \quad \Rightarrow \quad e^{xR\frac{2\phi}{\pi}} \leq e^{xR\sin\phi}$$

$$2\int_0^{\frac{\pi}{2}} d\phi R e^{-xR\sin\phi} B_l(R) \leq 2\int_0^{\frac{\pi}{2}} d\phi R e^{-\frac{2\phi}{\pi}xR} B_l = \frac{\pi}{x}\left[1 - e^{-xR}\right] B_l \leq \frac{\pi}{x} B_l \mapsto 0$$

When $x < 0$ we close the contour downwards using

$$| e^{ixz} | = e^{-(-x)R\sin(-\phi)}$$

and an analogous argument with $\phi \mapsto -\phi$. The argument is equivalent, but the winding number for closing downwards is minus unity as the contour is clockwise.

Example:

$$f(x) = \frac{1}{x^2+1} \quad B_l = \frac{1}{R^2-1} \mapsto \frac{1}{R^2} \mapsto 0$$

$$\tilde{f}(k) = \int_{\mathbf{R}} \frac{dx}{\sqrt{(2\pi)}} e^{-ikx} \frac{1}{x^2+1}$$

$$\mapsto \theta(-k)\int_{C_u(R)} \frac{dz}{\sqrt{(2\pi)}} e^{-ikz} \frac{1}{z^2+1} + \theta(k)\int_{C_d(R)} \frac{dz}{\sqrt{(2\pi)}} e^{-ikz} \frac{1}{z^2+1}$$

where $C_u(R)$ is the finite contour ranging from -R to R along the real axis and then around the upper semi-circle at radius R to close up and where $C_d(R)$ is the analogue contour using the lower semi-circle at radius R. Using the theta-functions is a trick to allow one to write the complete answer down using the separation into the choice of sign for k explicitly. We can then calculate the appropriate residue for the pole that is actively captured by the contour

$$\tilde{f}(k) = \theta(-k)\left[\frac{2\pi i}{2i}\frac{e^{k}}{\sqrt{(2\pi)}}\right] + \theta(k)\left[-\frac{2\pi i}{-2i}\frac{e^{-k}}{\sqrt{(2\pi)}}\right] = \left(\frac{\pi}{2}\right)^{\frac{1}{2}} e^{-|k|}$$

Failed Example:

$$f(x) = e^{-\frac{1}{2}x^2}$$

The bound *fails* for this example. To work at radius R it is best to use polar coordinates and then $z = Re^{i\phi} = R\cos\phi + \mathrm{i}R\sin\phi$ gives us

$$e^{-\frac{1}{2}z^2} = e^{-\frac{R^2}{2}\cos 2\phi} e^{-i\frac{R^2}{2}\sin 2\phi}$$

and although the phase is no problem, the real part blows up in the interval $\phi \in \{-\frac{\pi}{4}, \frac{\pi}{4}\}$, and we need a new idea.

This new idea is another type of closed contour, which we will notate as $C[a, b; R]$ with $b > a$. This is a rectangular contour with two long sides, $ia + \{-R, R\}$ and $ib + \{-R, R\}$, which limit to being parallel to the real axis, but pass through the imaginary points, ia and ib respectively. The other two sides go from $\pm R + \mathrm{i}a$ to $\pm R + \mathrm{i}b$ and complete a closed contour that tracks around the rectangle in an anti-clockwise direction. When using this contour we need to show that the two short sides limit to zero as $R \mapsto \infty$ and then the integral along ia can be moved up to ib at the cost of any residues that are in the rectangle. The changing of R is clearly just the moving of the contour using Cauchy's theorem and is clearly safe if there are no residues in the vicinity. We will now use this contour to find the Fourier transform of our Gaussian

$$\tilde{f}(k) = \int_{\mathbf{R}} \frac{dx}{\sqrt{(2\pi)}} e^{-ikx} e^{-\frac{1}{2}x^2} = \int_{\mathbf{R}} \frac{dx}{\sqrt{(2\pi)}} e^{-\frac{1}{2}(x+ik)^2} e^{-\frac{1}{2}k^2}$$

where we completed the square. Now we want to use $z = x + ik$ to get

$$\tilde{f}(k) = e^{-\frac{1}{2}k^2} \int_{ik+\mathbf{R}} dz e^{-\frac{1}{2}z^2}$$

The new contour with $C[0, k; R]$ if $k > 0$ and $C[k, 0; R]$ if $k < 0$ allows us to translate the contour back to the real axis, if we can show that the small sides to not contribute. The lines are $\pm R + iy$, where y is finite, and then

$$e^{-\frac{1}{2}z^2} = e^{-\frac{1}{2}(R^2 - y^2)} e^{-iRy}$$

and so the function is swamped by the large R and vanishes in the limit. We can then use real analysis to do the integral since

$$I = \int_{\mathbf{R}} dx e^{-\frac{1}{2}x^2} \quad \Rightarrow \quad I^2 = \int_{\mathbf{R}} dx e^{-\frac{1}{2}x^2} \int_{\mathbf{R}} dy e^{-\frac{1}{2}y^2} = \int_0^\infty d\rho \rho e^{-\frac{1}{2}\rho^2} \int_0^{2\pi} d\phi = 2\pi$$

$$\Rightarrow \quad I = \sqrt{(2\pi)}$$

and

$$\tilde{f}(k) = e^{-\frac{1}{2}k^2}$$

and the odd looking $\sqrt{(2\pi)}$ in the Fourier transform starts to make sense as the Fourier transform maps onto itself.

Examples: Particles in a box
You have met particles in a box in your quantum mechanics courses and you should know that for a box of width $2a$, the wavefunctions are

$$\psi_n(x) = \frac{1}{a^{\frac{1}{2}}} \sin\left[\frac{n\pi}{2a}(x + a)\right] \equiv \frac{1}{a^{\frac{1}{2}}} \sin[K_n(x + a)] \qquad E_n = \frac{\hbar^2 n^2 \pi^2}{8ma^2} \equiv \frac{\hbar^2 K_n^2}{2m}$$

where the vanishing of the wavefunction at $x{=}a$ quantises the problem and provides the parameterised momenta K_n. What is the momentum-space wavefunction? As usual, determining this wavefunction is real analysis but performing the inverse transform is complex analysis. We can immediately calculate

$$\tilde{\psi}_n(k) = \frac{1}{(2\pi a)^{\frac{1}{2}}} \int_{-a}^{a} dx e^{-ikx} \sin[K_n(x+a)]$$

$$= \frac{1}{(2\pi a)^{\frac{1}{2}}} \int_{-a}^{a} dx \frac{1}{2i} \left(e^{i(K_n-k)(x+a)} e^{ika} - e^{i(-K_n-k)(x+a)} e^{ika} \right)$$

$$= \frac{1}{(2\pi a)^{\frac{1}{2}}} \frac{1}{2i} \left[\frac{e^{i(K_n-k)(x+a)}}{i(K_n-k)} e^{ika} - \frac{e^{i(-K_n-k)(x+a)}}{i(-K_n-k)} e^{ika} \right]_{-a}^{a}$$

$$= \frac{1}{(2\pi a)^{\frac{1}{2}}} \frac{1}{2i} \left[\frac{e^{i(K_n-k)2a}-1}{i(K_n-k)} e^{ika} - \frac{e^{i(-K_n-k)2a}-1}{i(-K_n-k)} e^{ika} \right]$$

$$= \frac{1}{(2\pi a)^{\frac{1}{2}}} \left[\frac{\sin[(K_n-k)a]}{i(K_n-k)} e^{iK_n a} - \frac{\sin[(K_n+k)a]}{i(K_n+k)} e^{-iK_n a} \right]$$

and we recognise the Sinc function as the physical answer.

Now we come to the complex variable theory, the inverse transform is

$$\psi_n(x) = \frac{1}{a^{\frac{1}{2}}} \int_{-\infty}^{\infty} \frac{dk}{2\pi i} e^{ikx} \left[\frac{\sin[(k-K_n)a]}{k-K_n} e^{iK_n a} - \frac{\sin[(k+K_n)a]}{k+K_n} e^{-iK_n a} \right]$$

and at first glance there appear to be poles at $k{=}\pm K_n$ and the residue theorem appears relevant, but the Sinc function is analytic everywhere and there are no poles! What we need to do is to split up the Sinc function and *make* some poles!

$$\psi_n(x) = \frac{1}{a^{\frac{1}{2}}} \int_{-\infty}^{\infty} \frac{dk}{2\pi i} \frac{1}{2i} \left[\frac{e^{ik(x+a)} - e^{ik(x-a)} e^{2iK_n a}}{k-K_n} - \frac{e^{ik(x+a)} - e^{ik(x-a)} e^{-2iK_n a}}{k+K_n} \right]$$

and now we would like to split the integral into four parts and do each one independently, by closing the contour and using the residue theorem, but we cannot because the contour passes through the would be pole. To resolve this problem we need to move the contour and the previous contours $C[-d,0;R]$ or $C[0,d;R]$ allow us to move the contour to either -id+$\mathbf{R}$ or id+$\mathbf{R}$ respectively. The Sinc function is entire and so there are no residue issues and the denominator vanishes fast enough to be safe. The best choice is usually to go down, because when we close the contour down there are no poles so the answer vanishes and when we close the contour upwards we catch all the poles and the closed contour is anti-clockwise so the winding number is not an issue. We close upwards when the real number is positive and this supplies a theta-function. We obtain from the residue theorem

$$\psi_n(x) = \frac{1}{a^{\frac{1}{2}}} \frac{1}{2i} \Big[\theta(x+a) e^{iK_n(x+a)} - \theta(x-a) e^{iK_n(x-a)} e^{2iK_n a}$$

$$-\theta(x+a)e^{-iK_n(x+a)} + \theta(x-a)e^{-iK_n(x-a)}e^{-2iK_n a}\Bigg]$$

and in both cases we can recognise the sin function to get

$$\psi_n(x) = \frac{1}{a^{\frac{1}{2}}}\sin[K_n(x+a)]\,[\theta(x+a) - \theta(x-a)]$$

the starting wavefunction but with the top-hat function included. To complete the discussion we ought to explain what happens if we take the contour upwards and then we capture the poles when the real number is negative. The closed contour is now clockwise so the winding number is minus unity and we generate

$$\psi_n(x) = \frac{1}{a^{\frac{1}{2}}}\frac{1}{2i}\Bigg[- \theta(-x-a)e^{iK_n(x+a)} + \theta(a-x)e^{iK_n(x-a)}e^{2iK_n a}$$

$$+\theta(-x-a)e^{-iK_n(x+a)} - \theta(a-x)e^{-iK_n(x-a)}e^{-2iK_n a}\Bigg]$$

and in both cases we can recognise the sin function again to get

$$\psi_n(x) = \frac{1}{a^{\frac{1}{2}}}\sin[K_n(x+a)]\,[\theta(a-x) - \theta(-x-a)]$$

and since

$$\theta(z) + \theta(-z) = 1$$

this is just another representation of the top-hat function.

Useful contours

So far we have encountered four useful contours that you should be able to use. Firstly we have $C_u(R)$, a semi-circle of radius R, oriented upwards, which allows us to turn real integrals, along the real axis, into finite closed contours around this semi-circle. If we have an ordinary integral, then the integrand must vanish, along the curved part of the contour, faster than $\frac{1}{R}$, in order that the length of the integral is beaten by the decay of the function. When doing Fourier transforms we only need the function to vanish in the limit on the curved part of the contour, as the oscillating phase provides an additional $\frac{1}{R}$ using Jordan's lemma. Secondly we have $C_d(R)$, which is completely analogous to $C_u(R)$, except that the winding number is negative unity. Thirdly we have $C[a,b;R]$, which allows us to move contours from $ia+\mathbf{R}$ to $ib+\mathbf{R}$, with the cost of any residues between, and we saw that this can turn a complex integral into a real integral if either a or b vanishes. Finally, we had the segement of the circle, $\phi \in \{0,\Phi\}$ and $\rho \leq R$, which allows us to use rotational symmetry to calculate real integrals from the origin off to infinity.

The Exponential function

So far we have not really considered any explicit functions, but with the advent of Fourier transforms it is clear that the exponential function must be properly understood. The exponential function is *entire,* which is mathematician speak for analytic

everywhere in the complex plane. When doing Fourier transforms, this means that the phase part of the integral can never yield any residues and so all the singularities are controlled by the functions. The more subtle aspect is the fact that

$$e^{z+2\pi i} = e^z$$

and that the value of the exponential is periodic for translations through 2π along the y-axis. If we then look at a single region, $x \in (-\infty, \infty)$ and $y \in (-\pi, \pi]$, then the polar coordinates idea provides

$$e^z = e^x e^{iy}$$

and each point is uniquely mapped to once by this region. The consequence of this is that the entire complex plane, with the exception of the origin, is mapped to countable infinitely many times by the exponential function as we range over the complex plane. This has a broad range of consequences.

It is possible to generate poles with the exponential function by taking linear superpositions, and perhaps the most interesting example is

$$\coth z = \frac{\cosh z}{\sinh z} = \frac{e^z + e^{-z}}{e^z - e^{-z}} = \frac{e^{2z}+1}{e^{2z}-1} = 1 + \frac{2}{e^{2z}-1}$$

and is intimately related to the Bose function in statistical physics. It is clear that the periodicity is inherited and that

$$\coth(z + i\pi) = \coth z$$

but with half the period. There is clearly a pole at the origin, with residue unity, and then there are periodic copies of this pole at $z{=}in\pi$, for all integer n both positive and negative.

Performing sums using the Coth function

This section exists because it is both mathematically pretty and very instructive in the subtleties inherent in the sections on moving and closing contours. The logic of this section is to consider the integral

$$\oint_C \frac{dz}{2\pi i} \coth z f(z)$$

over some closed contour C. The residue theorem applied to this contour will naturally yield two types of residues, those coming from the $\coth z$ and those coming from $f(z)$. There is also the possibility that the poles are common to both functions but initially we will prohibit this in the interest of simplicity. We saw in the last section that $\coth z$ has poles at all the points, $\{in\pi\}$ for integer n, all with residue unity and so, if the contour winds once around a region in an anti-clockwise direction, then we would get a contribution of

$$\sum_{n \in C} f(in\pi)$$

from this source. Any poles in the function $f(z)$ would then provide

$$\sum_{r\in C} R(r)\coth r$$

where the poles are at positions r, and $R(r)$ is the residue of the pole. This is just the residue theorem applied directly. The logic is then to choose $f(z)$ to provide the desired sum over integer values, and this replaces the sum with a sum over the residues of $f(z)$ multiplied by $\coth r$. In order to put this idea into practice we need to control the contour C so that the integral vanishes and this corresponds to the previous section on closing the contour by making the contour very large and catching everything and using the properties of the function $f(z)$ at infinity to eliminate the answer in the limit.

Control over the $\coth z$ function comes from real analysis and thinking about the lines

$$\coth\left(in\pi+\frac{i\pi}{2}+x\right)=\tanh(x)$$

for each integer n. As x varies we map out the lines parallel to the x-axis but half way between two neighbouring poles. The key is that the real function $\tanh x$ is bounded between plus and minus unity and so the behaviour can be controlled by the integral of $\tanh x f\left(in\pi+\frac{i\pi}{2}+x\right)$. The anti-symmetry of the $\tanh x$ function then provides

$$\int_0^R dx\tanh(x)\left[f\left(in\pi+\frac{i\pi}{2}+x\right)-f\left(in\pi+\frac{i\pi}{2}-x\right)\right]$$

as the required contribution.

The eventual contour is created from a string of contours of the form $C[n\pi-\frac{\pi}{2},n\pi+\frac{\pi}{2};R]$, which is a rectangular contour of unit height centred on the n'th pole of $\coth z$ and of width $2R$ which we want to extend off to infinity. The logic is, for each of these small rectangular contours, to tend $R\mapsto\infty$ first and to demonstrate that the unit length edge pieces at $\pm R$ vanish in the limit. The $\coth z$ function tends to unity, with exponentially small corrections, $\sim e^{-2R+iy}$, and so we only need that $f(z)\mapsto 0$ as $x\mapsto\pm\infty$ to allow this. The second step is then to add all the rectangles together to leave one line at a very large $N+\frac{1}{2}$ and a second at a corresponding -N'-$\frac{1}{2}$ and then in the limit N and N' become infinite to show that these line integrals vanish. It is this final issue that can prove problematic.

Let us try to use this idea to generate

$$S_m=\sum_{n=1}^{\infty}\frac{1}{n^m}$$

After a bit of thought one might suggest the function

$$f(z)=\frac{1}{z-a}$$

as natural for this task, because the singularities in the $\coth z$ provide

$$\sum_n \frac{1}{in\pi - a}$$

and the singularity in $f(z)$ provides

$$\coth a$$

leading to

$$\coth a = -\sum_n \frac{1}{in\pi - a} = \frac{1}{a} + \sum_{n=1}^{\infty} \left(\frac{1}{in\pi + a} - \frac{1}{in\pi - a} \right) = \frac{1}{a} + \sum_{n=1}^{\infty} \frac{2a}{a^2 + n^2\pi^2}$$

and then a Taylor's expansion in a allows

$$\coth a = \frac{1}{a} + 2a \sum_{m=0}^{\infty} (-1)^m a^{2m} \frac{1}{\pi^{2m+2}} S_{2m+2}$$

which is indeed correct, it is just the small problem that the line integrals diverge logarithmically and the current argument is bogus!

The root cause of this problem is that the sum

$$S_1 = \sum_{n=1}^{\infty} \frac{1}{n}$$

does not converge and so if we were to extend the upwards line to infinity we would necessarily require a divergent answer and the mathematics knows this, the logarithmic divergence being the symptom. It is the lower line that cancels this divergence, since it is oppositely oriented, and so we need to fix $N'{=}N$ and add the two lines to get

$$\int_0^R dz \tanh z \frac{4az\left(N + \frac{1}{2}\right)}{\left[(z-a)^2 + \pi^2\left(N + \frac{1}{2}\right)^2\right]\left[(z+a)^2 + \pi^2\left(N + \frac{1}{2}\right)^2\right]}$$

which is then convergent and vanishes as $N \mapsto \infty$.

Clearly, the choice

$$f(z) = \frac{1}{z^2 - a^2}$$

would not have suffered these convergence issues, but you need to appreciate the subtlety to be wary of it.

Integration: Logic

Cauchy's theorem drives Complex Variable Theory, but something vanishing is difficult to learn much from, so the residue theorem is the critical calculational tool. The residue theorem allows one to handle extractable singularities. The physical picture is that a closed contour may be collapsed down to nothing using Cauchy's theorem, but

we need to leave little circles around each of the extractable singularities. There can be multiple circles about the same singularity circling in either direction and this leads to the winding number. A single anti-clockwise circle generates $2\pi i$ times the residue. Using Complex Variable Theory as a calculational tool then has two major skills; firstly finding residues and secondly closing finite contours to make closed contours so that the residue theorem can be applied. Closing the contour is about finding extensions to the initial open contour, using either symmetry or the vanishing of the integral in the limit as one approaches infinity. It is almost invariably true that messing about with the real analysis is more difficult than applying the residue theorem.

We have Laplace and Fourier transforms as examples, and currently the physical problems under control are nuclear decay, forced harmonic oscillators and momentum-space wavefunctions. Although we have not stressed it, momentum space wavefunctions are sometimes *entire* and do not have poles; do not say there is a pole just because you can see one! Check that the singularity does not cancel out by L'Hopital before you claim a singularity...

Electrostatics

Static electric fields in the absence of magnetic fields are controlled by two Maxwell equations,

$$\nabla \times \mathbf{E}(\mathbf{r}) = \mathbf{0} \quad \nabla.\mathbf{E}(\mathbf{r}) = \frac{\rho(\mathbf{r})}{\epsilon_0}$$

and if we restrict to free space with no charge then

$$\nabla \times \mathbf{E}(\mathbf{r}) = \mathbf{0} \quad \Rightarrow \quad \mathbf{E}(\mathbf{r}) = \nabla V(\mathbf{r})$$

for some potential $V(\mathbf{r})$, in a simply connected domain, combined with

$$\nabla.\mathbf{E}(\mathbf{r}) = 0 \quad \Rightarrow \quad \nabla^2 V(\mathbf{r}) = 0$$

and so the potential satisfies Laplace's equation. The fact that the real part and the imaginary part of an analytic function satisfy Laplace's equation gives us some interesting solutions to two-dimensional electrostatics problems. The technical idea behind these solutions is *conformal mapping.*

Incompressible, irrotational flow

In fluid mechanics, conservation of fluid is controlled by the continuity equation

$$\frac{\partial \rho}{\partial t}(\mathbf{r}) + \nabla.\mathbf{v}(\mathbf{r}) = 0$$

where $\rho(\mathbf{r})$ is the density of the fluid and $\mathbf{v}(\mathbf{r})$ is the velocity of the fluid. When the fluid does not react strongly to pressure, so a liquid and not a gas, then we can take the approximation that the fluid is incompressible and the density does not change as a function of time. The continuity equation then reduces to the vanishing of the divergence of the velocity. The local angular momentum of the fluid is controlled by the vorticity, $\nabla \times \mathbf{v}(\mathbf{r})$, and so if the fluid is currently not locally rotating and we ignore viscosity,

which generates such local rotation, then we can make the further approximation to incompressible, $\nabla.\mathbf{v}(\mathbf{r})=0$, and irrotational, $\nabla \times \mathbf{v}(\mathbf{r})=\mathbf{0}$, flow and then

$$\nabla \times \mathbf{v}(\mathbf{r}) = \mathbf{0} \quad \Rightarrow \quad \mathbf{v}(\mathbf{r}) = \nabla V(\mathbf{r})$$

for some potential $V(\mathbf{r})$, in a simply connected domain, combined with

$$\nabla.\mathbf{v}(\mathbf{r}) = 0 \quad \Rightarrow \quad \nabla^2 V(\mathbf{r}) = 0$$

and so the potential satisfies Laplace's equation. The fact that the real part and the imaginary part of an analytic function satisfy Laplace's equation gives us some interesting solutions to two-dimensional fluid flow problems. The technical idea behind these solutions is *conformal mapping.*

Conformal Mappings

In this section we will be thinking about analytic functions, now notated by $Z(z)$, as mappings of the complex plane onto itself. The property that we will be using is that the real and imaginary part of an analytic function satisfy Laplace's equation in two-dimensions and so these quantities can represent either equipotentials, in the case of electrostatics, or streamlines in the case of fluids. The physical logic for electrostatics is that a metal surface is automatically an equipotential, because it cannot support an electric field parallel to the surface (the electrons move in such a field in such a way as to cancel such a field out as you learned in second year electromagnetism). We may replace any equipotential with a metal surface and deduce the electric potential in the free space surrounding this surface. The physical logic for fluid flow is that a solid surface cannot be crossed by the fluid and so we can replace any of the streamlines by a solid surface and deduce the fluid flow in the free space surrounding the solid surface. The difficulty is to find the particular analytic function that provides the metal or solid surface that one wants to solve. We will work in reverse and analyse some simple analytic functions and determine which problems they solve; pot luck!

Before we look at the general theory, we will look at the simplest case, multiplying by a complex number, a say,

$$Z(z) = az$$

It is best to use polar coordinates in this case, and using $z = \rho e^{i\phi}$, $Z = Re^{i\Phi}$ and $a = Ae^{i\alpha}$ we find

$$Z = Re^{i\Phi} = az = Ae^{i\alpha}\rho e^{i\phi} = A\rho e^{i(\phi+\alpha)}$$

We find a change of scale, $R{=}A\rho$, by A and a rotation, $\Phi{=}\phi{+}\alpha$, through angle α.

For an analytic function, $Z(z)$, we have Taylor's theorem and so close to z_0 we have

$$Z(z) = Z(z_0) + (z - z_0)\frac{dZ}{dz}(z_0) + O((z - z_0)^2) \quad \Rightarrow \quad Z(z) - Z(z_0) \sim (z - z_0)\frac{dZ}{dz}(z_0)$$

and the derivative of the function at the point z_0 ploys the role of a local change of scale and a rotation. This is the meaning of the term *conformal,* that the local axes remain at right-angles under the transformation and angular differences are preserved.

Obviously, as we move about, the value of the derivative can change and so the scale and rotation angle smoothly move around allowing interesting behaviour to develop. Note that if the derivative vanishes then the next term goes like z^2 and the mapping is no longer an elementary rotation and the mapping is no longer conformal. Points where the derivative vanishes are special and are called stagnation points in the fluid analogue. Singularites and zeroes play an analogous role in conformal maps unlike integration where singularities are paramount and zeroes are irrelevant.

We tend to work in Cartesian coordinates, where the original choices of x and y being constant correspond to either a uniform electric field or uniform flow, and the transformed variables, X and Y, correspond to a more interesting situation. The simplest non-linear example is

$$Z(z) = z^2 \qquad \Rightarrow \quad X + iY = (x + iy)^2 = x^2 - y^2 + 2xyi \quad \Rightarrow \quad X = x^2 - y^2, Y = 2xy$$

To find the new equipotentials/streamlines, we view the old versions as either x is fixed and y could be anything, or y is fixed and x could be anything. In the first case we eliminate y and in the second case we eliminate x. So either

$$X = x^2 - \frac{Y^2}{4x^2}$$

or

$$X = \frac{Y^2}{4y^2} - y^2$$

In each case we view the equation as defining an equipotential and as we vary the original coordinate we map out the equipotentials. The conformal idea guarantees that these curves always cross each-other at right-angles and either set of curves can play the role of streamlines or equipotentials. In both cases we recognise parabolae, and in the first case they wrap around the negative X-axis and in the second case they wrap around the positive X-axis. The limiting contour, $x \mapsto 0$, is the natural target for replacement by either a metal surface or a solid wall and so we are looking at the electric field generated by the tip of a thin metal plate, or fluid flow around the tip of a long-thin promontory. Setting x=0 gives us that Y=0, and then X=-y^2 tells us that X is negative; the limiting contour. Setting y=0 also gives Y=0, and then X=x^2 tells us that X is positive; the other limiting contour.

The issue of boundary conditions is both subtle and important. The flow far from the promontory is controlled by the mapping far away, and this is non-linear. We do not have uniform flow far away and this is physically a bit worrying. For the first case, if you look far off in the positive X direction the flow is parallel to the Y-axis whereas if you look far off in the negative X direction then the flow is parallel to the X direction and in opposite directions either side of the promontory; The flow far away is highly non-trivial.

It is not much of a physical stretch to invert the problem and to imagine that Z is the complex number and that z is the mapped space and find out which problem would be solved from this inverse mapping. This amounts to considering either

$$x^2 - y^2 = X$$

where Y could be anything, mapping out the original lines of fixed X, or

$$2xy = Y$$

where X could be anything, mapping out the original line of fixed Y. These form perpendicular families of hyperbolae, and the natural physical problem that they elucidate is flow around a corner, or alternatively the electric potential in the vicinity of a metal plate bent into a right-angle. The limiting contour is now $X{=}0$ in the first case, giving the two line $y{=}\pm x$, for the second case $Y{=}0$ giving the two axes as the limiting contour.

If we do not think about the mapping aspect at all, then we can view an analytic function as a source of solutions to Laplace's equation and consequently we can view $X(x, y)$ and $Y(x, y)$ directly as the potential in either of our two physical problems. This logic provides the solution to the inverse mapping problem above.

The subtlety here is that the inverse conformal mapping is

$$Z(z) = z^{\frac{1}{2}}$$

and is not analytic at the origin in the same sense that $Z(z){=}z^2$ is, this is the first example of m not being an integer in the original derivation of solutions to Laplace's equation and he first example that has a branch structure and needs a branch cut.

Another very interesting example comes from the previous Green's function example

$$Z(z) = \frac{1}{2}\left(z + \frac{1}{z}\right)$$

which gives

$$2X = x + \frac{x}{x^2 + y^2} \qquad 2Y = y - \frac{y}{x^2 + y^2}$$

where we consider the inverse mapping. The critical contour is $Y{=}0$, and corresponds to either $y{=}0$ or $x^2{+}y^2{=}1$, but how do we decide which? The key here is the mapping aspect to the problem, we need to establish how the X-axis is mapped back into the original space. For this we need the explicit inverse mapping

$$z = Z \pm \left[Z^2 - 1\right]^{\frac{1}{2}}$$

and we want to insert $Z \mapsto X$. Far from the origin we have $z{=}X \pm X$, but when $-1 < X < 1$ we end up with a complex number $e^{\pm i\phi}$, with $X \equiv \cos\phi$ being the natural parameterisation. The choice of sign is absolutely crucial; We start out on the real axis, either close to infinity or close to the origin. We track towards the unit circle and once we meet it we track along the unit circle either on the upper semi-circle or the lower semi-circle. Once we arrive at the real axis again, we track either off to infinity or off to the origin. Each of these choices is the choice of sign in the square-root. The solution that we want is to start at minus infinity, track to the unit circle, go either way around the unit circle and finally track off to plus infinity. The real-axis we treat as part of the system but the unit circle we put in as a metal cylinder or a solid cylinder for the two physical problems. Varying Y then gives us the equipotentials from a metal cylinder in

an electric field or fluid flow around a solid cylinder. The choice of sign is conspicuous from

$$x^2 + y^2 = \frac{y}{y - 2Y} = 1 + \frac{2Y}{y - 2Y}$$

and if $Y > 0$ then either we need $y > 2Y$ and then $x^2 + y^2 > 1$ is also required, or we need $y < 0$ and $x^2 + y^2 < 1$ is also required. We are wholly outside the cylinder or wholly inside the cylinder.

When outside the cylinder, the boundary conditions issue is well behaved here, because far away from the origin we have $Z(z) \mapsto \frac{z}{2}$ and we consequently have uniform flow.

Another useful example comes from

$$Z^2 = z^2 - 1$$

which gives

$$X^2 - Y^2 = x^2 - y^2 - 1 \qquad XY = xy$$

where y=0 is the critical contour. The second equation tells us that we must be on one of the two axes and as we track x along the real axis, the first tells us which. When $x < -1$ or $x > 1$ we need to use the X-axis and when $-1 < x < 1$ we need to use the Y-axis. The sign choice is an issue again. As we track x from minus infinity to infinity, we start out on the X-axis at either $\pm\infty$, we track to the origin and then we move out along the Y-axis to either ± 1. We then go back to the real axis and track out to either $\pm\infty$ where we end. The appropriate choice of signs comes from a small y and eliminating x to get

$$X^2 - Y^2 = \frac{X^2 Y^2}{y^2} - y^2 - 1 \quad \Rightarrow \quad (Y^2 - y^2)(X^2 + y^2) = y^2$$

as the final potential. It is clear that $Y^2 > y^2$ and so the contour is always either above or below the x-axis. Physically, this final example corresponds to a solid promontory jutting out from a perpendicular solid wall with fluid flowing parallel to the solid wall but having to avoid the promontory. Close to the tip of the promontory we have Z=i+δZ and so locally

$$(i + \delta Z)^2 = z^2 - 1 \mapsto 2i\delta Z = z^2$$

and we have a rotated copy of the initial conformal mapping.

The previous boundary conditions issues are partially resolved by this example. Far from the promontory we have $Z(z) \mapsto \pm z$ and so we have uniform flow, but close to the tip of the promontory we have the fluid flowing in the very non-linear way. The distant flow is uniform, parallel to the shore-line, but locally the transformation allows non-trivial, non-linear behaviour. The most useful physical maps are the ones that provide the correct physical behaviour far away from where the interesting behaviour occurs.

Each mapping has an inverse mapping and a corresponding physical analogue, what is the problem associated with the previous example, because the new mapping appears at first glance almost identical to the previous, is it similar?

$$Z^2 = z^2 + 1$$

which corresponds to

$$X^2 - Y^2 = x^2 - y^2 + 1 \qquad XY = xy$$

The critical contour is still y=0 for which

$$XY = 0 \quad X^2 - Y^2 = x^2 + 1$$

and now X=0 is clearly not permitted and so Y=0 is necessary and $X^2 \geq 1$ is required. The critical contour involves the X-axis but with a gap between plus and minus unity. To find a general contour we eliminate x to find

$$X^2 - Y^2 = \frac{X^2 Y^2}{y^2} - y^2 + 1 \quad \Rightarrow \quad (X^2 + y^2)(y^2 - Y^2) = y^2$$

The most important contour occurs at y=1, for which

$$Y^2 = \frac{X^2}{X^2 + 1} < 1$$

which connects the stagnation point at the origin all the way to infinity at Y=$\pm$1. The flows are bounded by these lines and flow around the solid barriers and through the gap if between them and flow parallel to the X-axis in opposite directions for the two sides outside them. The physical problem is two flows in opposite directions parallel to the barriers.

Another important example, which is often used as a mapping, is

$$Z(z) = \frac{1}{z}$$

and in terms of Cartesian coordinates

$$X = \frac{x}{x^2 + y^2} \qquad Y = -\frac{y}{x^2 + y^2}$$

Treating this as the inverse map provides circles which pass through the origin

$$\left(x - \frac{1}{2X}\right)^2 + y^2 = \left(\frac{1}{2X}\right)^2 \qquad x^2 + \left(y + \frac{1}{2Y}\right)^2 = \left(\frac{1}{2Y}\right)^2$$

and the crucial trajectories are either $X \mapsto 0$ or $Y \mapsto 0$, which tend to the Cartesian axes. Neither of the two physical problems is very valuable, a charged cylinder on a flat surface, separated by an infinitely thin insulator and fluid flow between two cylinders through an infinitesimal gap are both a bit pathological. Polar coordinates are quite a useful description with

$$Z \equiv Re^{i\Phi} = \frac{1}{\rho} e^{-i\phi}$$

providing

$$R = \frac{1}{\rho} \qquad \Phi = -\phi$$

and the unit circle is mapped onto itself, but in a reversed direction and circles inside the unit circle are mapped onto circles outside the unit circle and vice-versa. Probably the most important fact is that the origin maps onto infinity and infinity maps onto the origin.

The final example we offer is the delightful

$$Z(z) = z + e^z$$

where we operate in Z and use $y{=}\pi$ and $y{=}\text{-}\pi$ as two critical contours. The first gives

$$Z = x + i\pi - e^x \quad \Rightarrow \quad Y = \pi \quad X = x - e^x$$

and the second offers

$$Z = x - i\pi - e^x \quad \Rightarrow \quad Y = -\pi \quad X = x - e^x$$

The real function x-e^x is maximal when $x{=}0$, at the value minus unity and ranges from minus infinity up to minus unity and back to minus unity, but in a very a-symmetric fashion. For fixed values of y between these two limits we find

$$X = x + \cos y e^x \quad Y = y + \sin y e^x$$

and we would like to eliminate x, which is difficult and unhelpful

$$\frac{X-x}{Y-y} = \frac{\cos y}{\sin y} \quad \Rightarrow \quad x = X + (y-Y)\cot y = \ln\left[\frac{Y-y}{\sin y}\right]$$

but when x is large and negative the exponential is irrelevant and so we have $Y \sim y$, but when x is large the exponential dominates and we have $\frac{Y}{X} \sim \tan y$ and a straight line at an angle of y to the X-axis. The two critical contours form a trench from which our fluid spews in all directions, or we can imagine a parallel plate capacitor and the field spewing out of the end.

Conformal Maps: Logic

Analytic functions, described by $Z(z)$ in this section, provide conformal maps; maps that maintain relative angles but allow a change of scale. As well as being useful maps, analytic functions are also solutions to Laplace's equation which crops up in Electrostatics and in incompressible, irrotational fluid flow. One can take $Z{=}X{+}iY$ as these solutions directly, the inverse problem, but we considered how either $x{=}x_0$ or $y{=}y_0$ are mapped onto new trajectories in Z-space. The behaviour at infinity is important, but not really under control, and needs to be understood if the physical examples are to be useful. We looked at specific examples which solved useful physical problems and encountered power-laws which do not have extractable singularites and are the topic of the next section.

Conformal maps are difficult to examine and will not be actively pursued in the exam.

Branch Cuts

We now have singularities with $m{=}\frac{1}{2}$ and although this function might be argued to be analytic, the singularity is definitely not extractable and the residue theorem is not applicable. The best way to think about this function is using polar coordinates

$$z = \rho e^{i\phi} \quad \Rightarrow \quad z^{\frac{1}{2}} = \rho^{\frac{1}{2}} e^{i\frac{\phi}{2}} \;\; or \;\; z^{\frac{1}{2}} = \rho^{\frac{1}{2}} e^{i\left(\frac{\phi}{2}+\pi\right)} = -\rho^{\frac{1}{2}} e^{i\frac{\phi}{2}}$$

and there are two choices of angle. Each choice is referred to as a particular branch of the function, which is inherited from multi-valued real functions. The difference between the branches is simply this choice of phase and locally the function is continuous and this choice of phase does not change. As we know, if we loop around the singularity once, we arrive back at the other branch and so we cannot use the whole complex plane to describe the function, we have to do something else. There are two choices, and in this first more elementary course we will use the easier 'branch cut' idea, whereas next term we will use the more sophisticated Riemann surface concept, which is by far the most useful.

To create a branch cut we simply leave out a line in the complex plane that connects to the singularity. The new subset of the complex plane is home to one branch of the function and the missing line marks a discontinuity between the two possible branches of the function. The position of the branch cut is a choice and is not unique. We can apply all the previous ideas in this restricted space, but contours are not allowed to cross the branch cut. Interestingly, one can include the branch-cut line into *both* sides of the complex plane, if we consider both lines to be distinct and to terminate the space. The values on the two lines are the two branch values and are different, for our square-root example they are of opposite signs. We can no longer consider the little circles to separate out the singularities, but we can employ the contour which would take us down to the little circle, but the two lines along the branch cut no longer cancel out. This contour is a new contour of use to us and is known as a *keyhole* contour. An example will clarify.

Example: Calculate

$$\int_0^\infty dx \frac{x^{\frac{1}{2}}}{x^2+1}$$

using a keyhole contour and repeat for

$$\int_0^\infty dx \frac{x^{-\frac{1}{2}}}{x^2+1}$$

A keyhole contour involves two separate, independent limits; we have the radius R for the large circle which is sent to infinity, and we have the radius r for the small circle which is sent to zero. We need to show that the behaviour of the function and the length of the contour combine to give a vanishing contribution. For our first example, at infinity, we have a length of $2\pi R$, and a limit bound of $B_l \sim \frac{R^{\frac{1}{2}}}{R^2}$ and the product gives $R^{-\frac{1}{2}}$, which is safe. For the second example we find $R^{-\frac{3}{2}}$ which is safer. On the small circle, at zero, we have a length of $2\pi r$, and a limit bound of $B_l \sim r^{\frac{1}{2}}$ and the

product gives $r^{\frac{3}{2}}$, which is safe. For the second example we find $r^{\frac{1}{2}}$ which is less safe but still good. These limits allow us to recognise the original target integral

$$\lim_{r\mapsto 0}\lim_{R\mapsto\infty}\int_r^R dx\frac{x^{\pm\frac{1}{2}}}{x^2+1}\mapsto\int_0^\infty dx\frac{x^{\pm\frac{1}{2}}}{x^2+1}$$

We choose to put the branch cut along the real axis in order to recognise the desired integral on the branch-cut line connected to the upper piece of complex plane. The integral on the other copy of the branch cut line, connected to the lower piece of complex plane, is in the opposite direction and has the other branch value. These two signs cancel and so we get a second copy of the integral and then the residue theorem can be applied to the finite contour to obtain the integral in terms of the residues. The denominator is the same as previous and can be dealt with using partial fractions

$$\frac{1}{x^2+1}=\frac{1}{2i}\left[\frac{1}{x-i}-\frac{1}{x+i}\right]$$

but the square-root needs special attention. We need to make sure that we pick the correct branch of the square-root and for this polar coordinates are the best

$$z^{\frac{1}{2}}=\rho^{\frac{1}{2}}e^{i\frac{\phi}{2}}\qquad z^{-\frac{1}{2}}=\rho^{-\frac{1}{2}}e^{-i\frac{\phi}{2}}$$

We can then deduce that

$$\int_0^\infty dx\frac{x^{\frac{1}{2}}}{x^2+1}=\frac{1}{2}\frac{2\pi i}{2i}\left[e^{i\frac{\pi}{4}}-e^{i\frac{3\pi}{4}}\right]=\frac{\pi}{\sqrt{2}}$$

and that

$$\int_0^\infty dx\frac{x^{-\frac{1}{2}}}{x^2+1}=\frac{1}{2}\frac{2\pi i}{2i}\left[e^{-i\frac{\pi}{4}}-e^{-i\frac{3\pi}{4}}\right]=\frac{\pi}{\sqrt{2}}$$

Note that transformations in real-analysis allow us to change the style of the integral dramatically. The transformation $x{=}y^2$ eliminates the branch singularity and permits

$$\int_0^\infty dx\frac{x^{\frac{1}{2}}}{x^2+1}=2\int_0^\infty dy\frac{y^2}{y^4+1}$$

and

$$\int_0^\infty dx\frac{x^{-\frac{1}{2}}}{x^2+1}=2\int_0^\infty dy\frac{1}{y^4+1}$$

which can be done by closing the contour using symmetry.

Both of these new integrals have the symmetry that suggest using a quarter of the circle as the segment. We can make a finite contour using $n\in\{0,2\}$

$$2\int_0^R dz\frac{z^n}{z^4+1}+2\int_{C(R)}dz\frac{z^n}{z^4+1}+2\int_{iR}^0 dz\frac{z^n}{z^4+1}=2\pi iR[i^{\frac{1}{2}}]$$

where we have applied the residue theorem. The length of the arc is $\frac{\pi}{2}R$ and the bound in the limit is $\frac{1}{R^4}$ or $\frac{1}{R^2}$ and both are safe. The transformation $z{=}iz'$ then allows

$$2\int_{iR}^{0} dz \frac{z^n}{z^4+1} = -2i^{1+n}\int_0^R dz' \frac{z'^n}{z'^4+1} = -i^{1+n}I$$

and all that remains is to calculate the residue. We can use Taylor's theorem on the denominator, and in the first case we get $4z^3$, whereas in the second case we get $2z$-$2z^{-3}$. For the first case we get

$$I = \frac{2}{1-i}2\pi i\frac{1}{4(\sqrt{i})^3} = \frac{\sqrt{i}}{1-i}\frac{2\pi i}{2(-1)} = \frac{\pi i}{\sqrt{i}-\frac{1}{\sqrt{i}}} = \frac{\pi}{2\sin\frac{\pi}{4}} = \frac{\pi}{\sqrt{2}}$$

and in the second case

$$I = \frac{2}{1-i^3}2\pi i\frac{1}{2\sqrt{i}-2(\sqrt{i})^{-3}} = \frac{2}{1+i}2\pi i\frac{\sqrt{i}}{2i-\frac{2}{i}} = \frac{\pi}{\sqrt{i}+\frac{1}{\sqrt{i}}} = \frac{\pi}{2\cos\frac{\pi}{4}} = \frac{\pi}{\sqrt{2}}$$

The Logarithm function

The logarithm function involves a branch singularity at the origin and a line needs to be drawn between the origin and infinity. The shape and orientation of the line is mathematically irrelevant but for applications the natural choices are straight radial lines. For these choices, using polar coordinates, $z{=}\rho e^{i\phi}$,

$$\ln z = \ln\rho + i\phi$$

The line is a particular choice of ϕ, ϕ_0 say, and circling the singularity picks up $2\pi i$ if the orientation is anti-clockwise. Every point on the plane is mapped to a unique point between $\phi_0 < \phi < \phi_0 + 2\pi$ but the line which marks the branch cut has two possible values, ϕ_0 and $\phi_0{+}2\pi$, which are continuous with the other values. If we integrate an analytic function multiplied by a logarithm on a keyhole contour

$$\int_{C(r,R)} dz(-1)\frac{\ln z}{2\pi i}f(z)$$

then, provided that the small and large circles yield a vanishing contribution in the limit, we can employ the residue theorem and deduce the value of

$$\int_0^\infty dx f(x)$$

through the residues of $f(z)$, weighted with $(-1)\frac{\ln z}{2\pi i}$.

Power laws

The function

$$Z(z) = z^m$$

where m is not an integer does not have an extractable singularity and needs to be described with a branch cut. Using polar coordinates, with $z{=}\rho e^{i\phi}$, we find

$$Z = \rho^m e^{im\phi}$$

and there is a discontinuity across the branch cut. If we use a straight line starting at the origin then we have a particular choice of ϕ, ϕ_0 say, that yields two choices with phases $e^{im\phi_0}$ and $e^{im\phi_0}e^{i2\pi m}$. If m is a rational number, $\frac{p}{q}$, then $e^{i2\pi m}$ is one of the q'th roots of unity, but if m is irrational we have an arbitrary phase.

Keyhole contours are useful with power laws as the difference between z^m on the two sides of the branch allows us to calculate real integrals. Perhaps the simplest example is

$$I_m = \int_0^\infty dz z^m \frac{1}{z+1}$$

In order to use the keyhole contour we need to show that the small circle and the large circle have vanishing contribution in the appropriate limit. At large R we find $2\pi R\frac{R^m}{R}$ and so we need $m < 0$. At small distance we have $2\pi r\frac{r^m}{1}$ and so we need $m{+}1{>}\ 0$. Combined we get $0 > m > -1$. The residue is $e^{i\pi m}$ and then applying the residue theorem give

$$(1 - e^{i2\pi m})I_m = 2\pi i e^{i\pi m} \quad \Rightarrow \quad I_m = \frac{\pi}{\sin(-m\pi)}$$

or more sensibly

$$\int_0^\infty dz z^{-m} \frac{1}{z+1} = \frac{\pi}{\sin \pi m}$$

for $0 < m < 1$.

We now need some physical examples where there are more than one branch point and for this the previous Bloch momentum Green's function is useful.

Bloch's Theorem

Fourier series can be physically introduced through condensed matter physics and periodic potentials. The periodic momentum is known as Bloch momentum and we will now introduce the simplest variant of Bloch's theorem that makes the connection. A ring of N equally spaced atoms in a perfect circle with one bond between each of the neighbours is described by the $N \times N$ matrix

$$\begin{bmatrix} 0 & -t & 0 & .. & .. & 0 & 0 & -t \\ -t & 0 & -t & .. & .. & 0 & 0 & 0 \\ 0 & -t & 0 & .. & .. & 0 & 0 & 0 \\ . & . & . & .. & .. & . & . & . \\ . & . & . & .. & .. & . & . & . \\ 0 & 0 & 0 & .. & .. & 0 & -t & 0 \\ 0 & 0 & 0 & .. & .. & -t & 0 & -t \\ -t & 0 & 0 & .. & .. & 0 & -t & 0 \end{bmatrix}$$

where t is the bonding energy or hybridisation, the basis is a fixed wavefunction on each site which minimises the atomic energy for the electron sitting on that atom and the

vector is the list of coefficients in the linear superposition of underlying orbitals which gives the final eigenstate of the system. If there are bonds between second-neighbours too, then the matrix becomes

$$\begin{bmatrix} 0 & -t_1 & -t_2 & .. & .. & 0 & -t_2 & -t_1 \\ -t_1 & 0 & -t_1 & .. & .. & 0 & 0 & -t_2 \\ -t_2 & -t_1 & 0 & .. & .. & 0 & 0 & 0 \\ . & . & . & .. & .. & . & . & . \\ . & . & . & .. & .. & . & . & . \\ 0 & 0 & 0 & .. & .. & 0 & -t_1 & -t_2 \\ -t_2 & 0 & 0 & .. & .. & -t_1 & 0 & -t_1 \\ -t_1 & -t_2 & 0 & .. & .. & -t_2 & -t_1 & 0 \end{bmatrix}$$

and the generalisation to longer range bonding should be clear. To solve this problem we need to diagonalise these matrices and the method that we use is *symmetry*. If two matrices commute with each-other then they can be simultaneously diagonalised. For us one matrix is the Hamiltonian matrix and the other will be the symmetry. The mathematical point is that the symmetry matrix may be easier to diagonalise than the original, and unique eigenstates of the symmetry will also be eigenstates of the Hamiltonian. Our symmetry is that of rotating the circle by one and recognising that the model is the same model. We call this symmetry the translation matrix and it takes the form

$$T = \begin{bmatrix} 0 & 1 & 0 & .. & .. & 0 & 0 & 0 \\ 0 & 0 & 1 & .. & .. & 0 & 0 & 0 \\ 0 & 0 & 0 & .. & .. & 0 & 0 & 0 \\ . & . & . & .. & .. & . & . & . \\ . & . & . & .. & .. & . & . & . \\ 0 & 0 & 0 & .. & .. & 0 & 1 & 0 \\ 0 & 0 & 0 & .. & .. & 0 & 0 & 1 \\ 1 & 0 & 0 & .. & .. & 0 & 0 & 0 \end{bmatrix}$$

and each atom is sent to its neighbour. The commutation is obvious once we recognise that the first matrix is

$$H = -t\left(T + T^{-1}\right)$$

and the second matrix is

$$H = -t_1\left(T + T^{-1}\right) - t_2\left(T^2 + T^{-2}\right)$$

and it should also be clear that if we have an eigenstate of T, with

$$T\mathbf{v}(\omega) = \omega\mathbf{v}(\omega)$$

then it is also an eigenstate of the two Hamiltonians with

$$E(\omega) = -t\left(\omega + \frac{1}{\omega}\right) \qquad E(\omega) = -t_1\left(\omega + \frac{1}{\omega}\right) - t_2\left(\omega^2 + \frac{1}{\omega^2}\right)$$

respectively. Our symmetry is also a permutation and permutations are very easy to diagonalise! If we write down $T\mathbf{v}(\omega)$=$\omega\mathbf{v}(\omega)$ and write in the first element of $\mathbf{v}(\omega)$ to

be unity, then, line by line applying the matrix relates one coefficient to another and we can deduce each coefficient in turn until the permutation closes and then the final equation provides the eigenvalues. For the current matrix this amounts to

$$\begin{bmatrix} 0 & 1 & 0 & .. & .. & 0 & 0 & 0 \\ 0 & 0 & 1 & .. & .. & 0 & 0 & 0 \\ 0 & 0 & 0 & .. & .. & 0 & 0 & 0 \\ . & . & . & .. & .. & . & . & . \\ . & . & . & .. & .. & . & . & . \\ 0 & 0 & 0 & .. & .. & 0 & 1 & 0 \\ 0 & 0 & 0 & .. & .. & 0 & 0 & 1 \\ 1 & 0 & 0 & .. & .. & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ \omega \\ \omega^2 \\ . \\ . \\ \omega^{N-3} \\ \omega^{N-2} \\ \omega^{N-1} \end{bmatrix} = \omega \begin{bmatrix} 1 \\ \omega \\ \omega^2 \\ . \\ . \\ \omega^{N-3} \\ \omega^{N-2} \\ \frac{1}{\omega} \end{bmatrix}$$

and we start out knowing only the two ones at the top by assumption, and the eigenvalue is assumed to be ω. The first row tells us that the next coefficient down is ω and we can also then fill that in on the right. Each subsequent row tells us that the next coefficient is ω times the previous coefficient, until we come to the final row which is special and gives the final $\frac{1}{\omega}$. This equation is true for any ω, but if we choose ω^N=1 then it becomes an eigenvector equation and we can deduce that the eigenvalues of T are the N'th roots of unity and complex variables appear for the first time. It is good to normalise your eigenvectors, so

$$v_j(\omega) = \frac{1}{\sqrt{N}}\omega^j$$

if we label the atoms by, $\{0, 1, ..., N-1\}$. You should also know that we can make a Unitary matrix which diagonalises the matrix by putting the eigenvectors in a line, as columns, and calling the resulting matrix U. For us

$$U_{j\omega} = v_j(\omega)$$

and the diagonalisation is achieved by

$$\sum_{jj'} U^\dagger_{\omega j} T_{jj'} U_{j'\omega'} = \delta_{\omega\omega'}\omega \quad \Rightarrow \quad U^\dagger T U = D$$

where D is a diagonal matrix composed of the N'th roots of unity as the elements. Pictorially, the idea is TU=UD

$$T\left[\mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_n\right] = \left[T\mathbf{v}_1, T\mathbf{v}_2, ..., T\mathbf{v}_n\right] = \left[\epsilon_1\mathbf{v}_1, \epsilon_2\mathbf{v}_2, ..., \epsilon_n\mathbf{v}_n\right]$$

$$= \left[\mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_n\right] \begin{bmatrix} \epsilon_1 & 0 & ... & 0 \\ 0 & \epsilon_2 & ... & 0 \\ . & . & ... & . \\ 0 & 0 & ... & \epsilon_n \end{bmatrix}$$

Using matrix algebra we are now able to find the Green's function alluded to early on in the course

$$G(E) = [E - H]^{-1} \quad \Rightarrow \quad U^\dagger G(E) U = [E - D]^{-1} = \frac{\delta_{\omega\omega'}}{E - E(\omega)}$$

where now D is the diagonal matrix of the eigenvalues of H, $E(\omega)$, and finally

$$G(E) = U\,[E - D]^{-1}\,U^{\dagger}$$

and then using indices

$$G_{jj'}(E) = \sum_{\omega} v_j(\omega)\frac{1}{E - E(\omega)}v_{j'}(\omega)^{\dagger} = \frac{1}{N}\sum_{\omega}\frac{\omega^{j-j'}}{E - E(\omega)}$$

This equation is an example of the *spectral representation,* and for every Hermitian matrix there is such a spectral representation in terms of the eigenvalues and eigenvectors. It is this equation and the residue theorem that makes this Green's function valuable! If you think about the energy as being a complex variable, then this equation is like partial fractions for matrices, and we have a collection of simple poles, one at each of the eigenvalues, and the residues of those poles are the eigenvectors in disguise

$$R(E(\omega)) = v_j(\omega)v_{j'}(\omega)^{\dagger}$$

and so the diagonalisation solution is coded into this Green's function and can be extracted using the residue theorem.

As an example, if one is doing statistical physics, then the total number, $\mathcal{N}$, and total energy, $\mathcal{E}$, of a non-interacting Fermi system is calculated as

$$\mathcal{N} = \sum_{n} f(E_n - \mu) \qquad \mathcal{E} = \sum_{n} E_n f(E_n - \mu)$$

and this can be calculated from our Green's function using

$$\mathcal{N} = \oint_C \frac{dz}{2\pi i} f(z-\mu) Trace\; G(E) \qquad \mathcal{E} = \oint_C \frac{dz}{2\pi i} z f(z-\mu) Trace\; G(E)$$

where the contour C captures all of the poles of the Green's function but none of the poles of the Fermi function, since

$$Trace\; G(E) = \sum_{n} \frac{1}{E - E_n}$$

as the eigenvectors are both orthogonal and normalised, or equivalently, the diagonalisation matrix, U, is unitary.

The final task in this formal section is to turn the sum over roots of unity into a complex integral and this is just the continuum limit. We can take this limit as a two stage process, going through Fourier series on the way, but here we will simply go straight there. The roots of unity are described by

$$\omega_n = e^{i\frac{2\pi n}{N}}$$

where $n \in \{0, 1, 2..., N-1\}$ guarantees one copy of each. The half-way procedure defines an angle k, using $k_n \equiv \frac{2\pi n}{N}$, in terms of which $\omega_n = e^{ik_n}$, and the direct method simply

uses the ω_n directly as complex numbers, $z_n \equiv \omega_n$, which lie on the unit circle. The subtlety here is that the angle k has to be restricted, in order not to multiply count the roots of unity, but the complex variable uniquely specifies these roots using the unit circle on an Argand diagram. The limit is to a very large number of atoms in the circle, which then looks locally like an infinite line, and so we want $N \mapsto \infty$. We use dn=1 to rewrite

$$G_{jj'}(E) = \frac{1}{N}\sum_{\omega}\frac{\omega^{j-j'}}{E-E(\omega)} = \frac{1}{N}\sum_{n=0}^{N-1} dn \frac{\omega^{j-j'}}{E-E(\omega)}$$

and then use

$$z = e^{i\frac{2\pi n}{N}} \quad \Rightarrow \quad dz = i\frac{2\pi}{N} dn e^{i\frac{2\pi n}{N}} \quad \Rightarrow \quad \frac{dn}{N} = \frac{dz}{2\pi i z}$$

to obtain

$$G_{jj'}(E) = \oint_C \frac{dz}{2\pi i z}\frac{z^{j-j'}}{E-E(z)}$$

where C is the unit circle, traversed once in an anti-clockwise direction. We have now fairly fully derived the Green's functions that occurred at the start of the course, but interestingly, if you look back, you will realise that we never actually performed these integrals, as they were used to provide complex functions and we had not done integration yet! Let us try to do these integrals using the residue theorem now.

We devoted some time to arguing that one should parameterise the roots and we eventually ended up with the integrand being

$$f_n(E) = \frac{z^n}{t(z-z_+)\left(z-\frac{1}{z_+}\right)} = \frac{1}{t\left(z_+ - \frac{1}{z_+}\right)}\left[\frac{z^n}{z-z_+} - \frac{z^n}{z-\frac{1}{z_+}}\right]$$

where z_+ is one of the roots and $z_- = \frac{1}{z_+}$ is the other. It should be clear that when $n \equiv j-j'$ is positive there are two simple poles and the residues are obvious, so as previously stated, the use of the residue theorem is easy, but which of the roots are inside the unit circle and which are outside? Polar coordinates tells us that

$$z_+ = \rho e^{i\phi} \quad \Rightarrow \quad \frac{1}{z_+} = \frac{1}{\rho}e^{-i\phi}$$

so if one of the roots is inside the unit circle, then the other is outside. This seems easy, we call z_+ the one inside and then we can deduce that

$$G_{jj'}(E) = \frac{z_+^{j-j'}}{t\left(z_+ - \frac{1}{z_+}\right)}$$

when $j > j'$ and we determine which one is inside when we look at the algebra for finding the roots. There is one major problem with this logic, what if the root is *on* the unit circle, then neither is inside and the integral is mathematically meaningless! What is even worse, is that we know that the eigenvalues are associated with the roots

of unity and so for any of the desired eigenstates we are guaranteed to be in trouble! This leaves two problems, firstly what to do when n is negative and there is a nasty singularity at the origin, and secondly what to do about roots on the contour. We will tackle the easy one first, negative n.

Now that we have done the conformal mappings part of the course we know, to some extent, how to handle

$$Z(z) = \frac{1}{z}$$

This map is conformal everywhere except the origin, which is not analytic, and is mapped off to infinity. The previous polar coordinates

$$z = \rho e^{i\phi} \quad \Rightarrow \quad \frac{1}{z} = \frac{1}{\rho} e^{-i\phi}$$

tells us that the inside of the unit circle is mapped onto the outside and vice-versa. We need to use this map as a change of variables and then

$$dZ = -\frac{dz}{z^2} \quad \Rightarrow \quad \frac{dZ}{Z} = -\frac{dz}{z}$$

and the mapping clearly shows that circling the unit circle in the anti-clockwise direction maps onto circling the unit circle in the clockwise direction for Z. This mapping allows us to deduce that

$$G_{jj'}(E) = \oint_C \frac{dz}{2\pi i z} \frac{z^{j-j'}}{E - E(z)} = \oint_C \frac{dZ}{2\pi i Z} \frac{\left(\frac{1}{Z}\right)^{j-j'}}{E - E\left(\frac{1}{Z}\right)}$$

and since $E\left(\frac{1}{z}\right)$=$E(z)$ in all cases, we can deduce that

$$G_{jj'}(E) = G_{j'j}(E)$$

and so we only need the positive case. Mathematically one can imagine extending the contour out to infinity rather than collapsing it down to the origin, but this transformation is a more controllable version of this because it maps infinity to the origin where we can easily investigate its behaviour using a Laurent series (if it exists).

The more disturbing problem is the failure of the theory when the singularities lie on the unit circle and this is controlled by the algebra of solving for the explicit positions of the poles. We previously did this, the quadratic equation

$$z^2 + \frac{E}{t} z + 1 = 0$$

is solved by

$$z_{\pm} = -\frac{E}{2t} \pm \frac{1}{2t} \left[E^2 - 4t^2\right]^{\frac{1}{2}}$$

and the roots are on the unit circle when $-2t < E < 2t$, precisely where the eigenvalues are. The key observation here is that we should be thinking of E as a complex variable, and then the only place we have a problem is when E is real and in the set $E \in [-2t, 2t]$,

everywhere else, one of the roots is inside the unit circle and one is outside. We can see this from the conformal map

$$Z(z) = \frac{1}{2}\left(z + \frac{1}{z}\right)$$

which we looked at earlier, and thinking of z as the root then E=-2tZ. When we looked at statistical physics we treated E as a complex variable and captured all the poles of $G(E)$ and so we did not have to determine $G(E)$ on those poles, only integrate on a contour surrounding them. This is part of the answer and the next course provides the rest. When E is *not* on the problem interval then we can determine the Green's function. When $E \in (-\infty, -2t)$ we find

$$z_+ = -\frac{E}{2t} - \frac{1}{2t}\left[E^2 - 4t^2\right]^{\frac{1}{2}}$$

and when $E \in (2t, \infty)$ we find

$$z_+ = -\frac{E}{2t} + \frac{1}{2t}\left[E^2 - 4t^2\right]^{\frac{1}{2}}$$

where the real square-root defaults to a positive sign and the first term is partly cancelled by the second term to find the root inside. The diagonal piece of $G(E)$ is the most important and so for $E \in (-\infty, -2t)$ we find

$$G_{jj}(E) = \frac{(-1)}{[E^2 - 4t^2]^{\frac{1}{2}}}$$

and for $E \in (2t, \infty)$ we find

$$G_{jj}(E) = \frac{1}{[E^2 - 4t^2]^{\frac{1}{2}}}$$

Considering the limit $E \mapsto \pm\infty$ shows that these signs are consistent with what one would expect.

The total energy and total number

For a general Hamiltonian matrix, H, the Green's function has a spectral representation

$$G(E)_{jj'} \equiv [E - H]^{-1}_{jj'} = \sum_n \frac{v_j^{(n)} \bar{v}_{j'}^{(n)}}{E - E_n}$$

and if we think about E as being a complex variable then the diagonalisation problem is equivalent to finding the poles of this function and finding the eigenstates

$$\sum_{j'} H_{jj'} v_{j'}^{(n)} = E_n v_j^{(n)}$$

is equivalent to finding the residues of these poles. As was previously pointed out, the $Trace$ of the Green's function is particularly useful

$$Trace\, G(E) = \sum_n \frac{1}{E - E_n}$$

because the residue theorem allows us to find the total number and energy of a fermionic system by calculating

$$\mathcal{N} = \int_C \frac{dz}{2\pi i} f(z-\mu) Trace\, G(z) \qquad \mathcal{E} = \int_C \frac{dz}{2\pi i} z f(z-\mu) Trace\, G(z)$$

where the contour C captures all the poles of the Green's function but none of the poles of the Fermi function. For our Bloch example the trace of our Green's function is, (N is the original size of the ring), when E is greater than $2t$

$$\frac{1}{N} Trace\, G(E) = \frac{1}{[E^2 - 4t^2]^{\frac{1}{2}}}$$

and has two branch singularities, one at E=-2t and the other at E=2t, and is the perfect candidate for thinking about two branch singularities. Before we can do this, we need to analyse the singularities in the Fermi function and be clear about the contour C.

The Fermi function is

$$f(z-\mu) \equiv \frac{1}{e^{\beta(z-\mu)} + 1}$$

and so it inherits its properties from the exponential function. It should be clear that

$$f\left(z + \frac{2\pi i}{\beta}\right) = f(z)$$

and that the whole complex plane is mapped to in an appropriate strip. We can deduce the positions of the poles from -1=$e^{i(2n+1)\pi}$ and so there are countably infinite numbers of poles when

$$\beta(z_n - \mu) = i(2n+1)\pi \quad \Rightarrow \quad z_n = \mu + i\frac{(2n+1)\pi}{\beta}$$

known as the *Matsubara frequencies,* these poles lie along an imaginary axis passing through the chemical potential. Since the poles of the Green's function are all real, the contour C needs to capture the real axis but miss all the Matsubara frequencies. We can consider the contour C to wrap around the real axis, being infinitesimally above and infinitesimally below, and closing below the lowest energy and closing above the highest energy (which may not exist, but the Fermi function will cut it off if required).

For us the relevant quantity is

$$G(E) = \frac{1}{[E^2 - 4t^2]^{\frac{1}{2}}}$$

but we need to understand how to position the branch cut and, indeed, how this choice is to be made. Our energies are bounded $-2t \leq E \leq 2t$ and so if we go to very large E

the Hamiltonian becomes irrelevantly small and the Green's function limits to $\frac{1}{E}$. This suggests that the more useful representation is

$$G(E) = \frac{1}{E\left[1 - \frac{4t^2}{E^2}\right]^{\frac{1}{2}}}$$

and then close to infinity the limit is as expected. For this choice the branch cut needs to avoid infinity and then the only natural choice is to connect up the two branch singularities with a branch cut between them on the real axis from -2t to 2t, does this make sense? The physical logic is to stay very close to the real axis and limit to being very close to one of the branch-points. Close to E=2t the argument of the square root is

$$[(E-2t)(E+2t)]^{\frac{1}{2}}$$

$E \mapsto 2t + \delta z$ only really affects the first term which becomes $\delta z^{\frac{1}{2}}$ and using polar coordinates this becomes $\rho^{\frac{1}{2}} e^{i\frac{\phi}{2}}$. Circling the branch point above gives $e^{i\frac{\pi}{2}}$=i, but circling the branch-point below gives $e^{-i\frac{\pi}{2}}$=-i and so the two values are different and the branch structure makes sense. We can then proceed along the branch cut until we arrive at the second singularity at E=-2t. The mapping $E \mapsto -2t + \delta z$ now only really affects the second term which becomes $\delta z^{\frac{1}{2}}$, as before, and using polar coordinates this becomes $\rho^{\frac{1}{2}} e^{i\frac{\phi}{2}}$. The second term is not affected by this, but this term has the discontinuity and is i on the upper branch and -i on the lower branch. If we circle above the second branch point we get a second i which gives i^2=-1 in total and if we circle below the second branch point then we get a second -i which gives $(-i)^2$=-1 in total and this connects up to the upper branch with no discontinuity. The minus one might initially be a source of concern, but this is just the sign of E which is supposed to flip if we employ the more natural quantity

$$E\left[1 - \frac{4t^2}{E^2}\right]^{\frac{1}{2}}$$

designed to be good at infinity. The cut between the two branch points does indeed give us the natural description at infinity.

These issues give us access to the main problem with dealing with branch cuts. The notation we use looks like a real square root *but it is not!* When we take a square-root it is best to use polar coordinates for which

$$z = \rho e^{i\phi} \quad \Rightarrow \quad z^{\frac{1}{2}} = \pm \rho^{\frac{1}{2}} e^{i\frac{\phi}{2}}$$

and the angle is active and changes as we move around in the complex plane. What we just did is track this angle by staying close to the real-axis. When we are on the real axis we saw that sometimes the answer was imaginary, and sometimes it was real, but the sign when it was real was subtle and we could not simply think of the square-root symbol as being equivalent to the real analogue. In practice, we choose our branch-cut and then we work out what the signs on the real axis are and then we know the meaning of the complex square-root symbol. It is always true that the function is analytic as we traverse the complex plane and so we never transit from one branch to another without crossing the branch cut.

We can now think about calculating the total number and energy using our Green's function. We find

$$\frac{\mathcal{N}}{N} = \int_C \frac{dz}{2\pi i} f(z-\mu) \frac{1}{\left[z^2 - 4t^2\right]^{\frac{1}{2}}} \qquad \frac{\mathcal{E}}{N} = \int_C \frac{dz}{2\pi i} z f(z-\mu) \frac{1}{\left[z^2 - 4t^2\right]^{\frac{1}{2}}}$$

and we can collapse the contour down to wrap around the branch cut, with the anti-clockwise contour giving $\left[z^2 - 4t^2\right]^{\frac{1}{2}} \mapsto i\left[4t^2 - z^2\right]^{\frac{1}{2}}$ on the top piece and $\left[z^2 - 4t^2\right]^{\frac{1}{2}} \mapsto -i\left[4t^2 - z^2\right]^{\frac{1}{2}}$ on the bottom piece. The top piece has the contour going in the wrong direction and so reversing the direction cancels the sign and makes both terms contribute the same giving eventual real integrals

$$\frac{\mathcal{N}}{N} = \int_{-2t}^{2t} \frac{dz}{\pi} \frac{f(z-\mu)}{\left[4t^2 - z^2\right]^{\frac{1}{2}}} \qquad \frac{\mathcal{E}}{N} = \int_{-2t}^{2t} \frac{dz}{\pi} \frac{z f(z-\mu)}{\left[4t^2 - z^2\right]^{\frac{1}{2}}}$$

There are two lessons available from this calculation, firstly, the Green's function gives us access to the density of states, and for this example

$$\rho(E) = \frac{1}{\pi} \frac{\theta[4t^2 - E^2]}{\left[4t^2 - E^2\right]^{\frac{1}{2}}}$$

and secondly that Cauchy's theorem allows one to collapse a contour down on to a branch cut, but the integral along the branch cut gives you a real analysis integral and cannot be tackled immediately using the residue theorem.

The density of states

For a finite system we can count the number of particles using the θ-function

$$N(E) = \sum_n \theta(E - E_n)$$

and if the energy, E_n, is below E the term in the sum contributes unity, whereas if the energy is above E the term contributes zero. This quantity is simply the number of energies below E. The density of states is defined to be

$$\rho(E) = \frac{dN}{dE}(E)$$

which is a bit pathological for this finite case, because almost always this quantity vanishes and at an energy level the quantity is discontinuous and the derivative does not exist. Using distributions we can recognise the δ-function and formally

$$\rho(E) = \sum_n \delta(E - E_n)$$

To make a connection with the usual smooth density of states we need to take the continuum limit and allow the number of solutions to diverge so that the number appears continuous. Physically, we need to average over a tiny energy window, chosen to be much

larger than the separation of energies but tending to zero in the limit that the system size diverges. In the previous section we observed that

$$\mathcal{N} = \int_C \frac{dz}{2\pi i} f(z-\mu) Trace\, G(z)$$

and for a finite system we catch all of the poles and so from the residue theorem

$$\mathcal{N} = \sum_n f(E_n - \mu)$$

and in the limit $T \mapsto 0$ the Fermi-function becomes a θ-function,

$$\lim_{T\mapsto 0} f(E-\mu) = \theta(\mu - E)$$

and we have the previous definition of $N(\mu)$. In terms of the density of states

$$N(\mu) = \int_{-\infty}^{\mu} dE\rho(E) = \int_{-\infty}^{\infty} dE\rho(E)\theta(\mu - E) = \lim_{T\mapsto 0}\int_{-\infty}^{\infty} dE\rho(E) f(E-\mu)$$

and we can see the connection to the density of states alluded to in the previous section. As we take the limit of more and more atoms the number and density of states both increase and so we need to count the number per atom, which is the previous division by N in the last section and then the contour in the equation

$$\frac{\mathcal{N}}{N} = \lim_{T\mapsto 0}\int_C \frac{dz}{2\pi i} f(z-\mu) G_T(z) \quad G_T(z) \equiv \frac{1}{N} Trace\, G(z)$$

can be collapsed down to the real axis to give

$$\frac{\mathcal{N}}{N} = \lim_{T\mapsto 0}\int_{-\infty}^{\infty} dE f(E-\mu)\frac{1}{2\pi i}\lim_{\delta\mapsto 0}\left[G_T(E-i\delta) - G_T(E+i\delta)\right]$$

where for our anti-clockwise contour the bottom limb (E-$i\delta$) is in the usual direction, from minus infinity to infinity, but the top limb (E+$i\delta$) is in the wrong direction and needs an additional minus sign. We can read off that

$$\rho(E) = \frac{1}{2\pi i}\lim_{\delta\mapsto 0}\left[G_T(E-i\delta) - G_T(E+i\delta)\right]$$

This provides an explanation for where branch cuts usually appear in physical problems. For a Green's function, wherever the spectrum exists there is an automatic discontinuity in the imaginary part of the function which physically corresponds to the density of states. The real-axis constitutes an automatic branch-cut wherever the spectrum exists!

There is a real-analysis method of thinking about these issues which is more usually supplied in books. It is elementary that

$$\frac{1}{E - E_n - i\delta} = \frac{E - E_n + i\delta}{(E-E_n)^2 + \delta^2}$$

and it is also elementary that

$$\lim_{\delta \mapsto 0} \frac{\delta}{(E - E_n)^2 + \delta^2} = \pi \delta(E - E_n)$$

and then

$$\frac{1}{\pi} \lim_{\delta \mapsto 0} \Im \, Trace \, G(E - i\delta) = \sum_n \delta(E - E_n) = \rho(E)$$

which makes sense even before the continuum limit is considered.

Impurity Theory

Small local distortions in Hamiltonians can be mathematically exactly controlled by impurity theory. This theory allows the exact binding energy of bound states to be found and is useful in superconductivity where bound states between pairs of electrons form. The theory is centred on the previous Green's functions and there needs to be a solvable Hamiltonian, designated by H_0, and a local distortion, designated by H_1. For us these are simply matrices and the H_1 will start out being a 1×1 matrix. The simplest problem to tackle is the previous ring of atoms with one special atom somewhere around the ring, which has the same binding energy to the neighbours but a different atomic energy, -V say. The full Hamiltonian is then

$$H = H_0 + H_1$$

where H_1 is a 1×1 matrix with -V being the energy of the special atom, found on the diagonal at the appropriate site and H_0 is one of the previous Hamiltonian matrices, solvable using the roots of unity. We have the solvable Green's function

$$G_0(E) = [E - H_0]^{-1}$$

which we have already found, and the Green's function that we want

$$G(E) = [E - H_0 - H_1]^{-1}$$

which would have a pole at any new energy that is an eigenvalue of the new problem but not of the original. The initial matrix algebra is generic to all problems and need only be performed once and then used when needed.

We focus on $\Sigma(E)$, which is related to the previous Green's functions by

$$G(E) = G_0(E) + G_0(E)\Sigma(E)G_0(E)$$

because $\Sigma(E)$ is local, for us is a 1×1 matrix, and is much simpler than the full $G(E)$. Multiply on the right by $G_0(E)^{-1}$ and on the left by $G(E)^{-1}$ to get

$$G_0(E)^{-1} = G(E)^{-1} + G(E)^{-1}G_0(E)\Sigma(E)$$

Now use the facts that

$$G(E)^{-1} = E - H_0 - H_1 = G_0(E)^{-1} - H_1 \quad \Rightarrow \quad G(E)^{-1}G_0(E) = 1 - H_1 G_0(E)$$

to get

$$H_1 = (1 - H_1 G_0(E))\Sigma(E) \quad \Rightarrow \quad \Sigma(E) = [1 - H_1 G_0(E)]^{-1} H_1$$

and this final equation is the key being a 1×1 matrix for us, but in general being a finite matrix if H_1 is local. We previously found the diagonal piece of $G_0(E)$ and so if $E \in (-\infty, -2t)$ we find

$$\Sigma(E) = \frac{1}{1 - \frac{V}{[E^2 - 4t^2]^{\frac{1}{2}}}}(-V)$$

and if $E \in (2t, \infty)$ then

$$\Sigma(E) = \frac{1}{1 + \frac{V}{[E^2 - 4t^2]^{\frac{1}{2}}}}(-V)$$

new complex functions with more complicated branch structure.

Operator Inverses

Working with Green's functions means that taking an inverse becomes the precursor to diagonalisation and some technical proficiency is required in dealing with inverses. There is a sometimes useful result which involves the inverse of a 2×2 matrix of operators. Provided that the diagonal terms are invertible, then the full matrix can be inverted, without having to commute the operators. If

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} \equiv \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

where a, b, c and d are known and A, B, C and D are to be determined. Simply demanding that the multiple is unity provides two sets of two equations

$$aA + bC = 1 \quad cA + dC = 0 \quad \Rightarrow \quad C = -d^{-1}cA \quad \left[a - bd^{-1}c\right] A = 1$$

and

$$aB + bD = 0 \quad cB + dD = 1 \quad \Rightarrow \quad B = -a^{-1}bD \quad \left[d - ca^{-1}b\right] D = 1$$

and so, provided the inverses exist, then

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \begin{bmatrix} \left[a - bd^{-1}c\right]^{-1} & -a^{-1}b\left[d - ca^{-1}b\right]^{-1} \\ -d^{-1}c\left[a - bd^{-1}c\right]^{-1} & \left[d - ca^{-1}b\right]^{-1} \end{bmatrix}$$

This result is useful for us because the local operator can be described as

$$H_1 \equiv \begin{bmatrix} \eta & 0 \\ 0 & 0 \end{bmatrix}$$

if we split up our basis into states for which H_1 is non-zero and states for which H_1 is zero. A quite general Green's function in this basis

$$G_0(E) = \begin{bmatrix} \alpha & \beta \\ \beta^\dagger & \gamma \end{bmatrix}$$

then allows

$$\Sigma(E) = \left[1 - H_1 G_0(E)\right]^{-1} H_1 \equiv \begin{bmatrix} 1-\eta\alpha & -\eta\beta \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} \eta & 0 \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} [1-\eta\alpha]^{-1} & [1-\eta\alpha]^{-1}\eta\beta \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \eta & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} [1-\eta\alpha]^{-1}\eta & 0 \\ 0 & 0 \end{bmatrix}$$

and we only need the Green's function in the subspace where H_1 does not vanish; α. This is the mathematical reason that our problem turns into a 1×1 matrix problem and the only quantity that we need is $G_0(E)_{jj}$, the diagonal element.

At a technical level, one should diagonalise H_1 and then work in the subspace where the eigenvalues of H_1 are non-zero. For example, if there are two impurities which are second-neighbours, we do not need to deal with the atom between them, because the associated state has a zero eigenvalue for H_1. If there are N local impurities we need an $N \times N$ matrix, although the Green's function restricted to this subspace may be quite complicated and tricky to determine. In the subspace where H_1 is non-zero, it is also invertible and so we often use the final formula

$$\Sigma(E) = \left[H_1^{-1} - G_0(E)\right]^{-1} \equiv \left[\eta^{-1} - \alpha\right]^{-1}$$

to simplify calculations, where we work in the finite subspace where H_1 is non-zero.

Impurities in tight-binding theory

We previously found our Green's function for the case where the poles are parameterised by z_+, which is the pole inside the unit circle, as

$$G_0(E)_{jj'} = \frac{z_+^{|j-j'|}}{t\left(z_+ - \frac{1}{z_+}\right)}$$

which depicts as

$$G_0(E) = \frac{1}{t\left(z_+ - \frac{1}{z_+}\right)} \begin{bmatrix} . & . & . & . & . & . & . \\ . & 1 & z_+ & z_+^2 & z_+^3 & z_+^4 & . \\ . & z_+ & 1 & z_+ & z_+^2 & z_+^3 & . \\ . & z_+^2 & z_+ & 1 & z_+ & z_+^2 & . \\ . & z_+^3 & z_+^2 & z_+ & 1 & z_+ & . \\ . & z_+^4 & z_+^3 & z_+^2 & z_+ & 1 & . \\ . & . & . & . & . & . & . \end{bmatrix}$$

where

$$E = -t\left(z_+ + \frac{1}{z_+}\right)$$

which solves to give either

$$z_+ = \frac{1}{2t}\left(-E - \left[E^2 - 4t^2\right]^{\frac{1}{2}}\right)$$

if $E \in (-\infty, -2t)$ or

$$z_+ = \frac{1}{2t}\left(-E + \left[E^2 - 4t^2\right]^{\frac{1}{2}}\right)$$

if $E \in (2t, \infty)$. We are now in a position to find $\Sigma(E)$ for any configuration of impurities.

(1) One impurity gives

$$\Sigma(E) = \frac{1}{\frac{\pm 1}{[E^2-4t^2]^{\frac{1}{2}}} - \frac{1}{V}}$$

where the plus sign is for $E < -2t$ and the minus sign is for $E > 2t$. The energies of the system are poles in $G(E)$, and since we know that $G_0(E)$ has poles in $(-2t, 2t)$, any new bound-states caused by the impurity must be poles in $\Sigma(E)$. This then provides us with a new type of pole to find, in situations where there are branch singularities. It is clear that if $V > 0$ then any new pole occurs when $E < -2t$ and that if $V < 0$ new poles would occur when $E > 2t$. We can solve for the vanishing of the denominator by squaring

$$\pm\left[E^2 - 4t^2\right]^{\frac{1}{2}} = V \quad \Rightarrow \quad E^2 - 4t^2 = V^2 \quad \Rightarrow \quad E = \mp\left[4t^2 + V^2\right]^{\frac{1}{2}}$$

where we have chosen the signs carefully to be consistent with the constraints. There is always a bound state, no matter how small V is and if the impurity gains a bit of energy the state is below the existing states and if the impurity loses a bit of energy the state is above the existing states.

(2) Two neighbouring impurities of the same strength is described by a diagonal impurity matrix proportional to the identity, so we need to calculate

$$\Sigma(E) = \begin{bmatrix} \frac{\pm 1}{[E^2-4t^2]^{\frac{1}{2}}} - \frac{1}{V} & \frac{\pm z_+}{[E^2-4t^2]^{\frac{1}{2}}} \\ \frac{\pm z_+}{[E^2-4t^2]^{\frac{1}{2}}} & \frac{\pm 1}{[E^2-4t^2]^{\frac{1}{2}}} - \frac{1}{V} \end{bmatrix}^{-1}$$

or at the very least determine any new poles. Interestingly, for this case (and many others), the solving of the quadratic is not required to find the energies of the bound-states! If we keep with the parametrisation then we have

$$E = -t\left(z_+ + \frac{1}{z_+}\right)$$

and

$$\Sigma(E) = \begin{bmatrix} \frac{-1}{t\left(z_+ - \frac{1}{z_+}\right)} - \frac{1}{V} & \frac{-z_+}{t\left(z_+ - \frac{1}{z_+}\right)} \\ \frac{-z_+}{t\left(z_+ - \frac{1}{z_+}\right)} & \frac{-1}{t\left(z_+ - \frac{1}{z_+}\right)} - \frac{1}{V} \end{bmatrix}^{-1}$$

and so the singularities in $\Sigma(E)$ can be determined as values of z_+ which can be found and substituted into the equation for the energy. The subtleties are that the sign issue is resolved by this analysis and both signs are dealt with at the same time. The second subtlety is that the algebra happens to be easier. $\Sigma(E)$ has a clear permutation

symmetry, one can swap the two impurities and we have the same problem. This symmetry is controlled by the 2×2 matrix

$$T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

which is translational invariance on two sites. The two roots of unity are ± 1 and the matrix that diagonalises this problem is

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = U^\dagger = U^{-1}$$

and a general matrix with this symmetry is

$$\begin{bmatrix} a & b \\ b & a \end{bmatrix} = a + bT$$

and has eigenvalues $a \pm b$. For the current example the singularities are therefore controlled by

$$\frac{-1 \mp z_+}{t\left(z_+ - \frac{1}{z_+}\right)} = \frac{1}{V} = \frac{z_+}{t(1 \mp z_+)}$$

where the $\pm$ is associated with this symmetry and is independent of the previous $\pm$ which related to whether the energy was positive or negative. This final equation is trivial to solve and gives

$$z_+ = \frac{t}{V \pm t} \quad \Rightarrow \quad E = -V \mp t - \frac{t^2}{V \pm t}$$

If V is positive, then there is always at least one bound-state, but the second only appears if $V > 2t$.

Parameterisation and control of signs

In the previous section there are many choices of sign, $\pm$, and it is usually better not to make this choice of sign but to parameterise it. The central quantity was

$$\delta \equiv \left[E^2 - 4t^2\right]^{\frac{1}{2}}$$

and when $E <$-$2t$ it had one sign and when $E >2t$ it had the other. It is usually better to *define* the δ through the choice of which $z_\pm$ is inside the contour. It is also useful to eliminate unnecessary energy scales. For our problem we naturally extract t and then $E \equiv t\epsilon$, $V = t\nu$ and then the controlling quadratic becomes

$$z^2 + \epsilon z + 1 = 0$$

and we can define

$$z_+ = -\frac{\epsilon}{2} + \frac{\Delta}{2}$$

to be the root which is inside the contour. Clearly

$$\Delta^2 = \epsilon^2 - 4$$

and the diagonal element of the Green's function satisfies

$$tG_{00}(\epsilon) = \frac{1}{\Delta}$$

Previously we had to employ $\pm$ to write this down, but now the choice of sign is parameterised by the fact that z_+ is inside the contour. The calculation of $\Sigma(\epsilon)$ remains the same

$$\Sigma(\epsilon) = \left[\frac{1}{H_1} - G_{00}(\epsilon)\right]^{\frac{1}{2}} = \left[-\frac{1}{t\nu} - \frac{1}{t\Delta}\right]^{-1} = -\frac{t\nu\Delta}{\Delta+\nu}$$

The new pole is now hidden in the algebra, but the sign of Δ is now clear, the denominator must vanish and so if $\nu > 0$ then we need the negative sign and if $\nu < 0$ we need the positive sign. We can then unravel the algebra to find the solution by using

$$\Delta^2 = \epsilon^2 - 4 = \nu^2 \quad \Rightarrow \quad \epsilon = \pm\left[\nu^2 + 4\right]^{\frac{1}{4}}$$

and now Δ=-ν provides

$$z_+ = -\frac{\nu}{2} - \frac{\epsilon}{2}$$

and for the case $\nu > 0$ we need to choose ϵ to be negative to make z_+ inside the unit circle and for $\nu < 0$ we need to choose ϵ to be positive to make z_+ inside the unit circle; a different sign was actually chosen eventually! This level of subtlety is quite usual.

We can also calculate the wavefunction for our bound state and this comes from the residue of the Green's function

$$G(E)_{jj'} = G_0(E)_{jj'} + G_0(E)_{j0}\Sigma(E)G_0(E)_{0j'}$$

The new pole is in Σ, so up to normalisation, we need the residue of $\Sigma(E)$ and then the wavefunction is proportional to $G_{j0}(E)$ evaluated at the energy of the bound state. To calculate the residue we need to appreciate that there is an elementary pole and so we can use Taylor's theorem on the denominator to find the residue

$$R[\epsilon] = t\frac{N(\epsilon)}{\frac{dD}{d\epsilon}(\epsilon)}$$

where the t is the scale factor turning the multiplication by E-$t\epsilon^*$ into one for $\epsilon - \epsilon^*$. We can use the sign free definition

$$\Delta^2 = \epsilon^2 - 4 \quad \Rightarrow \quad 2\Delta\frac{d\Delta}{d\epsilon} = 2\epsilon$$

to deduce that

$$R[\epsilon] = -\frac{t^2\nu\Delta}{\frac{d\Delta}{d\epsilon}} = -\frac{t^2\nu\Delta^2}{\epsilon} = -\frac{t^2\nu^3}{\epsilon}$$

and the signs mean that this is positive definite. We can then deduce that

$$\psi(\epsilon)_j\psi^\dagger(\epsilon)_{j'} = -\frac{\nu^3}{\epsilon}\frac{1}{\Delta^2}\left[-\frac{\nu}{2}-\frac{\epsilon}{2}\right]^{|j|+|j'|} = -\frac{\nu}{\epsilon}\left[-\frac{\nu}{2}-\frac{\epsilon}{2}\right]^{|j|+|j'|}$$

One can even check normalisation, by setting $j'=j$ and summing

$$\sum_j \mid \psi \mid_j^2 = -\frac{\nu}{\epsilon}\left(-1+2\sum_{n=0}^{\infty}\left[-\frac{\nu}{2}-\frac{\epsilon}{2}\right]^{2n}\right)$$

as the middle term only appears once, and then the geometric series gives

$$\sum_j \mid \psi \mid_j^2 = -\frac{\nu}{\epsilon}\left(-1+\frac{2}{1-\left[\frac{\nu+\epsilon}{2}\right]^2}\right) = -\frac{\nu}{\epsilon}\left(\frac{4+(\nu+\epsilon)^2}{4-(\nu+\epsilon)^2}\right) = -\frac{\nu}{\epsilon}\left(\frac{4+\nu^2+\epsilon^2+2\nu\epsilon}{4-\nu^2-\epsilon^2-2\nu\epsilon}\right)$$

$$= -\frac{\nu}{\epsilon}\frac{2\epsilon^2+2\nu\epsilon}{-2\nu^2-2\nu\epsilon} = 1$$

where we used $\epsilon^2=4+\nu^2$. This brings us on to our final topic, which is.....

Algebra is very difficult!

In this section we will investigate bound-states in the model where there is second-neighbour hopping. The parameterisation idea is very powerful and the original diagonal Green's function

$$t_2G_{00}(E) = \int_C \frac{dz}{2\pi iz}\frac{1}{\frac{E}{t_2}+\frac{t_1}{t_2}\left(z+\frac{1}{z}\right)+\left(z+\frac{1}{z}\right)^2-2}$$

is easily converted into

$$t_2G_{00}(E) = \int_C \frac{dz}{2\pi iz}\frac{1}{z_+ + \frac{1}{z_+} - z_- - \frac{1}{z_-}}\left(\frac{1}{z+\frac{1}{z}-z_+-\frac{1}{z_+}}-\frac{1}{z+\frac{1}{z}-z_--\frac{1}{z_-}}\right)$$

and in terms of $r \equiv \frac{t_1}{t_2}$ and $\epsilon \equiv \frac{E}{t_2}$ the roots of the quadratic

$$x^2+rx+\epsilon-2=0$$

control the parameterisation

$$z_\pm + \frac{1}{z_\pm} = -\frac{r}{2} \pm \Delta \;\; ; \quad \Delta^2 = \frac{r^2}{4}+2-\epsilon$$

For a general complex energy, one of the pair $z_\pm$ and $\frac{1}{z_\pm}$ is inside the unit circle and the other is outside, and if we choose to define the $z_\pm$ to be the one inside, then we can do the integrals using the residue theorem to provide

$$t_2G_{00}(E) = \frac{1}{z_+ + \frac{1}{z_+} - z_- - \frac{1}{z_-}}\left(\frac{1}{z_+-\frac{1}{z_+}}-\frac{1}{z_--\frac{1}{z_-}}\right)$$

$$= \frac{1+z_+z_-}{1-z_+z_-}\frac{1}{\left(z_+ - \frac{1}{z_+}\right)\left(z_- - \frac{1}{z_-}\right)}$$

where the final result cancels out the obvious factor of $(z_+ - z_-)$ and is elementary algebra. Employing the residue theorem is really very easy! To continue we need to solve the two quadratic equations for $z_\pm$ and their inverses, and this gives

$$z_+, \frac{1}{z_+} = -\frac{r}{4} + \frac{\Delta}{2} \pm \frac{\Gamma_+}{2}$$

$$z_-, \frac{1}{z_-} = -\frac{r}{4} - \frac{\Delta}{2} \pm \frac{\Gamma_-}{2}$$

where

$$\Gamma_\pm^2 = \frac{r^2}{2} - 2 - \epsilon \mp r\Delta$$

which can be obtained by multiplying the two roots together. A crucial bit of inobvious algebra is that

$$(\Gamma_+\Gamma_-)^2 = \left(\frac{r^2}{2} - 2 - \epsilon\right)^2 - r^2\Delta^2 = (\epsilon+2)^2 - 4r^2$$

and then we are beset by the choice of signs and how do we pick the two smallest roots? As with parameterisation, it is best to put off the choice until later! The choice of sign for Δ is irrelevant because the answer has the symmetry that if you swap the z_+ with the z_- it is the same, and if we simply flip the sign of either $\Gamma_\pm$ we exchange the two roots and so we can choose later when we know more by flipping the appropriate sign. We will choose

$$z_- = -\frac{r}{4} - \frac{\Delta}{2} + \frac{\Gamma_-}{2} \qquad z_+ = -\frac{r}{4} + \frac{\Delta}{2} + \frac{\Gamma_+}{2}$$

under the assumption that all roots are real and positive and that the r dominates over the Δ. The other solutions are then

$$\frac{1}{z_-} = -\frac{r}{4} - \frac{\Delta}{2} - \frac{\Gamma_-}{2} \qquad \frac{1}{z_+} = -\frac{r}{4} + \frac{\Delta}{2} - \frac{\Gamma_+}{2}$$

which are useful because we can rewrite

$$\frac{1+z_+z_-}{1-z_+z_-} = \frac{\frac{1}{z_+} + z_-}{\frac{1}{z_+} - z_-} = \frac{-r - \Gamma_+ + \Gamma_-}{2\Delta - \Gamma_+ - \Gamma_-}$$

and then

$$\Gamma_-^2 - \Gamma_+^2 = 2\Delta r = (\Gamma_- + \Gamma_+)(\Gamma_- - \Gamma_+)$$

allows an inobvious cancellation down to

$$\frac{1+z_+z_-}{1-z_+z_-} = \frac{r}{\Gamma_- + \Gamma_+} = \frac{\Gamma_- - \Gamma_+}{2\Delta\Gamma_-\Gamma_+}$$

and the final result of

$$t_2 G_{00}(E) = \frac{r}{(\Gamma_- + \Gamma_+)\Gamma_+\Gamma_-} = \frac{\Gamma_- - \Gamma_+}{2\Delta\Gamma_+\Gamma_-}$$

and the denominator satisfies

$$\Delta^2(\Gamma_+\Gamma_-)^2 = \left(\frac{r^2}{4} + 2 - \epsilon\right)(\epsilon + 2 - 2r)(\epsilon + 2 + 2r)$$

and controls the branch structure for one choice of phase. This choice of phase is impressively subtle! There are three natural energies in our denominator; $\epsilon_0 \equiv$-2-2r, $\epsilon_1 \equiv$-2+2r and $\epsilon_2 \equiv \frac{r^2}{4} + 2$. The special case of r=4 finds ϵ_1=ϵ_2 and everywhere else, $\epsilon_2 > \epsilon_1$. The $\Gamma_\pm$ vanish at ϵ_0 and ϵ_1. which control the changes in phase, but at ϵ_2 we find Δ=0 and

$$\Gamma_\pm^2 = \frac{r^2}{4} - 4 = \frac{(r+4)(r-4)}{4}$$

and so if $r > 4$ the $\Gamma_\pm$ are real at this point and if $r < 4$ then they are imaginary. There is one additional energy, $\epsilon_3 = \frac{r^2}{2} - 2$, of interest, in terms of which

$$\Gamma_\pm^2 = \epsilon_3 - \epsilon \mp r\left(\epsilon_2 - \epsilon\right)^{\frac{1}{2}}$$

in terms of which we can understand the behaviour when ϵ is real.

Start out with the limit that $\epsilon \mapsto \infty$ and the details of the Hamiltonian are irrelevant. We find that

$$\Delta^2 \mapsto -\epsilon \quad \Gamma_\pm^2 \mapsto -\epsilon$$

and then the choice of signs requires that

$$-\Gamma_+ = \Gamma_- = \Delta \quad \Rightarrow \quad \frac{\Gamma_- - \Gamma_+}{2\Delta} = 1 = \frac{r}{\Gamma_- + \Gamma_+} \quad \Rightarrow \quad t_2 G(E) \mapsto \frac{1}{\epsilon}$$

as is required from the original inverse formula. When ϵ is large and negative the $\Gamma_\pm$ are real and have opposite signs, but when ϵ is large and positive the $\Gamma_\pm$ are complex, dominantly imaginary and are complex conjugates of each other. We recognise two quite different possibilities, $r > 4$, for which $\epsilon_3 > \epsilon_2 > \epsilon_1 > \epsilon_0$, and $r < 4$, for which $\epsilon_2 > \epsilon_1 > \epsilon_3 > \epsilon_0$. It is useful to define α=$\mid \epsilon_3 - \epsilon \mid$ and β^2=$\mid \epsilon_2 - \epsilon \mid$, where α and β are positive, to allow us to focus on the signs and phases.

For the case $r > 4$, the first region is $\epsilon > \epsilon_3$, for which

$$\Gamma_\pm^2 = -\alpha \mp ir\beta$$

and then ϵ=ϵ_3 marks where α changes sign. In the next region $\epsilon_3 > \epsilon > \epsilon_2$ we find

$$\Gamma_\pm^2 = \alpha \mp ir\beta$$

and the two $\Gamma_\pm$ remain conjugates and become more real. The next transition, when ϵ=ϵ_2, marks when β^2 changes sign, at which point the two $\Gamma_\pm$ are equal and are real. In the next and final two regions

$$\Gamma_\pm^2 = \alpha \pm r\beta$$

are both real and the next transition, when ϵ=ϵ_1, marks when α=$r\beta$ and Γ_+^2 changes sign and Γ_+ becomes pure imaginary. This is the branch-point and marks the start of the branch cut. At this point the $z_\pm$ reside on the unit circle and we cannot say which is inside the unit circle and which is outside, either of the two can be chosen, marking the two branch choices. The final transition ϵ=ϵ_0 also marks when α=$r\beta$ and Γ_+^2 changes sign back, but Γ_+ becomes real with the opposite sign to Γ_-. This marks the second branch points and closes the branch cut. One might ask what happened to the other singularity, ϵ=ϵ_2, and this is quite subtle. When this was relevant, the $\Gamma_\pm$ where complex conjugates and so this singularity cancelled!

For the case $r < 4$, the first region is $\epsilon > \epsilon_2$, for which

$$\Gamma_\pm^2 = -\alpha \mp ir\beta$$

as before, and now ϵ=ϵ_2 comes first. This is where β^2 changes sign and the two $\Gamma_\pm$ become pure imaginary. At this point $z_\pm$ reside on the unit circle and so this point is a branch singularity and we have entered the branch cut. For this case all three singularities are active and we have three branch points! In the next region, $\epsilon_2 > \epsilon > \epsilon_1$, we have

$$\Gamma_\pm^2 = -\alpha \mp r\beta$$

and at the next transition, ϵ=ϵ_1, we have α=$r\beta$ and Γ_-^2 vanishes and Γ_- is subsequently real. At the next transition ϵ=ϵ_3, α changes sign and we have

$$\Gamma_\pm^2 = \alpha \mp r\beta$$

with Γ_+ remaining pure imaginary and Γ_- being real. The final transition, when ϵ=ϵ_0 marks the change in sign for Γ_+^2, as in the last case, and both $\Gamma_\pm$ become real but with opposite signs.

You should be able to see that understanding the branch structure of even the simplest analytic function is subtle and taxing!

The density of states is non-zero when the spectrum exists. In real space we have

$$E = -2t_1 \cos k - 2t_2 \cos 2k \quad \Rightarrow \quad \epsilon = -2rc - 2(2c^2 - 1) = -\left(2c + \frac{r}{2}\right)^2 + 2 + \frac{r^2}{4}$$

where we used the previous rescaling and $c \equiv \cos k$. c=1 corresponds to ϵ=-2r-2=ϵ_0 and marks the bottom of the spectrum. The top of the spectrum is ϵ=2+$\frac{r^2}{4}$=ϵ_2 if c=-$\frac{r}{4}$ >-1, which is $r < 4$, and is ϵ=-2+2r=ϵ_1 otherwise, when $r > 4$. This suggests that when $r > 4$ we have a branch cut for $\epsilon \in (\epsilon_0, \epsilon_1)$ whereas when $r < 4$ we have one solution in the region $\epsilon \in (\epsilon_0, \epsilon_1)$ and two solutions in the region $\epsilon \in (\epsilon_1, \epsilon_2)$, in total sympathy with the previous analysis.

To calculate the bound-state energy we need to solve the 1×1 problem

$$G_0(E) = \frac{1}{H_1} \quad \Rightarrow \quad \frac{1}{t_2}\frac{r}{(\Gamma_+ + \Gamma_-)\Gamma_+\Gamma_-} = -\frac{1}{V}$$

and setting $\nu \equiv \frac{V}{t_2}$ this is

$$(\Gamma_+ + \Gamma_-)\Gamma_+\Gamma_- = -r\nu$$

We can eliminate the unknown signs by squaring and using

$$(\Gamma_+\Gamma_-)^2 = (\epsilon + 2)^2 - 4r^2 \equiv P^2$$

to get

$$r^2\nu^2 = P^2(\Gamma_+^2 + \Gamma_-^2 + 2\Gamma_+\Gamma_-) \quad \Rightarrow \quad \frac{r^2\nu^2}{2} = P^2(\frac{r^2}{2} - 2 - \epsilon + \Gamma_+\Gamma_-)$$

and square again to get

$$P^6 = \left[\frac{r^2\nu^2}{2} - P^2\left(\frac{r^2}{2} - 2 - \epsilon\right)\right]^2$$

which is a horrible quintic in ϵ+2. Algebra is very difficult, even this simple problem requires a computer!

Numerically solving the problem: Yet more signs!

The equation we need to solve is

$$(\Gamma_+ + \Gamma_-)\Gamma_+\Gamma_- = -r\nu$$

where independent of the signs the $\Gamma_\pm$ are controlled by

$$\Gamma_\pm^2 = \frac{r^2}{2} - 2 - \epsilon \mp r\Delta$$

and the sign-free definition of Δ is

$$\Delta^2 = \frac{r^2}{4} - 2 - \epsilon$$

The choice of signs that we made was

$$z_+ = -\frac{r}{4} + \frac{\Delta}{2} + \frac{\Gamma_+}{2}$$

$$z_- = -\frac{r}{4} - \frac{\Delta}{2} + \frac{\Gamma_-}{2}$$

and the two $z_\pm$ are *inside* the unit circle. In order to unravel the algebra we need to square our equation to eliminate the signs and this gives (dividing by $2P^2$ after squaring)

$$\frac{r^2}{2} - 2 - \epsilon + P = \frac{r^2\nu^2}{2P^2} \quad \Rightarrow \quad P = \epsilon + 2 - \frac{r^2}{2} + \frac{r^2\nu^2}{2P^2}$$

this equation controls the sign of P and then we square again to make

$$P^2 = \left[\epsilon + 2 - \frac{r^2}{2} + \frac{r^2\nu^2}{2P^2}\right]^2 = (\epsilon+2)^2 - 4r^2$$

which is a quintic equation in ϵ and needs to be solved using the computer. Once we have found ϵ we can then determine the sign of P using its controlling equation. Once we have found P we can use the fact that

$$\Gamma_+ + \Gamma_- = -\frac{r\nu}{P} \qquad \Gamma_+\Gamma_- = P$$

to build the quadratic equation for $\Gamma_\pm$

$$\Gamma^2 + \frac{r\nu}{P}\Gamma + P = 0 \quad \Rightarrow \quad \Gamma_\pm = -\frac{r\nu}{2P} \pm \left[\frac{r^2\nu^2}{4P^2} - P\right]^{\frac{1}{2}}$$

and now we have the signs, but which one is which? Clearly the quadratic equation cannot tell the difference, so we need to go back to $z_\pm$ and which one is inside to answer this question! We know that

$$z_+ = -\frac{r}{4} + \frac{\Delta}{2} + \frac{\Gamma_+}{2} \qquad \frac{1}{z_+} = -\frac{r}{4} + \frac{\Delta}{2} - \frac{\Gamma_+}{2}$$

$$z_- = -\frac{r}{4} - \frac{\Delta}{2} + \frac{\Gamma_-}{2} \qquad \frac{1}{z_-} = -\frac{r}{4} - \frac{\Delta}{2} - \frac{\Gamma_-}{2}$$

and only one choice of $\Gamma_\pm$ will guarantee that the two are inverses of each other. We can then check whether both of the $z_\pm$ are inside the unit circle. There are solutions to the quintic that provide unphysical solutions where both $z_\pm$ are not inside the unit circle, but there are two solutions where they are. The two solutions correspond to the two possible signs of ν and one solution is a bound-state below the spectrum and the other is a bound-state above the spectrum.

In practical applications we have two cases; (1) r>4 and (2) r<4 as was previously observed.
(1) r>4 and the spectrum is $\epsilon \in [\epsilon_0, \epsilon_1]$ and

$$\epsilon_3 > \epsilon_2 > \epsilon_1 > \epsilon_0$$

and when $\epsilon_2 > \epsilon$ we have real solutions for Δ and $\Gamma_\pm$ and when $\epsilon > \epsilon_2$ we have imaginary solutions for Δ and complex solutions for $\Gamma_\pm$. The boundary occurs when

$$\nu = -\frac{2}{r}\left(\frac{r^2}{4} - 4\right)^{\frac{3}{2}}$$

above the top of the spectrum.
(2) r<4 and the spectrum is $\epsilon \in [\epsilon_0, \epsilon_2]$ and

$$\epsilon_2 > \epsilon_1 > \epsilon_3 > \epsilon_0$$

and, above the spectrum, when $\epsilon > \epsilon_2$, we always have imaginary solutions for Δ and complex solutions for $\Gamma_\pm$.

Summary for Complex Variable Theory

The main engine for the course is Cauchy's theorem

$$\int_C dz f(z) = 0$$

if $f(z)$ is analytic in the simply-connected domain bounded by the closed curve C. For functions with extractable singularities this leads to the residue theorem

$$\int_C dz f(z) = 2\pi i \sum_{z_e} W[z_e; C] R[z_e]$$

where $W[z_e; C]$ is the winding number, the number of times the contour, C, circles the extractable singularity, z_e, in an anticlockwise direction and $R[z_e]$ is the residue of the pole at the extractable singularity, z_e. Analytic points, z_a, have a Taylor's expansion

$$f(z) = \sum_{m=0}^{\infty} a_m (z - z_a)^m$$

which converges to the function in a neighbourhood of the point and extractable singularities, z_e, have a Laurent expansion

$$f(z) = \sum_{m=-\infty}^{\infty} a_m (z - z_e)^m$$

which converges to the function in a neighbourhood of the point, and the residue is $R[z_e]$=a_{-1} in this expansion.

You need to be able to find and characterise the singularities in analytic functions and for extractable singularities you need to be able to find the residues. Residues are often found using partial fractions and the cover-up rule, but there is also Taylor's theorem for multiple poles, an n'th order pole has a residue which can be calculated from

$$R[z_e] = \frac{1}{(n-1)!} \left(\frac{d}{dz} \right)^{n-1} [(z - z_e)^n f(z)]_{z=z_e}$$

and sometimes. for an elementary pole, the denominator is a better starting point because for such a pole

$$D[z] = (z - z_e) d[z]$$

then $d[z_e]$ contributes to the residue and

$$d[z_e] = \frac{d}{dz} D[z] \mid_{z=z_e}$$

and so the residue is

$$R[z_e] = \frac{N[z_e]}{\frac{dD}{dz}(z_e)}$$

This is most valuable when there are multiple roots and the denominator is easy to differentiate.

Branch singularities involve non-integer power-laws and cannot be extracted using the residue theorem. Branch singularities are connected to each-other with branch-cuts, lines where the function has a discontinuity across the line, and the best that can be done with Cauchy's theorem is to collapse the contour down ono this line and rerepresent the complex integral as a real integral. This integral may or may not be susceptible to further complex variable ideas, but usually not! Contours may not pass through branch cuts!

Nuclear decay chains and forced harmonic oscillators are the physical examples that are used to test your ability to find singularities, residues and to apply the residue theorem.

Analytic functions satisfy Cauchy-Riemann equations which also imply Laplace's equation. Each real solution to Laplace's equation leads to an analytic function that can be found from Cauchy-Riemann and integration. Analytic functions are also conformal mappings and lead to useful solutions to electrostatics and incompressible, irrotational fluid flow in two-dimensions.

Periodic potentials lead to a Fourier series and a Green's function description of the energetics. We can calculate these quantities using the residue theorem. Impurity theory is an extension to this theory which allows us to construct localised bound-states from localised potential fluctuations. Surprisingly, the residue theorem is quite straightforward but the algebra of dealing with the parameterised motion and properties of the singularities is quite difficult due to branch structure.

Branch singularities cannot be controlled by the residue theorem and Cauchy's theorem allows collapse onto the branch cut and answers in terms of real integrals along the branch cut. Keyhole contours allow the calculation of real integrals which involve non-integer power-laws and the logarithm function.

The next course deals with Fourier transforms, analytic continuation and Riemann surfaces.